Coordinate Measuring Machines and Systems

Hocken R.J., P.H. Pereira (2012)
Coordinate Measuring Machines and Systems
Chapters: 4, 6, and 7

Learning Targets

The student knows:

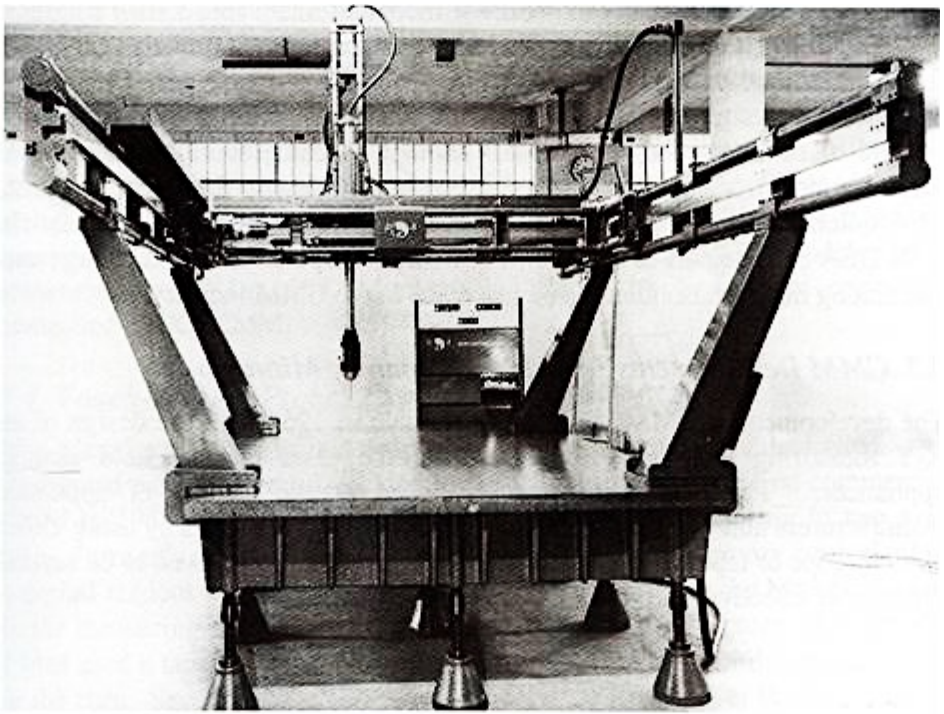
- What a coordinate measuring machine is;
- What a probing system is;
- What coordinate measuring systems are;
- What the process capability is.

COORDINATE MEASURING MACHINE - CMM

CMM: *Measuring system with the means to move a probing system and capability to determine spatial coordinates on a workpiece surface*. (ISO 10360-1)

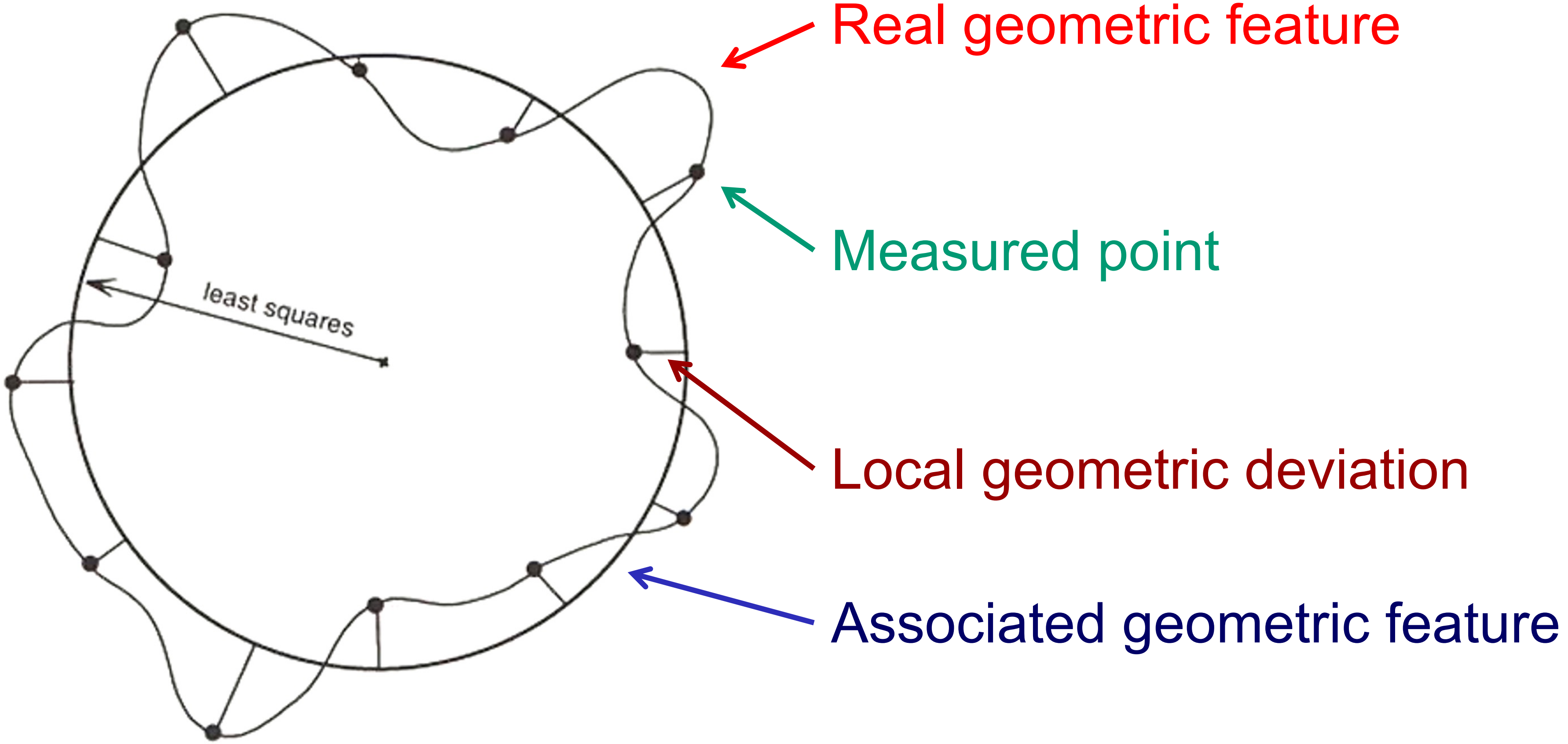
A CMM is a machine that gives a physical realization of a three-dimensional rectilinear Cartesian coordinate system.

COORDINATE MEASURING MACHINE - CMM

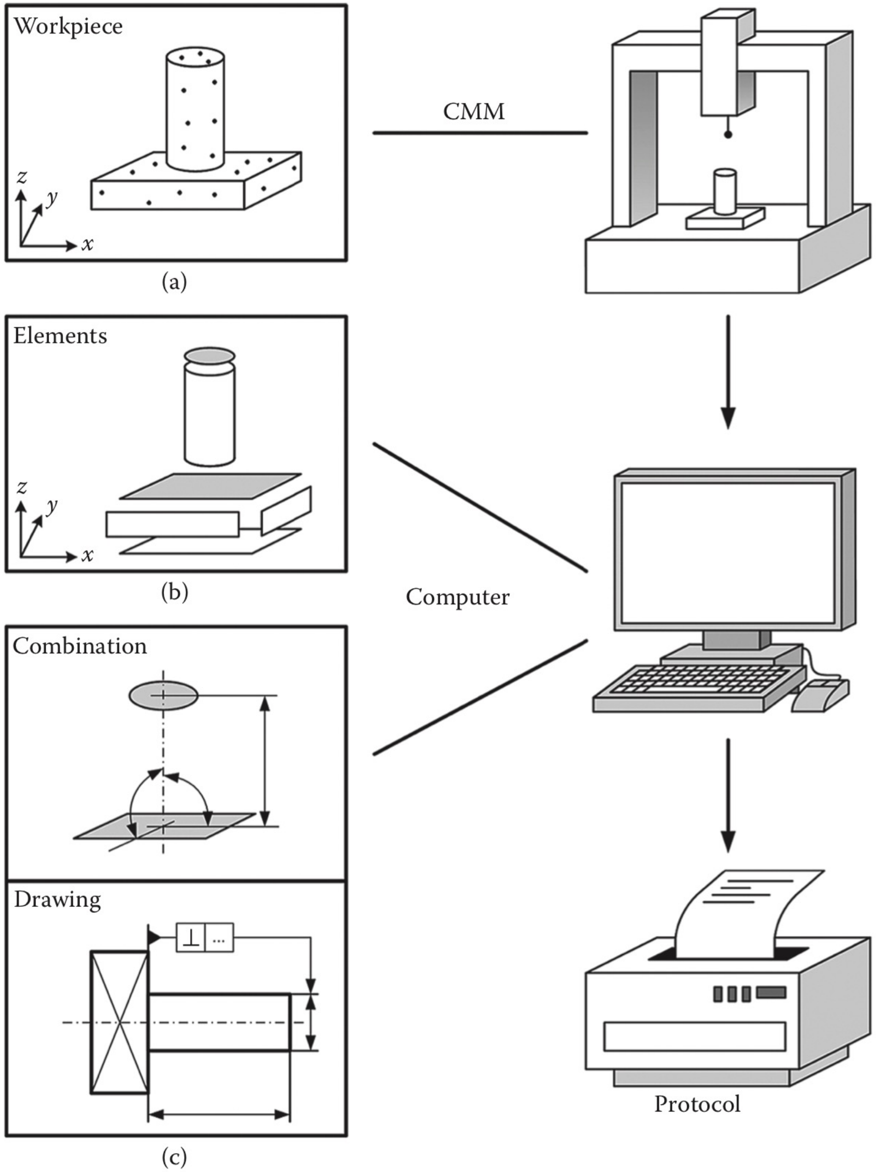


DEA 3D CMM (2500 x 1600 x 600)
European Machine Show
October 1963

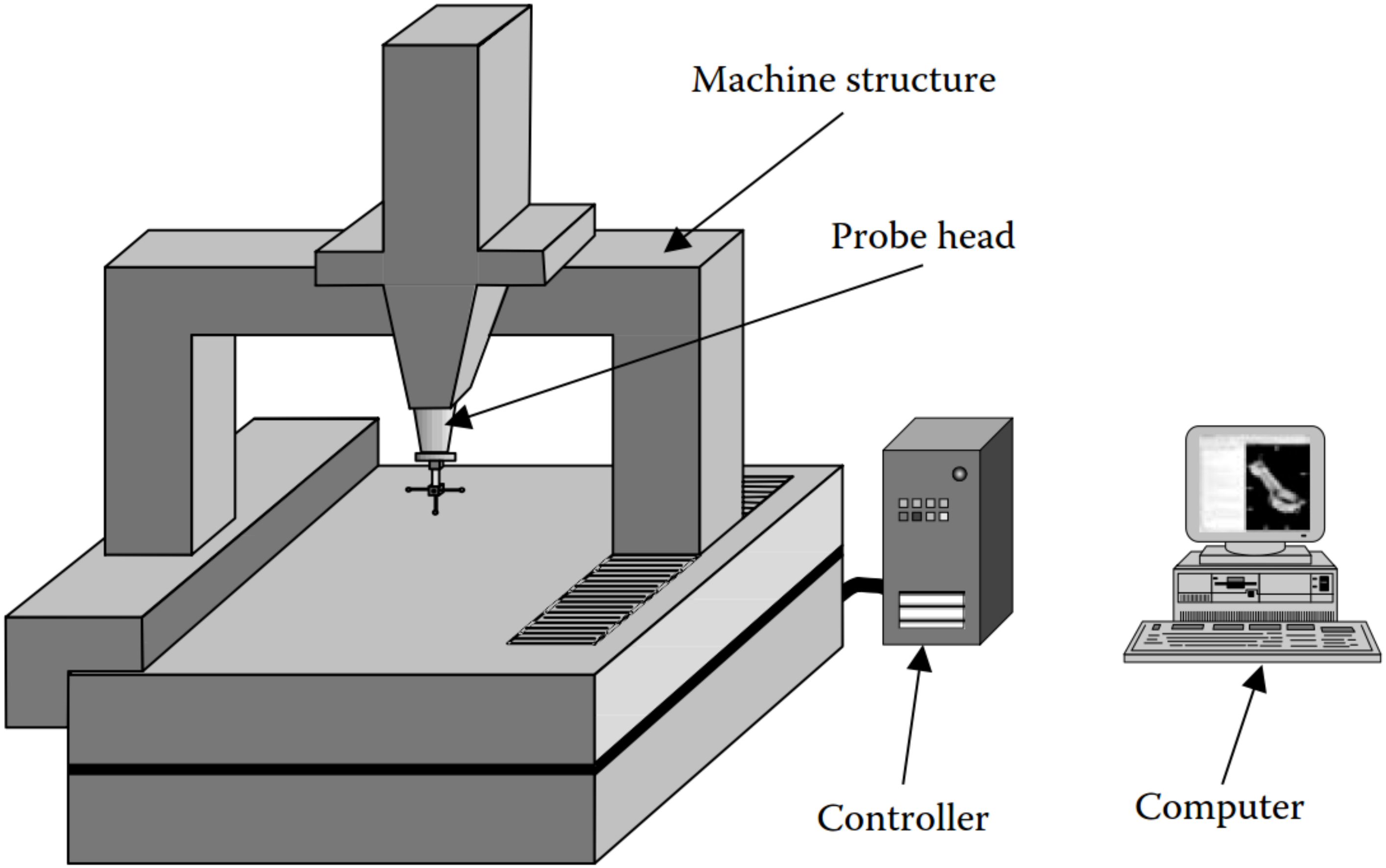
CMM: Working Principle



WORKING PRINCIPLE



COORDINATE MEASURING MACHINE - CMM



CMM: Mechanical Structure

Fundamental requirements

- Static and dynamic stiffness
- Workpiece accessibility
- Ease of operation
- Ease of workpiece load/unload

Ideal mechanical characteristics

- Dimensional stability (short and long term)
- Static and dynamic stiffness
- Lightweight
- High damping capacity
- Low coefficient of thermal expansion
- High thermal conductivity

CMM: Structures

Structures (ISO 10360-1)

A - Cantilever

B - Bridge

C - Gantry

D - Column

E - Horizontal-arm

A

B

C

D

E

Articulated Arm

CMM: Mechanical Structure

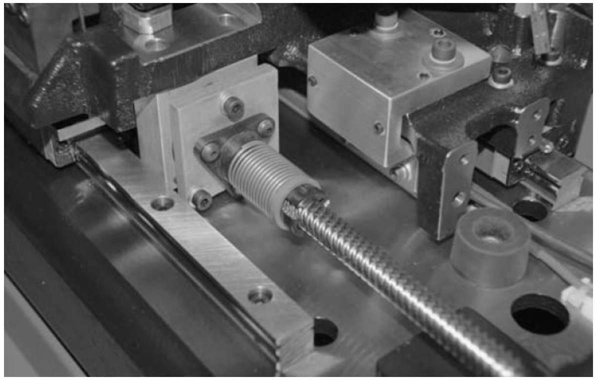
Materials:

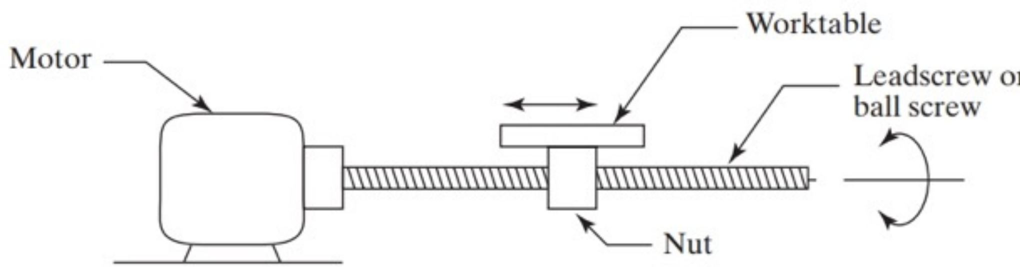
- Steel
- Cast Iron
- Granite (diabase)
- Ceramic
- Aluminum
- Composite (CFRP)

Components:

- Positioning System
- Guideways and bearings
- Displacement transducers

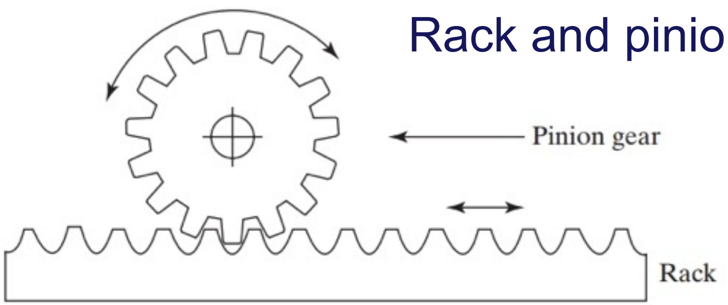
Rotary-to-Linear Motion Conversion

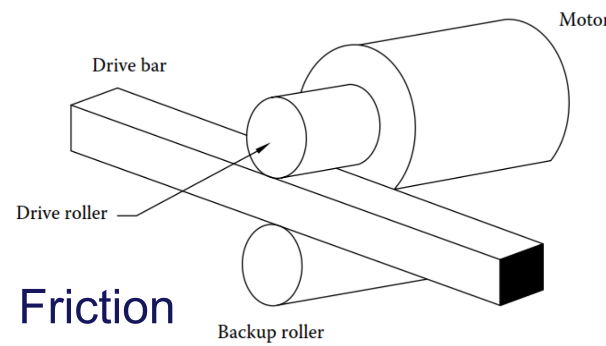




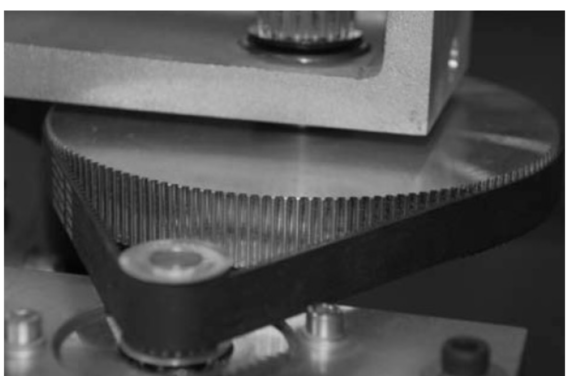
Ball screw

Rack and pinion

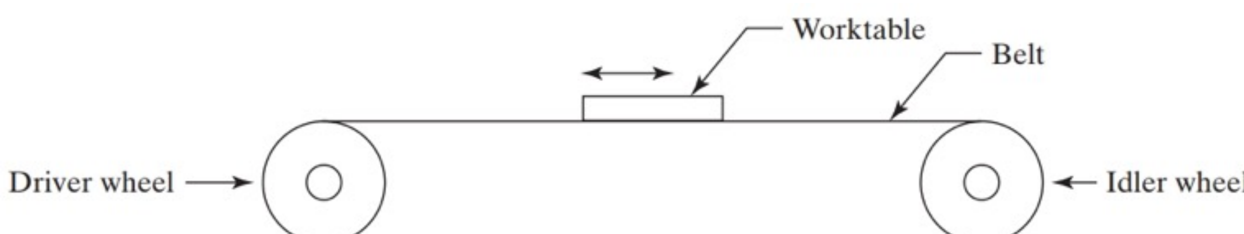




Friction

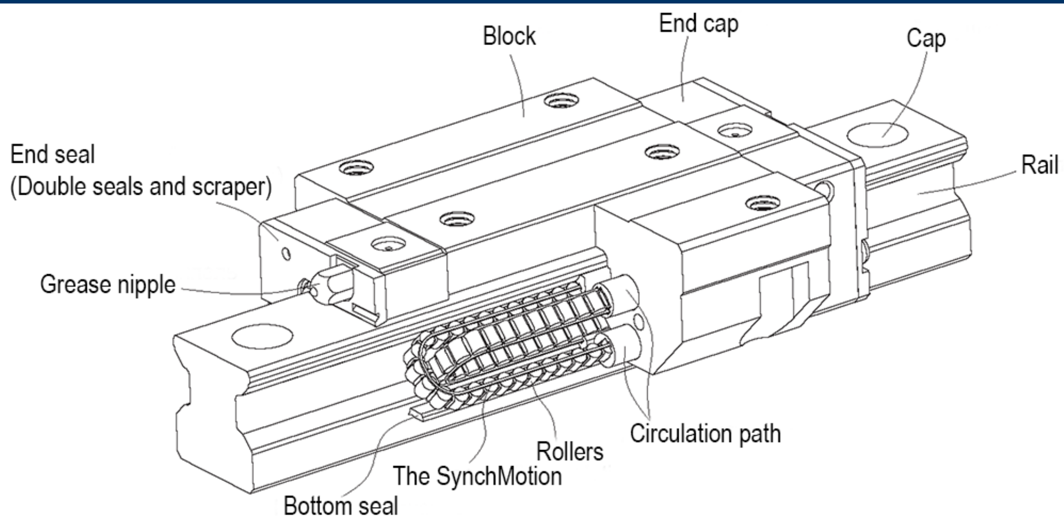


Pulley System



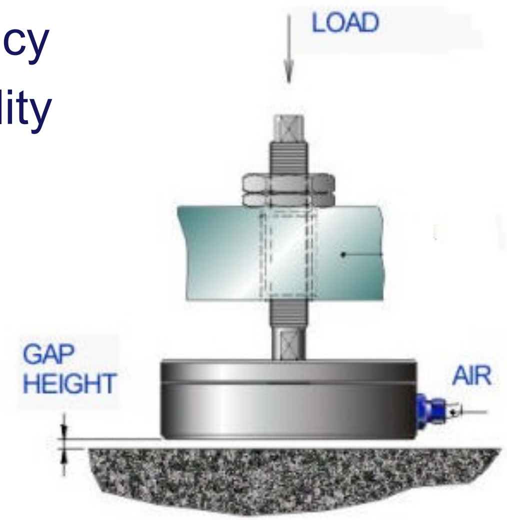
CMM: Guideways and Bearings

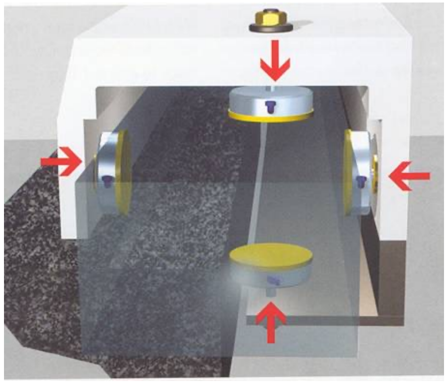
Hard Bearings: Max load

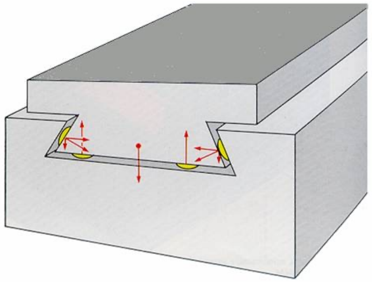


Air Bearings:

- Accuracy
- Durability



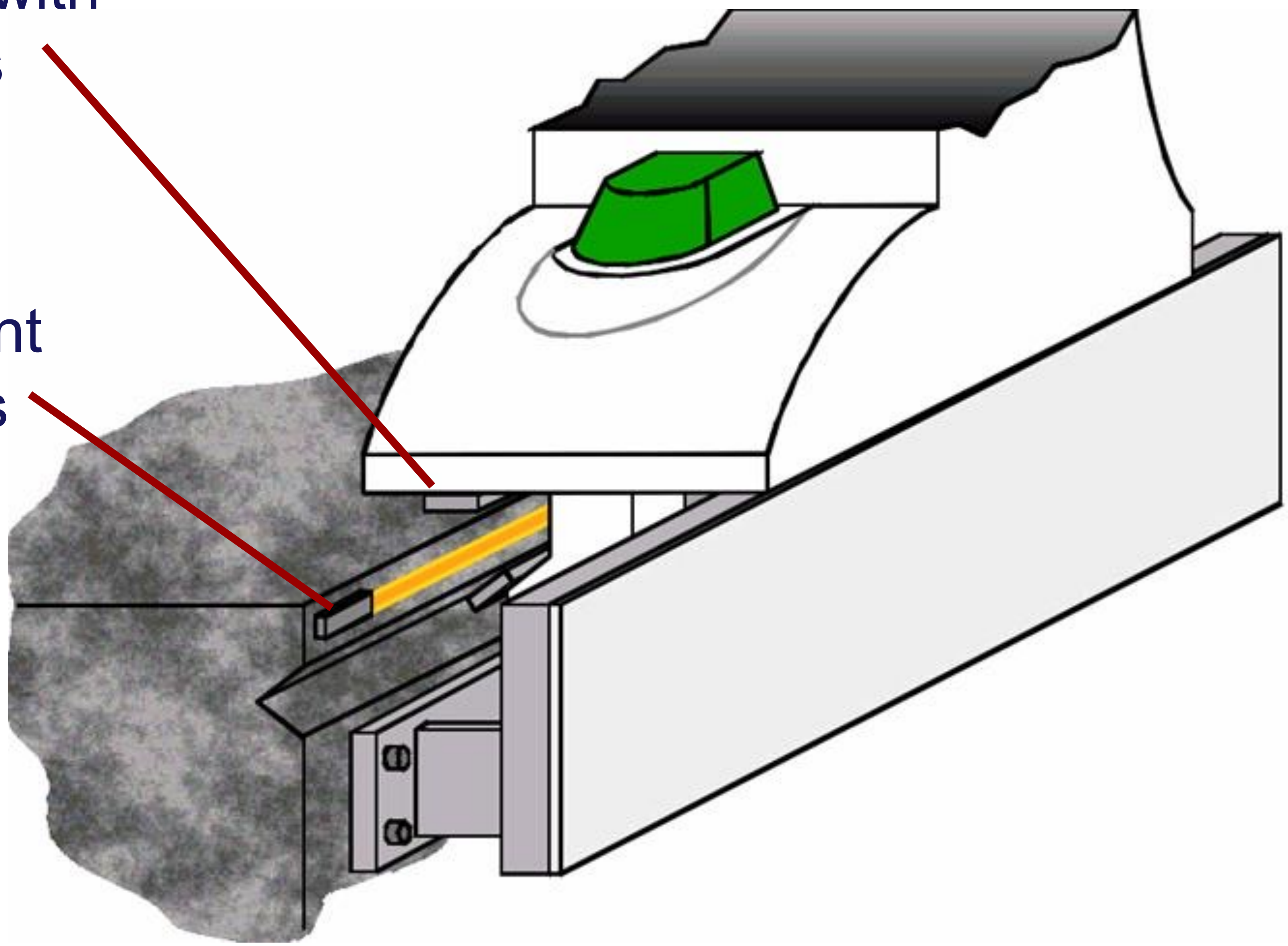




CMM: Displacement Transducers

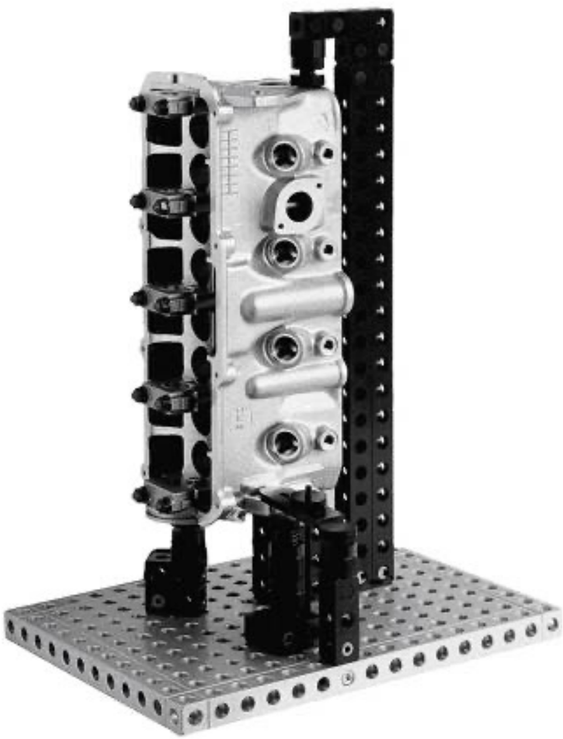
Guideways with Air Bearings

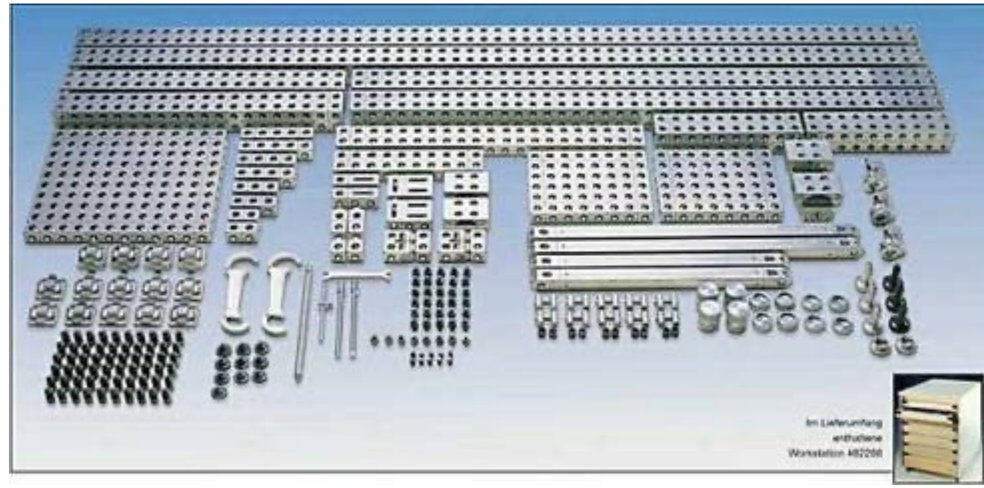
Displacement Transducers

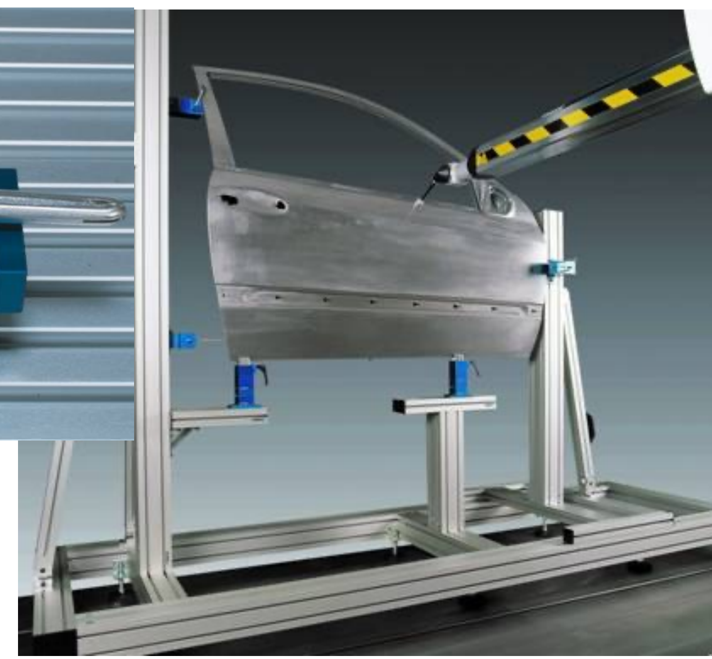


CMM: FIXTURING

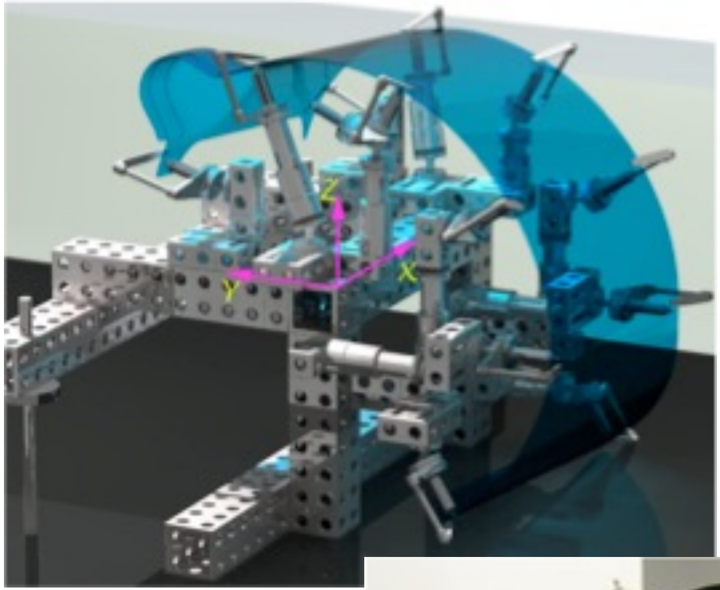
Modular fixturing

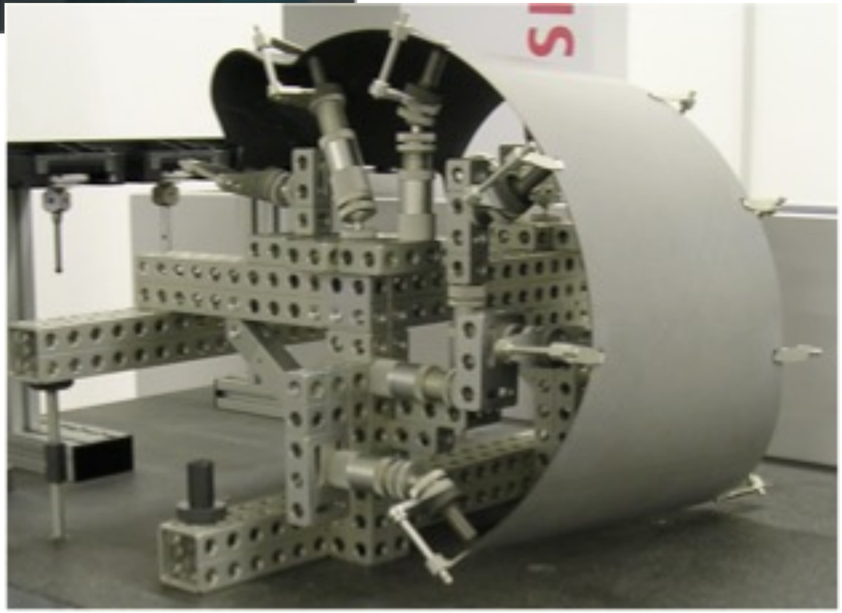


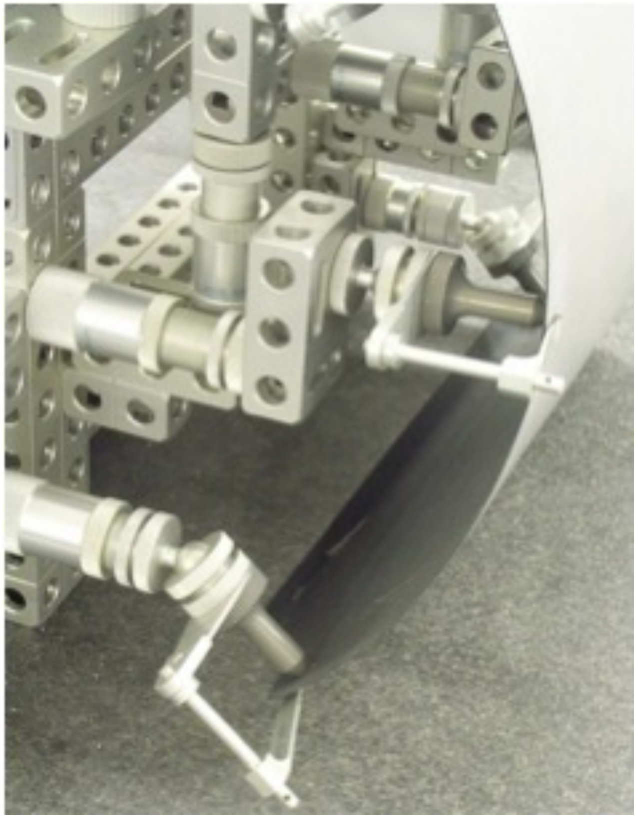




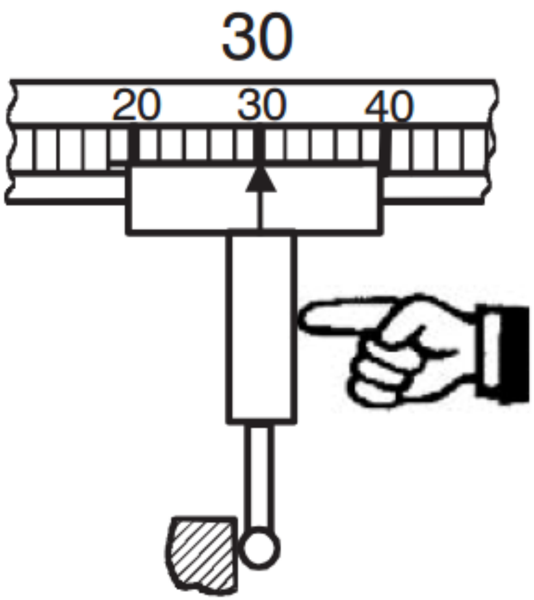
CMM: FIXTURING



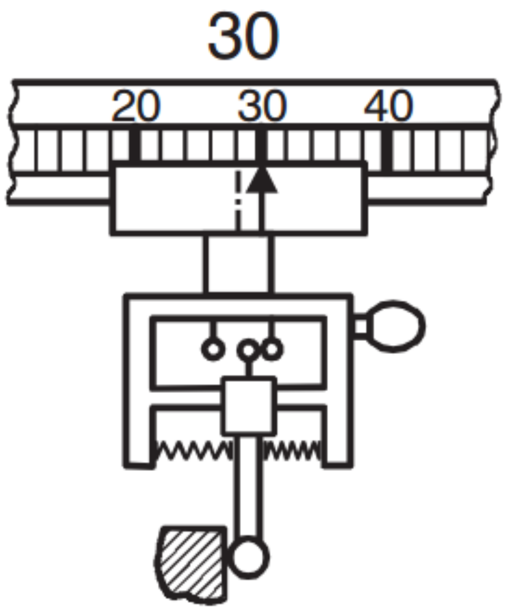




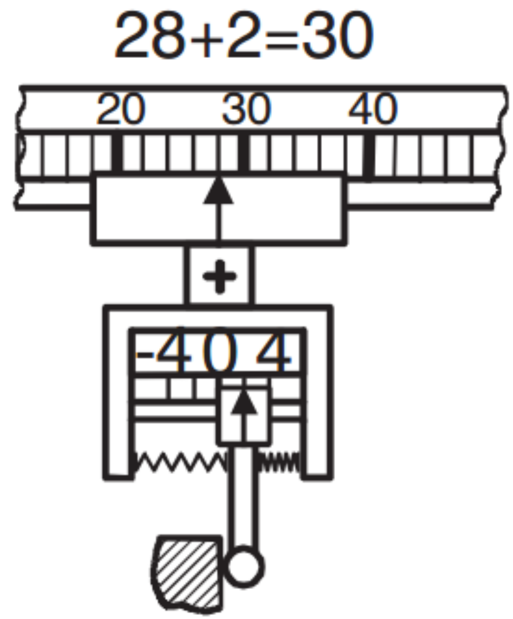
CMM: Contact Probe



Hard probe: the operator detects the contact



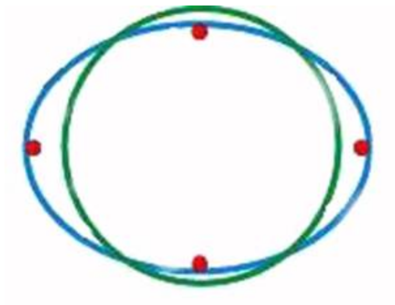
Touch-trigger probe: the machine detects the contact

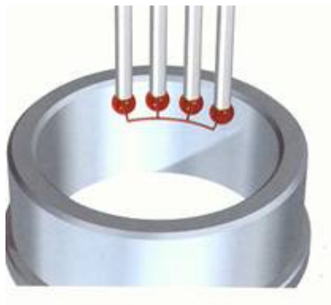


Measuring probe: the machine measures the displacement of the tip

CMM: Contact Probe

Point to point





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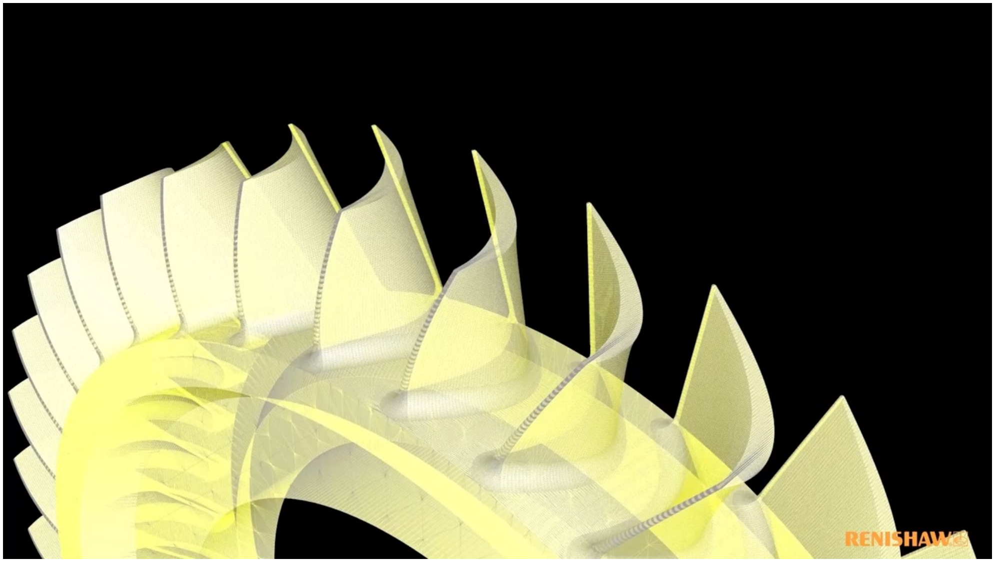
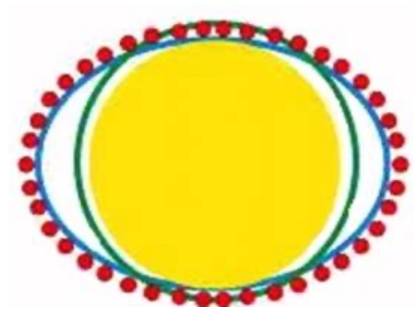
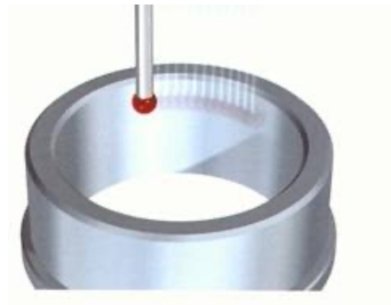


Scanning





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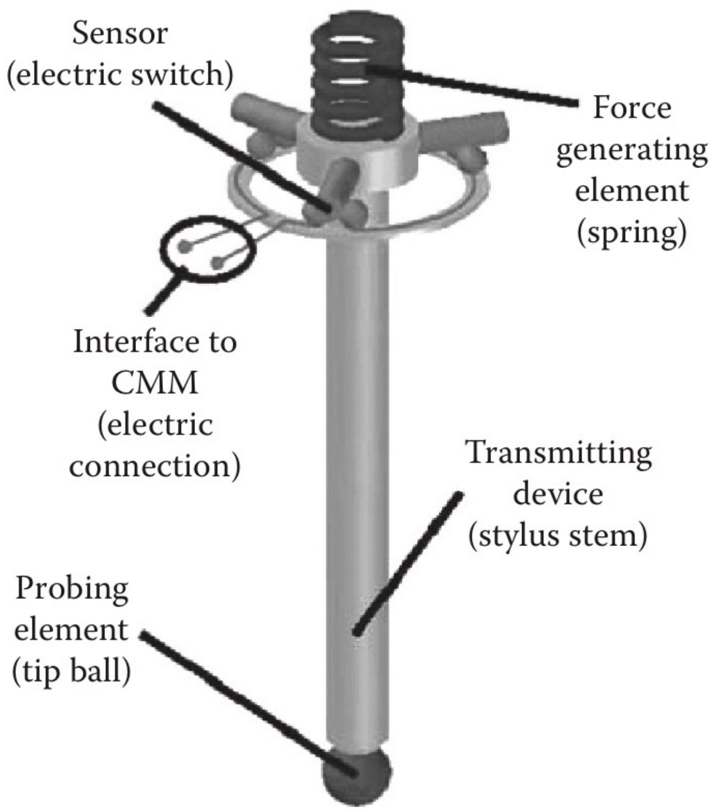
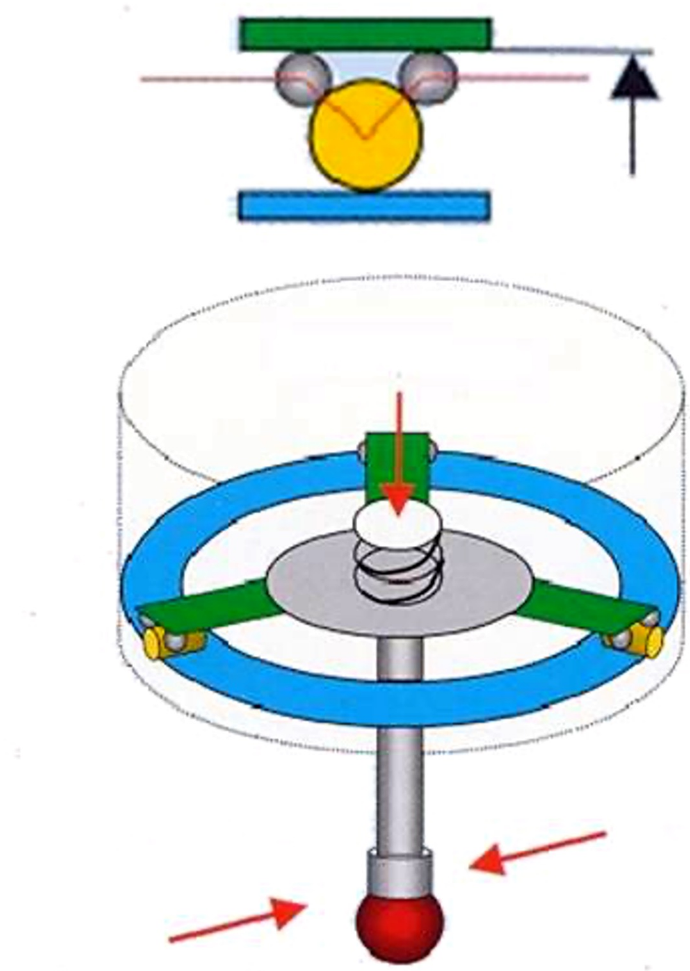
CMM: Touch-Trigger Probe

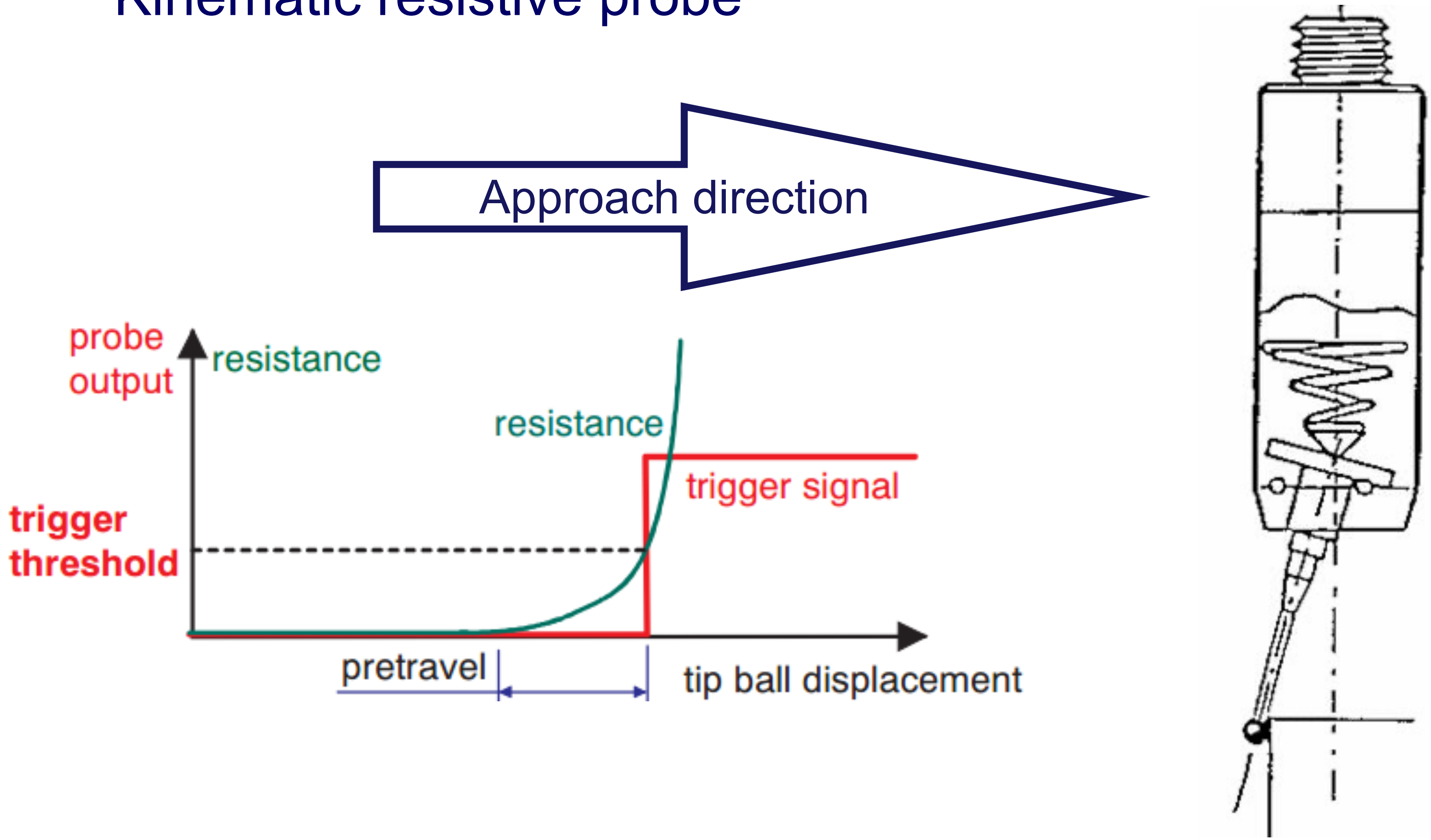
Kinematic resistive probe





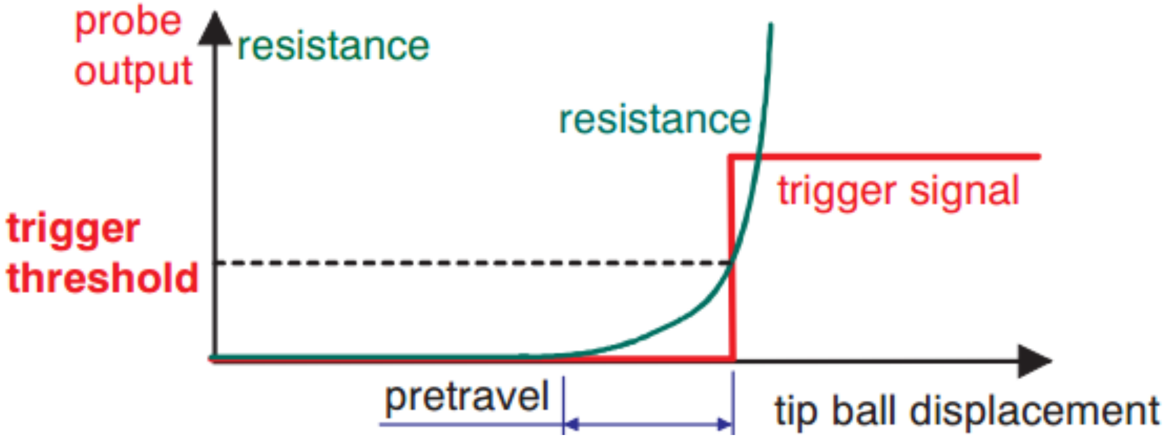
CMM: Touch-Trigger Probe

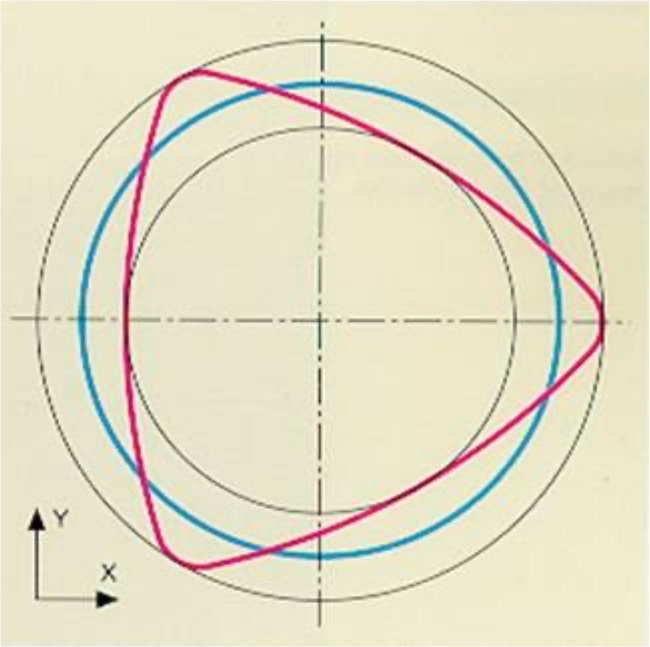
Kinematic resistive probe



CMM: Touch-Trigger Probe

Kinematic resistive probe

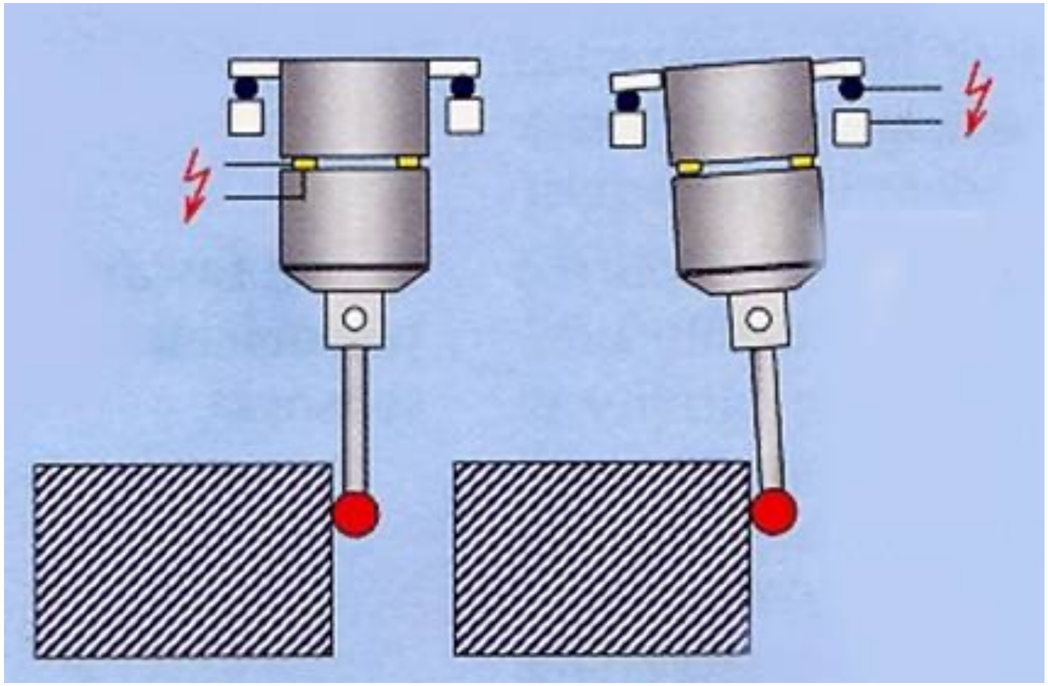


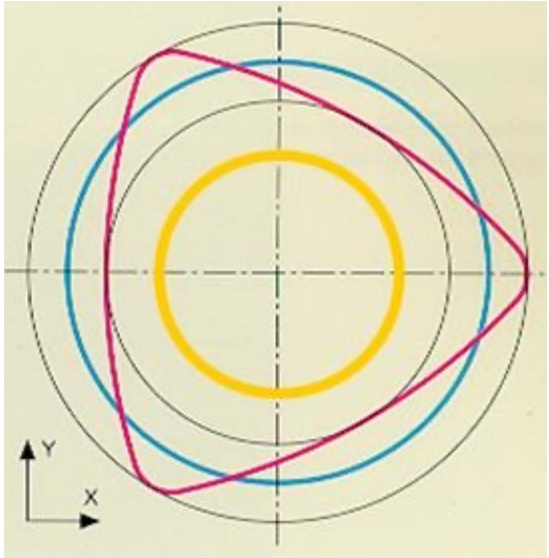


pre-travel

CMM: Touch-Trigger Probe

Piezo-electric probe



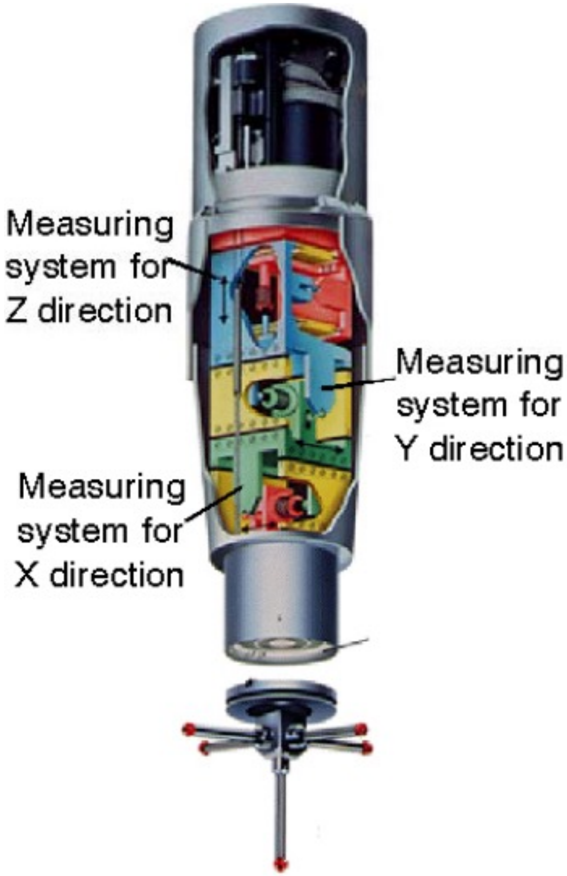


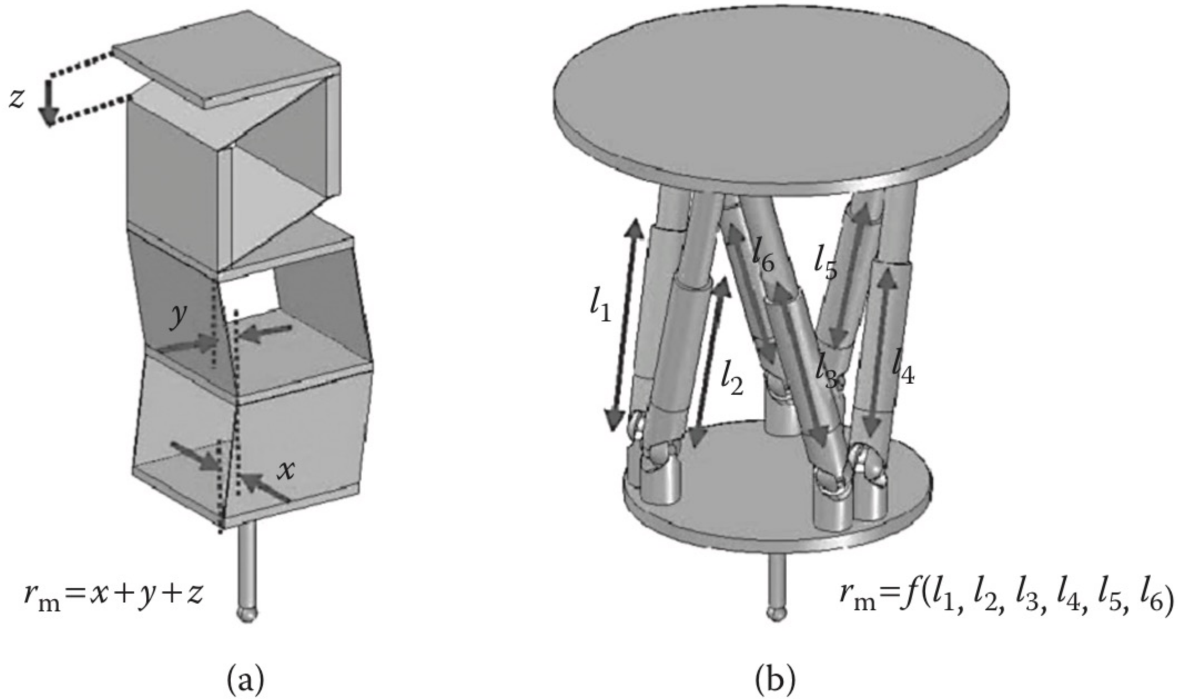
Ideal

Electro-mechanic

Piezo-electric

CMM: Measuring Probes



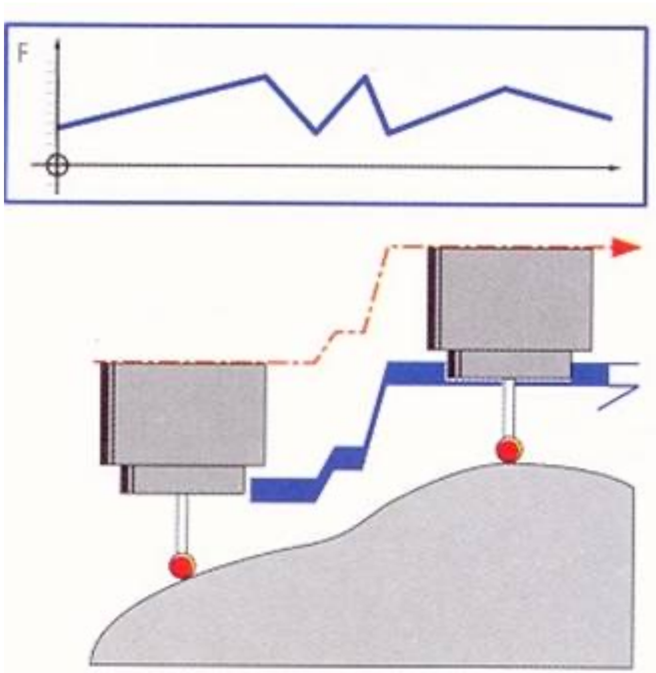


Serial Kinematic

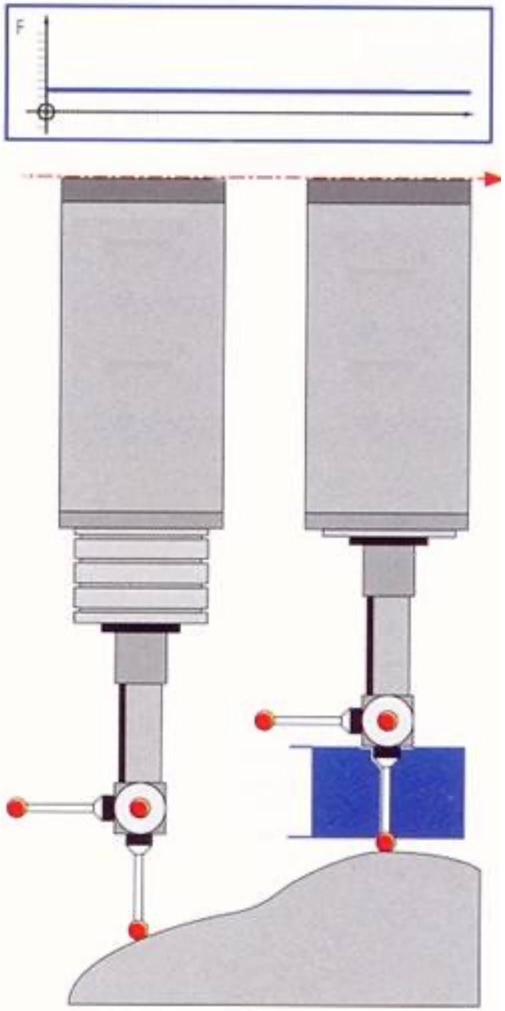
Parallel Kinematic

CMM: Measuring Probes

Passive probe

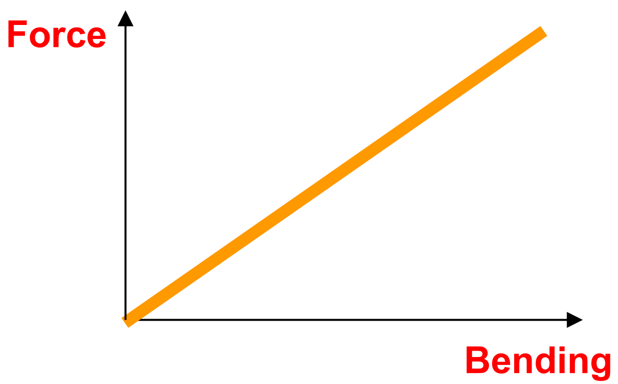


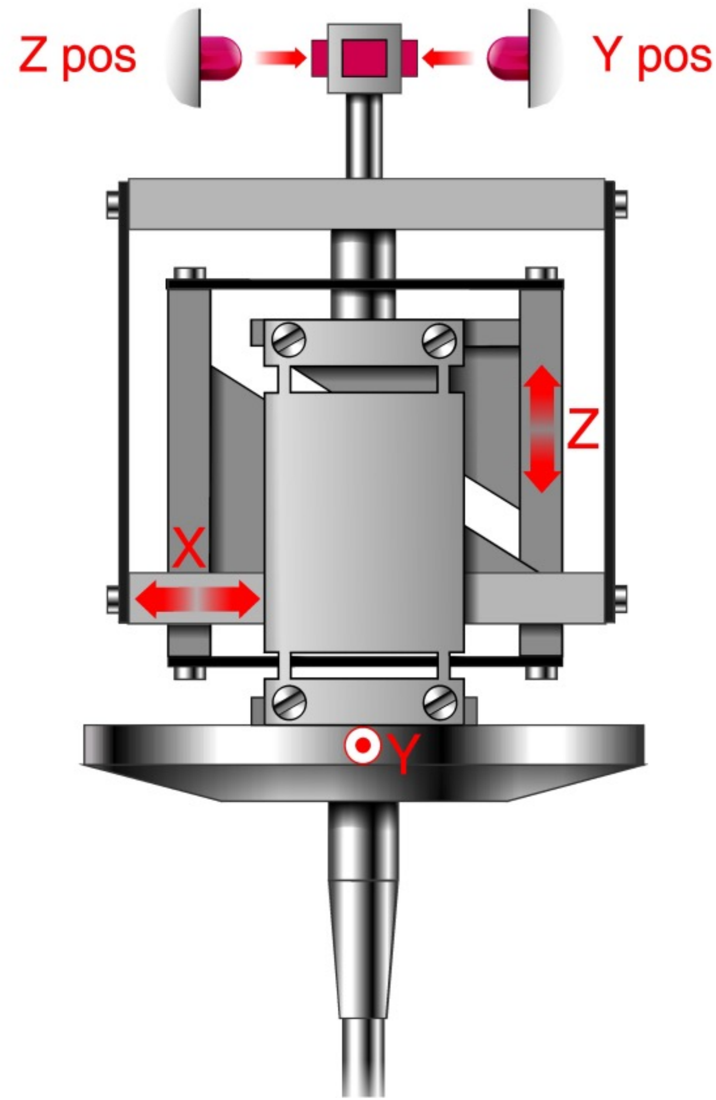
Active probe



CMM: Measuring Passive Probes

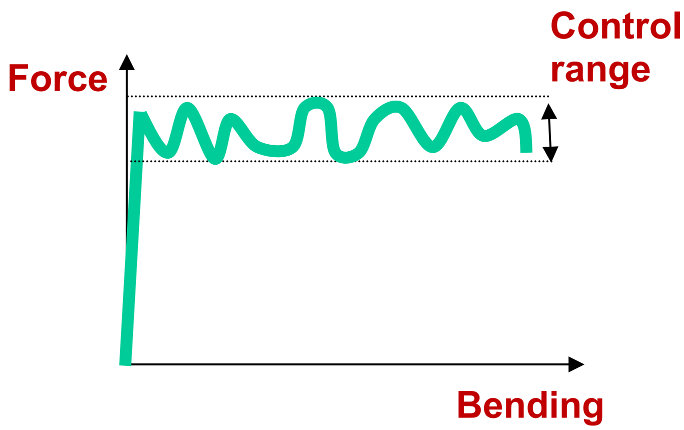
- No actuation
- Contact force created by elastic elements
- Force proportional to bending
- Short movement range
- Light and robust structure, but not as touch-trigger structure

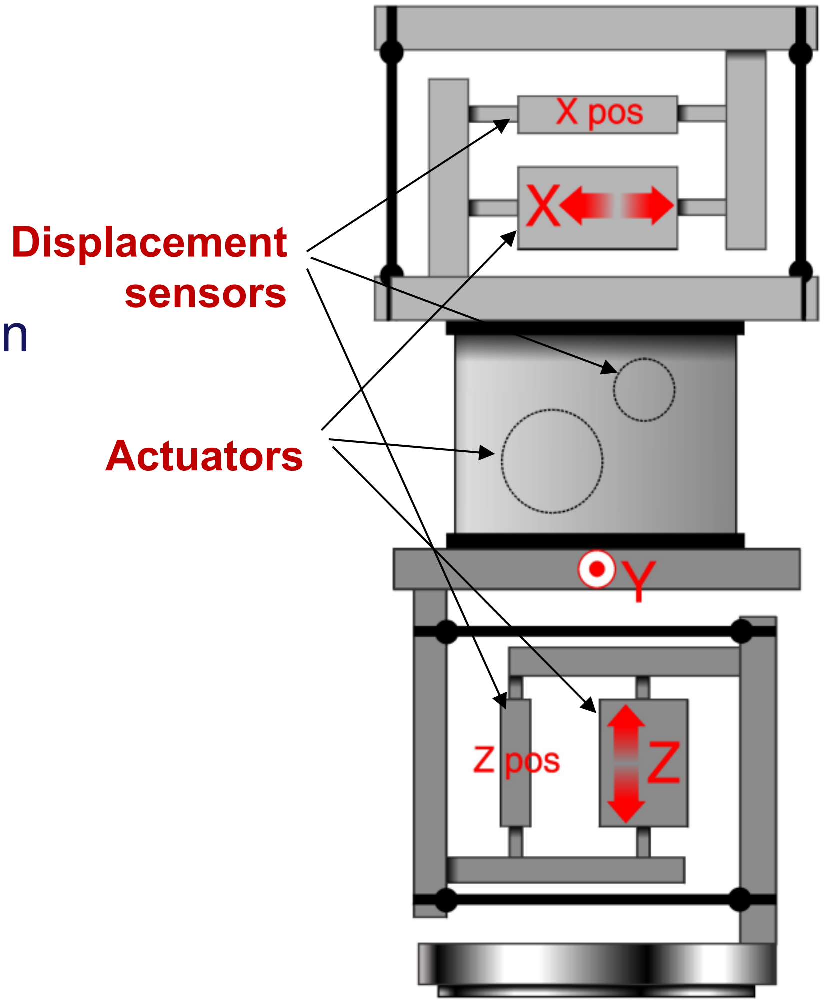




CMM: Measuring Active Probes

- Actuators for each axis
- Modulated but not constant force
- Bending can change as required
- Wider movement range
- Possibility of unknown surface scan





CMM: Probing Mode

The paradox of the scan

CMMs can move quickly, however, most of the measures are at low speed, less than 15 mm/s.

Why?

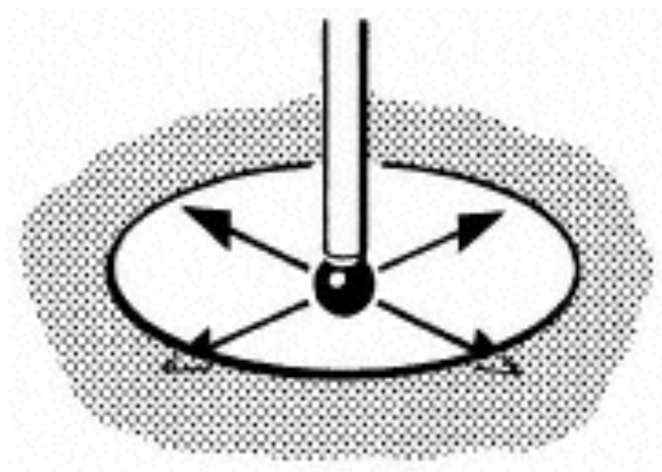
The scan generates dynamic forces in the structure of the CMM and the sensor itself, impacting on performance.

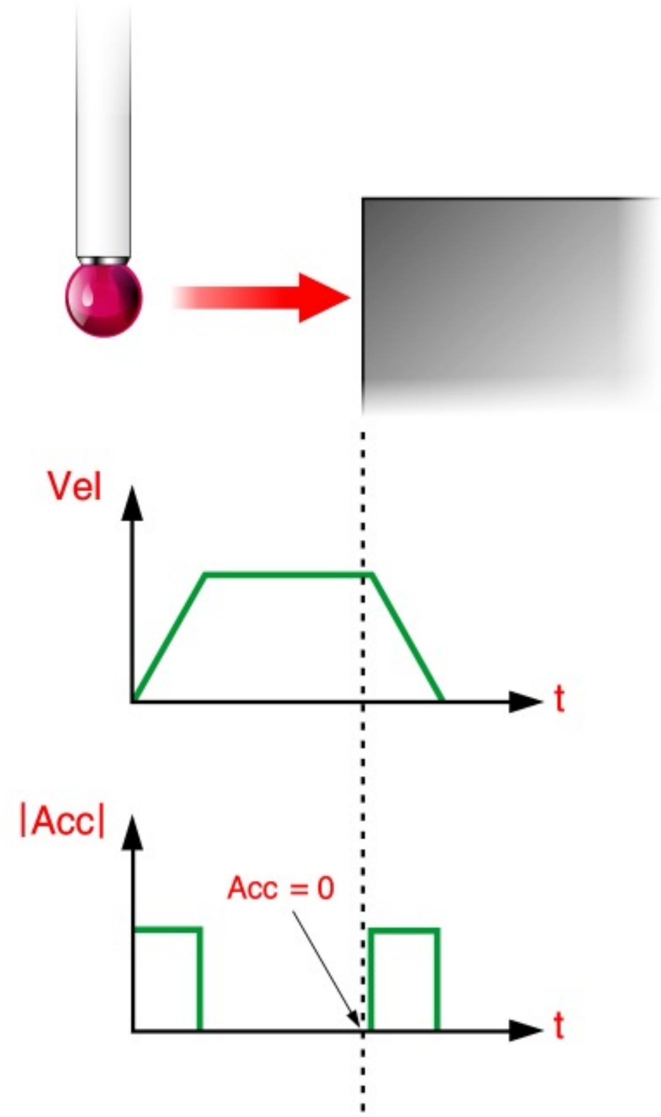
The dynamic errors are related to the acceleration of the machine and the probe during the movement of the machine itself on the workpiece surface.

CMM: Probing Mode

The **discrete-point probing** is performed at a **constant speed** - the acceleration is zero at the moment of contact.

So there are no **inertial forces** on the machine nor on the probe.



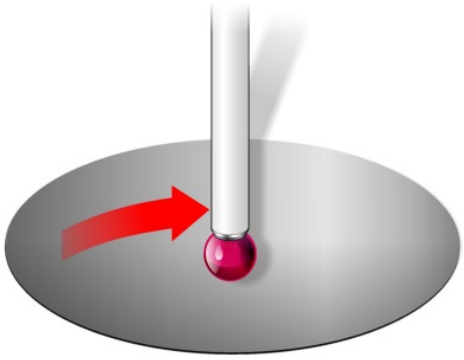


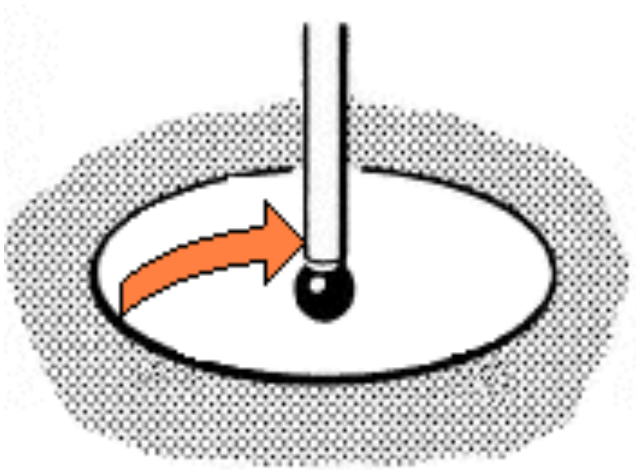
CMM: Probing Mode

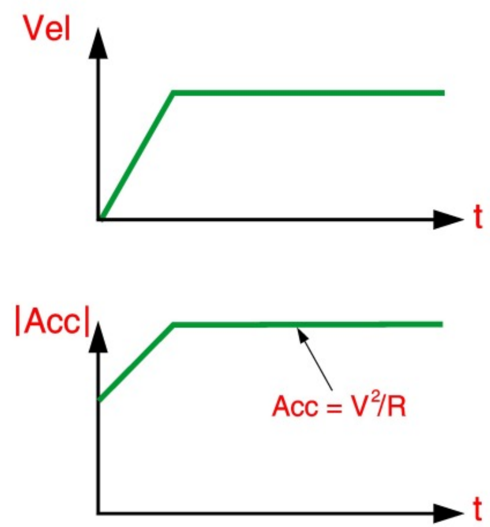
The scan requires continuously variable speed vectors while the stylus moves along a curved surface.

This generates variable inertial forces, which makes the machine to inflect.

Vibration may also be relevant during the scan.

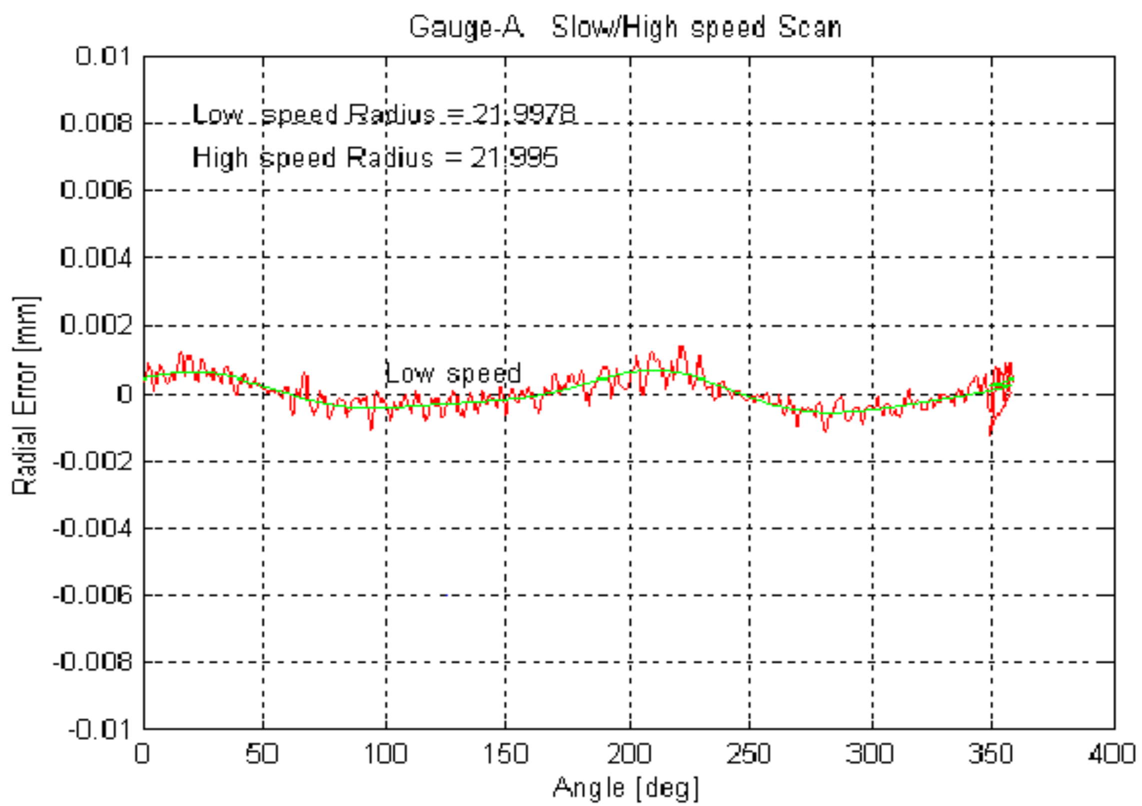






CMM: PROBING MODE

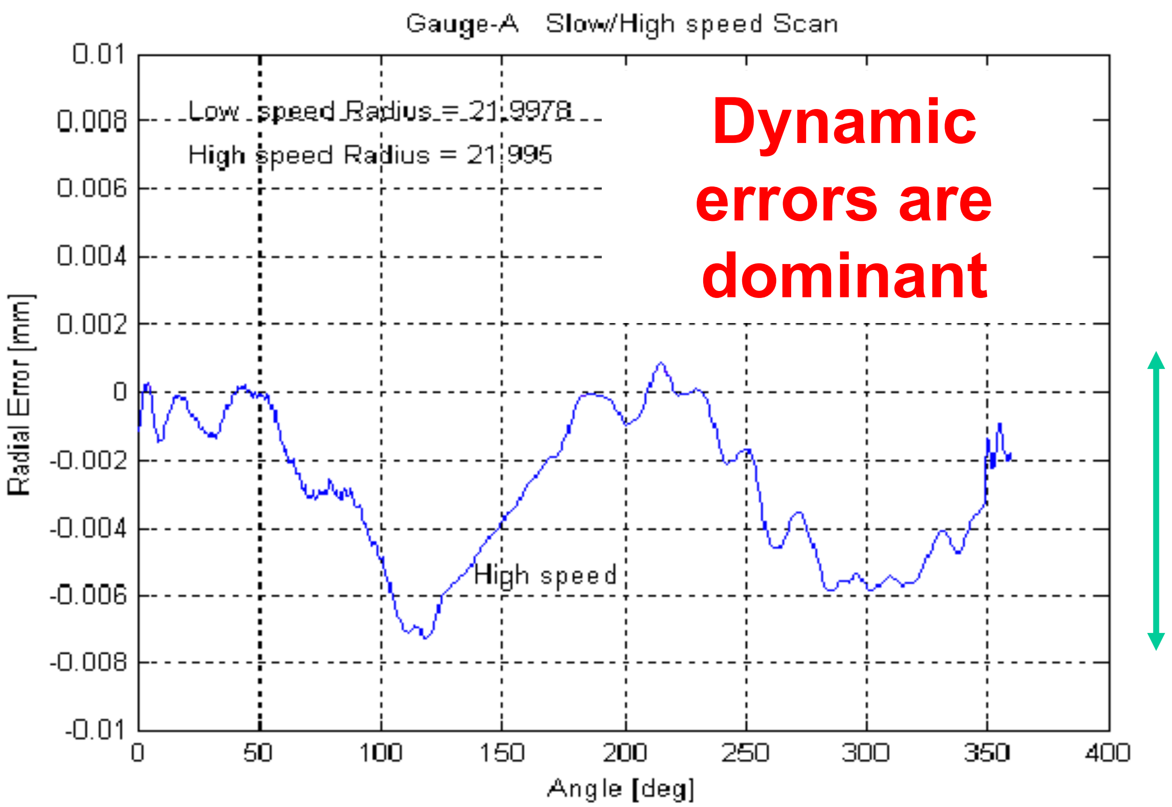
Example: Ø50 ring measurement mm at **10 mm/s** with a CMM with MPEe di **2,5 + L/250**



Form error **2μm**

CMM: PROBING MODE

Example: Ø50 ring measurement mm at **50 mm/s** with a CMM with MPEe di **2,5 + L/250**



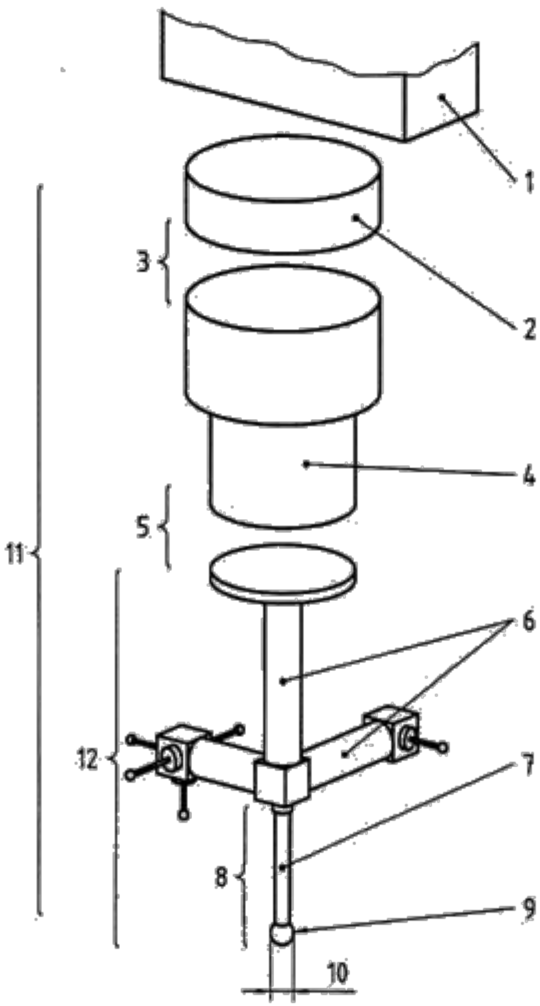
Form error
8µm

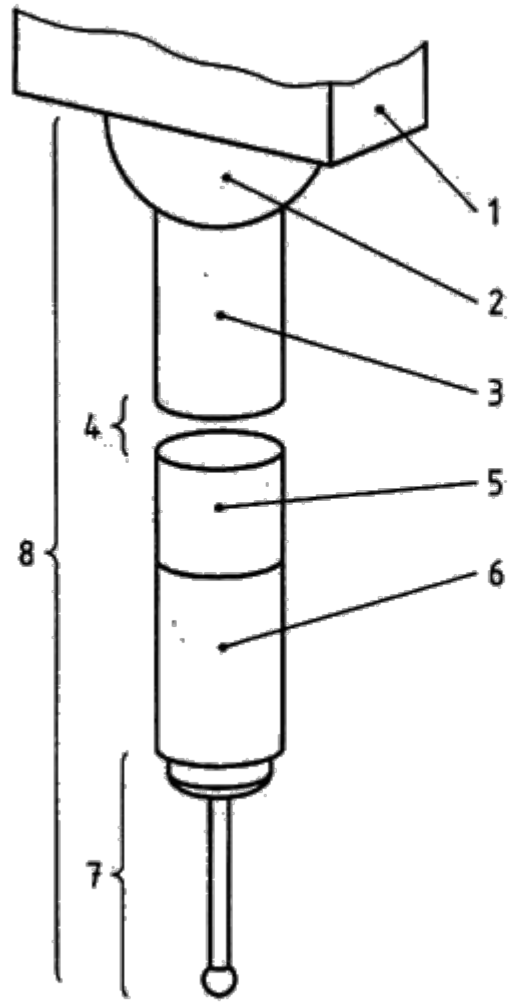
CMM: Probing System

ISO 10360-1

Probing System

1 Ram
2 Probe extension
3 Probe changing system
4 Probe
5 Stylus changing system
6 Stylus extension
7 Stylus shaft
8 Stylus
9 Stylus tip
10 Tip diameter
11 Probing system
12 Stylus system





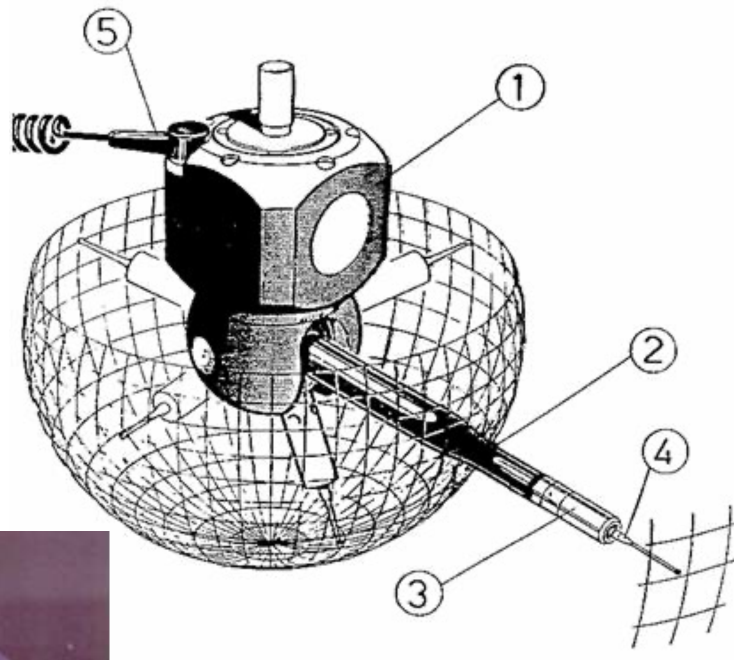
ISO 10360-1

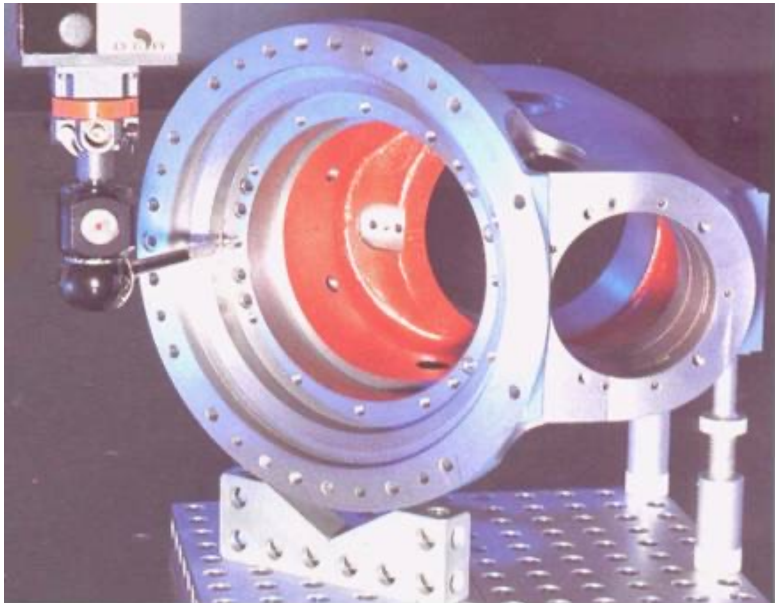
Articulating Probing System

1 Ram
2 Articulating system
3 Probe extension
4 Probe changing system
5 Probe
6 Stylus extension
7 Stylus
8 Articulating Probing system

CMM: Probing System

articulating probing system



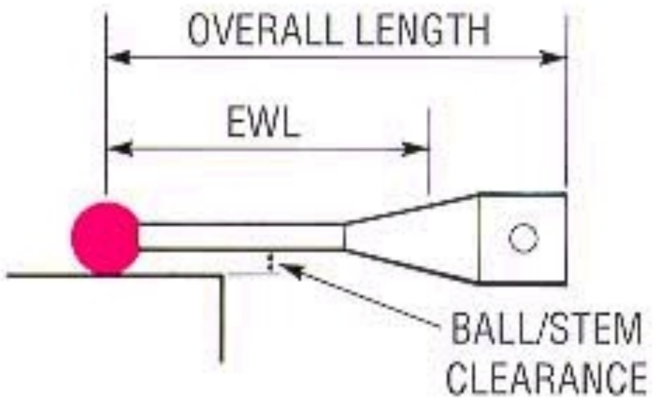


CMM: Probing System

Why should the stylus be changed?

To optimize the probing repeatability by choosing a stylus characterized by :

- **Minimum length**
- **Maximum stiffness**
- **Minimum articulation**



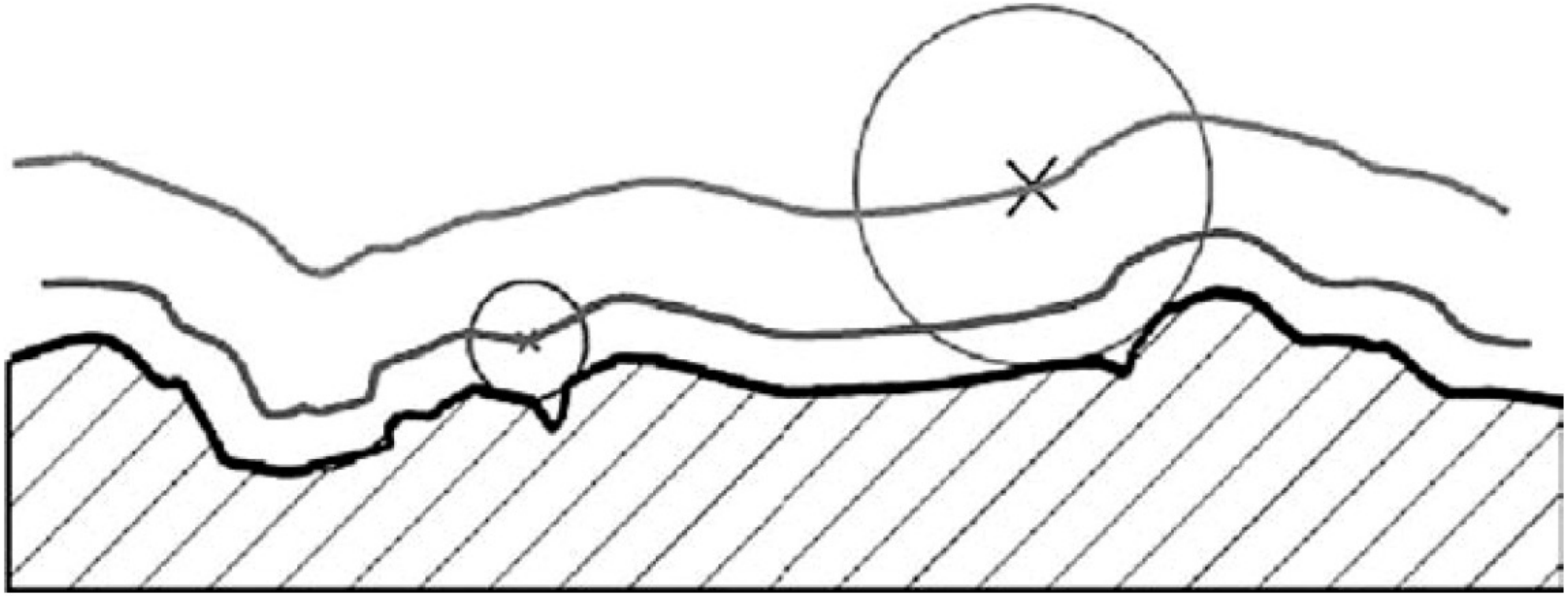
Results:

Repeatability as the stylus length changes

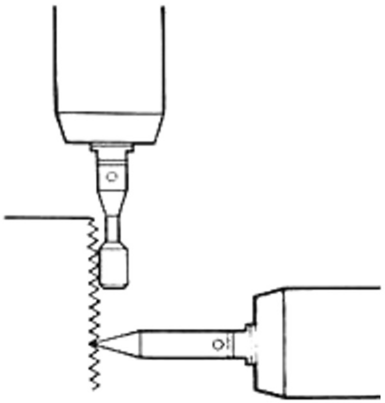
Length	**10mm**	**50mm**
Unidirectional repeatability	0,30 µm	0,40 µm
2D form error	±0,40 µm	±0,80 µm
3D form error	±0,65 µm	±1,00 µm

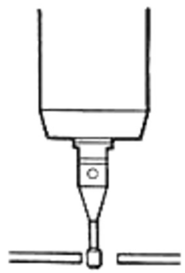
CMM: Probing System

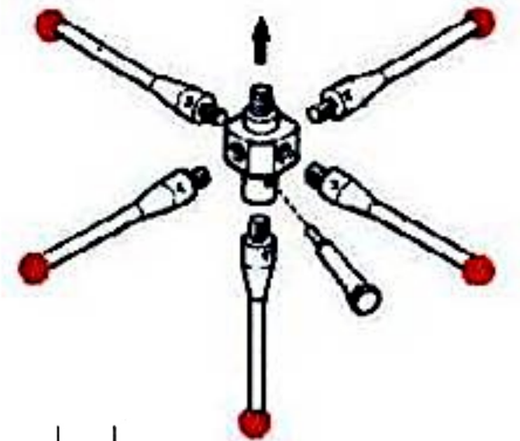
Mechanical filtering effect of the tip ball.

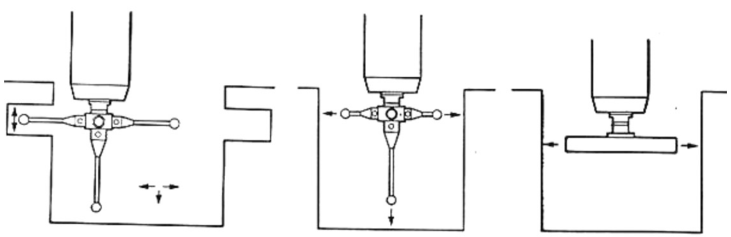


CMM: Probing System Configuration







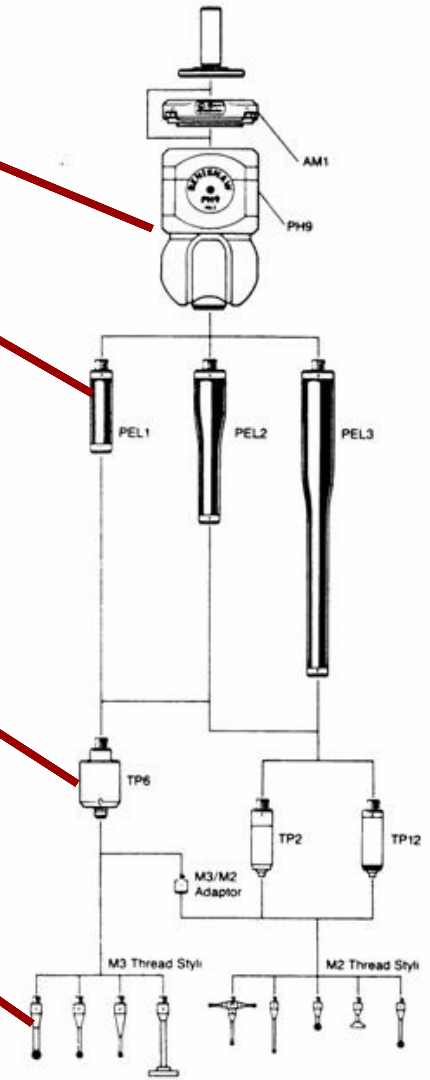


head

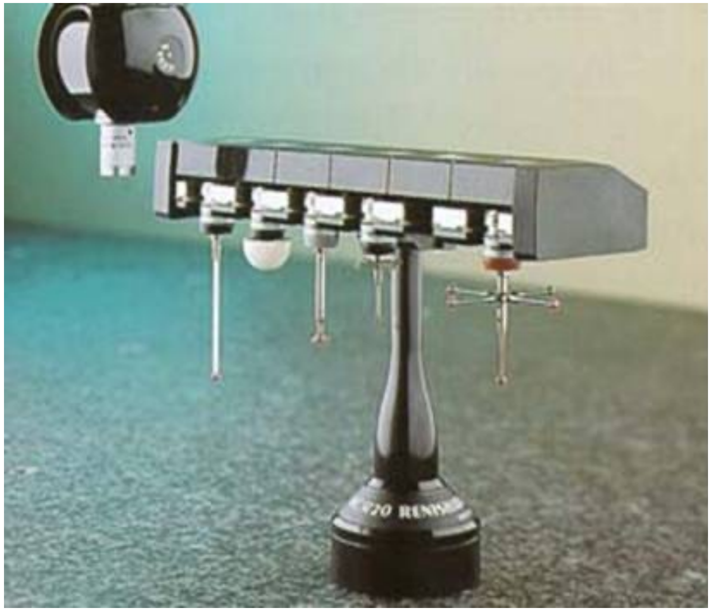
probe extensions

probes

stylus or styli group

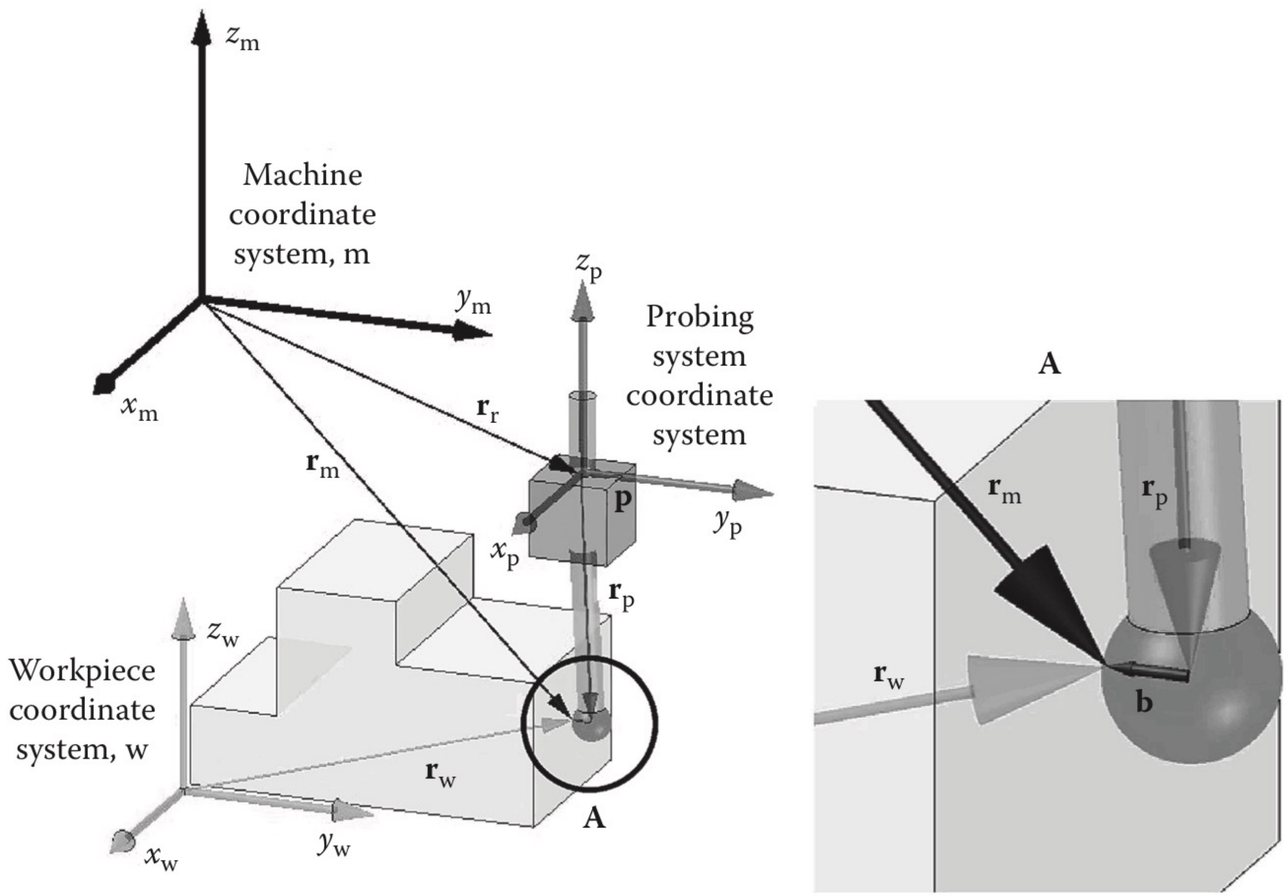


CMM: Automatic Tool Changer

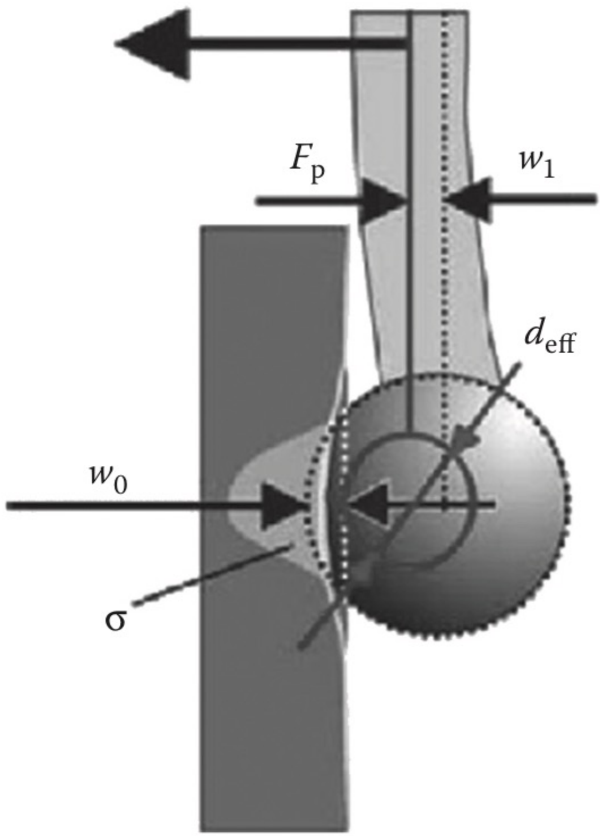


CMM: Probing System Qualification

The probing point real coordinates $\boldsymbol{p}$ can be derived exactly if the probing matrix $\boldsymbol{P}$ and the position measuring matrix $\boldsymbol{M}$ are precisely known and are invertible.



CMM: Probing System Qualification



Effective tip ball diameter (d_{eff})

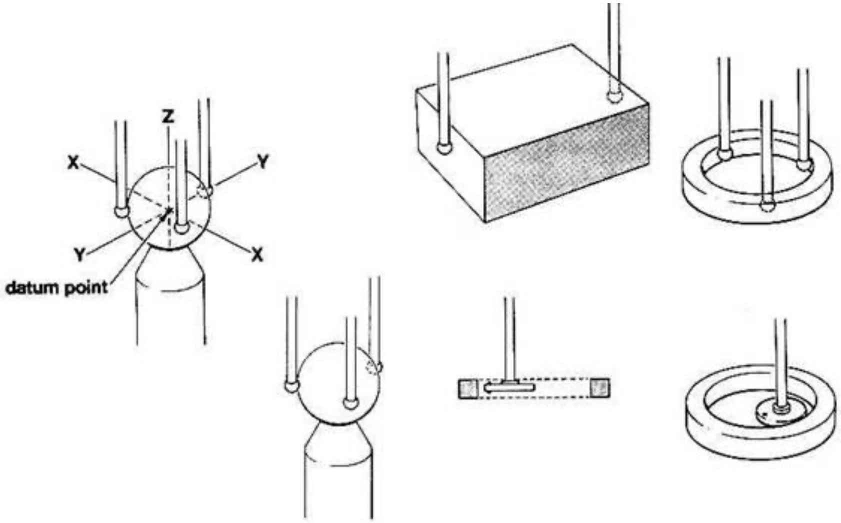
F_p: Probing force;

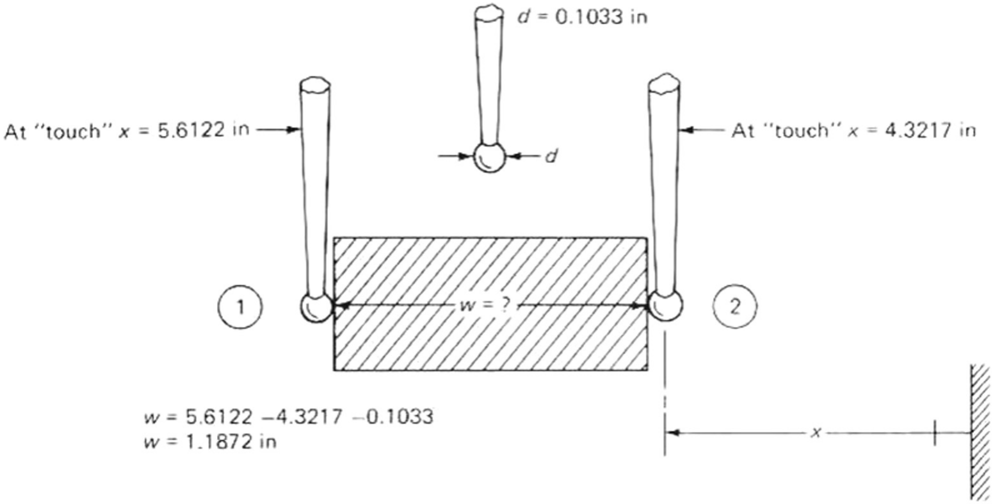
w_0: Elastic deformation of tip ball and workpiece;

w_1: Elastic deformation of stylus stem;

σ: Hertzian contact stress.

CMM: Probing System Qualification





CMM: Measurement Management

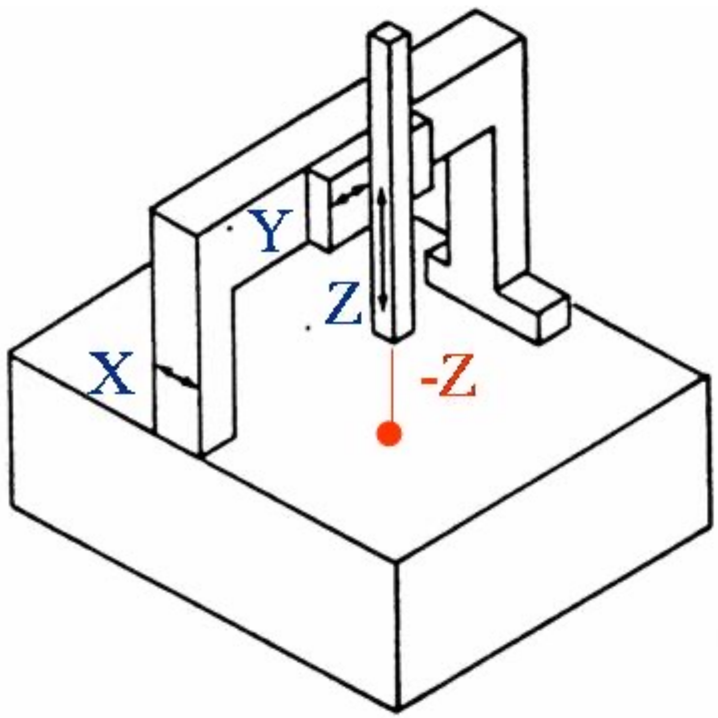
- Operator directed
- CNC discrete-point probing
- Automatic scan of known surfaces
- Automatic scan of unknown surfaces

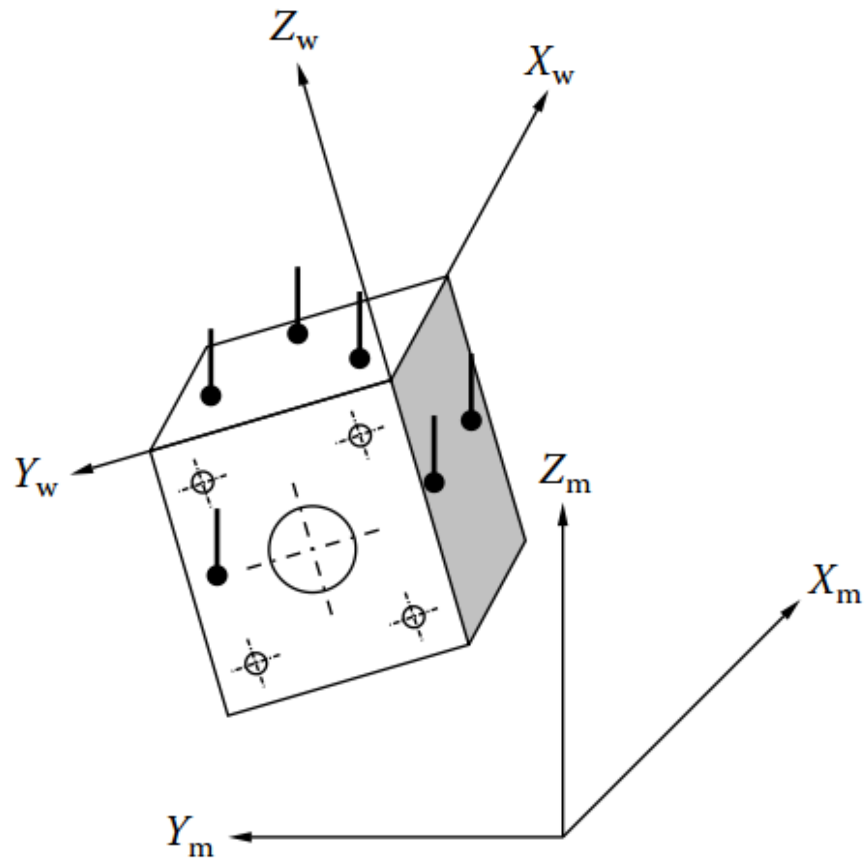
CMM: Control System

Main functions:

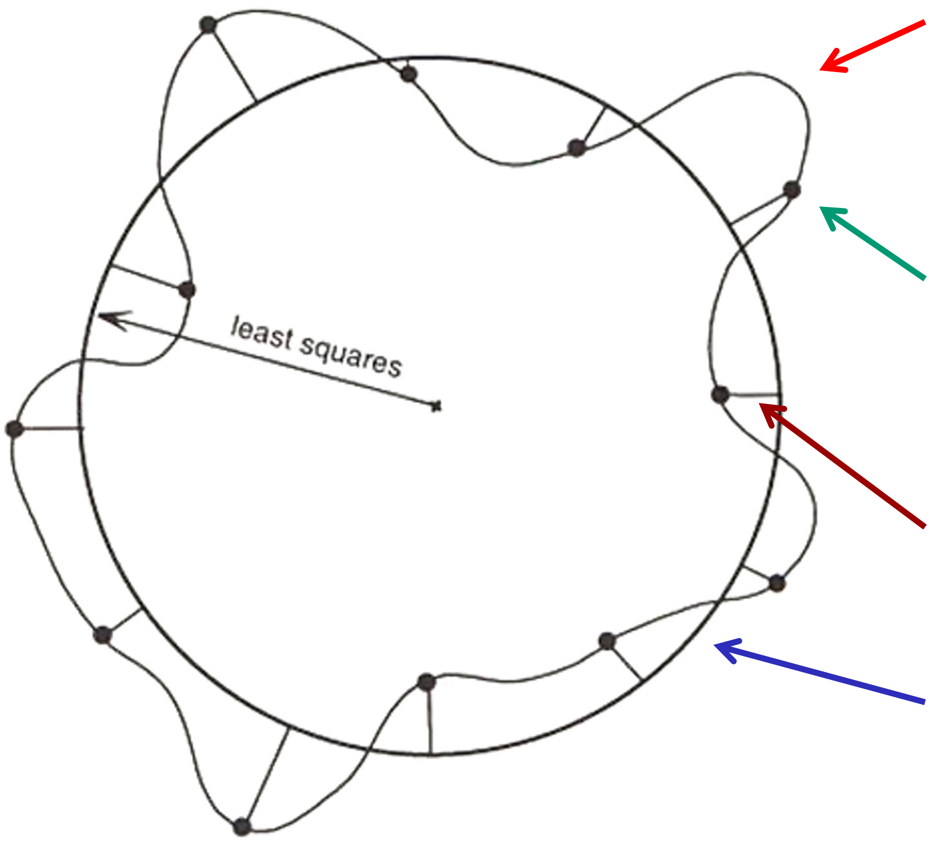
- Mathematical Alignment
- Error compensation
- Geometric fitting of measured points
- Local and global deviation analysis
- Storing and presenting measurement results

MATHEMATICAL ALIGNMENT





FITTING



Real geometric feature

Measured point

Local geometric deviation

Associated geometric feature

FITTING

The optimal fit will be determined by the sizes of the residuals. The residuals d_i are the (orthogonal) distances from the n points to the geometry that must be fit.

L_p-norm equation:

$$\min L_p = \min \left[\frac{1}{n} \sum_{i=1}^{n} |d_i|^p \right]^{\frac{1}{p}}$$

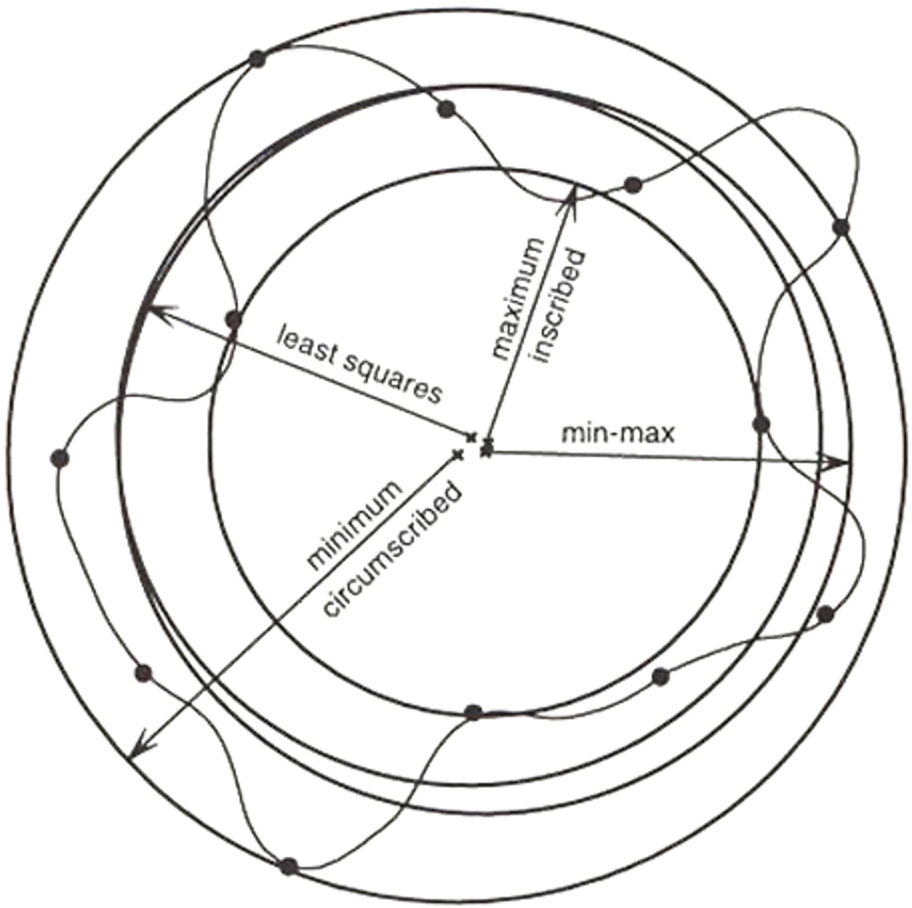
L_2 fitting (least squares):

$$\min L_2 = \min \sqrt{\frac{1}{n} \sum_{i=1}^{n} |d_i|^2}$$

L_∞ fitting (min-max):

$$\min L_\infty = \min \max |d_i|$$

Example: circle fitting



FITTING

Other commonly adopted fit objectives (e.g. fitting of a circle)

Let's consider a circle of radius r and r_i is the distance of the measured point $\boldsymbol{P}_i$ from the center of the fitted circle.

Minimum circumscribed: minimize the radius r

$$\min r$$
$$s.t.: r_i \leq r \quad \forall i \in \{1,2,\dots,n\}$$

Maximum inscribed: maximize the radius r (circle)

$$\max r$$
$$s.t.: r_i \geq r \quad \forall i \in \{1,2,\dots,n\}$$

FITTING

Other commonly adopted fit objectives (e.g. fitting of a circle)

Let's consider a circle of radius r and r_i is the distance of the measured point $\boldsymbol{P}_i$ from the center of the fitted circle.

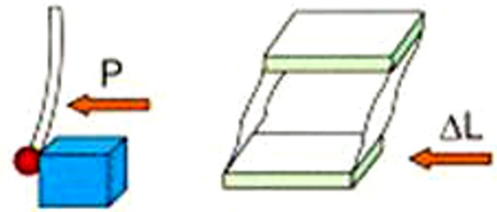
Minimum Zone:

All points belong to the minimum possible tolerance zone

MZ of a circle is a ring with internal radius r_{int} and external radius r_{ext}

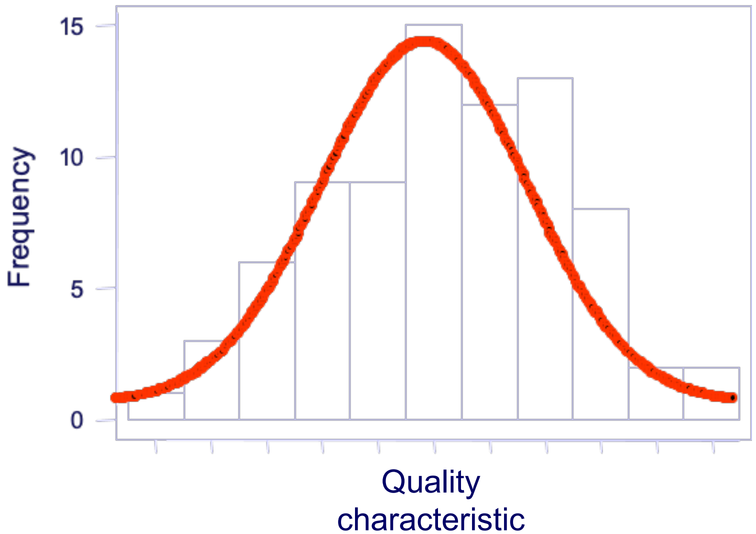
$$
\begin{aligned}
&\min(r_{ext} - r_{int}) \\
&s.t.: r_i \geq r_{int} \quad \forall i \in \{1,2,\dots,n\} \\
&\qquad\;\; r_i \leq r_{ext} \quad \forall i \in \{1,2,\dots,n\}
\end{aligned}
$$

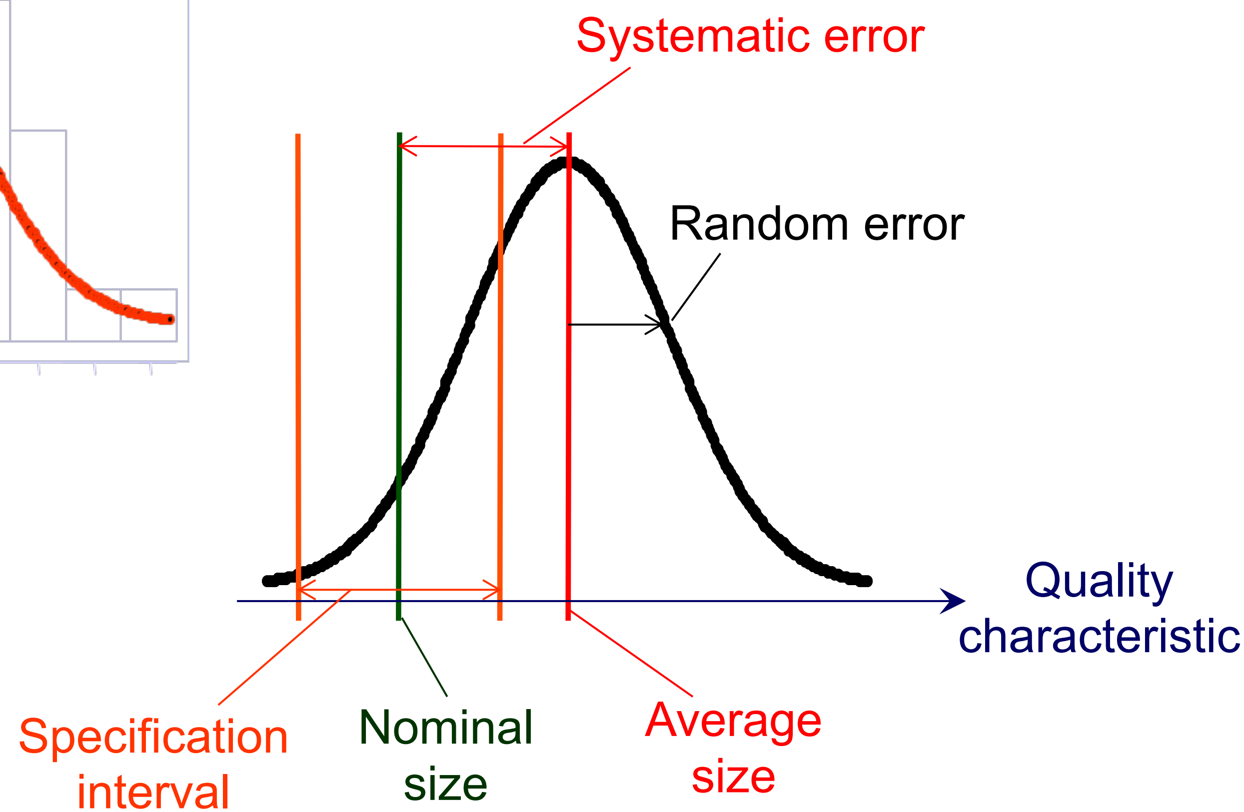
ERROR COMPENSATION

Type of compensation	Compensation measures	
Linear compensation	Temperature-related linear expansion, linear scale error	
CAA (Computer Aided Accuracy)	Volumetric, static component deviations, thermally induced table bend	
Probe system	Vectorial bend of styli caused by probing forces, component deviations of measuring systems and spring parallelograms	
D-CAA (Dynamic CAA)	Dynamic deviations of overall machine construction due to vectorial accelerations	

CMM: CONTROL SYSTEM

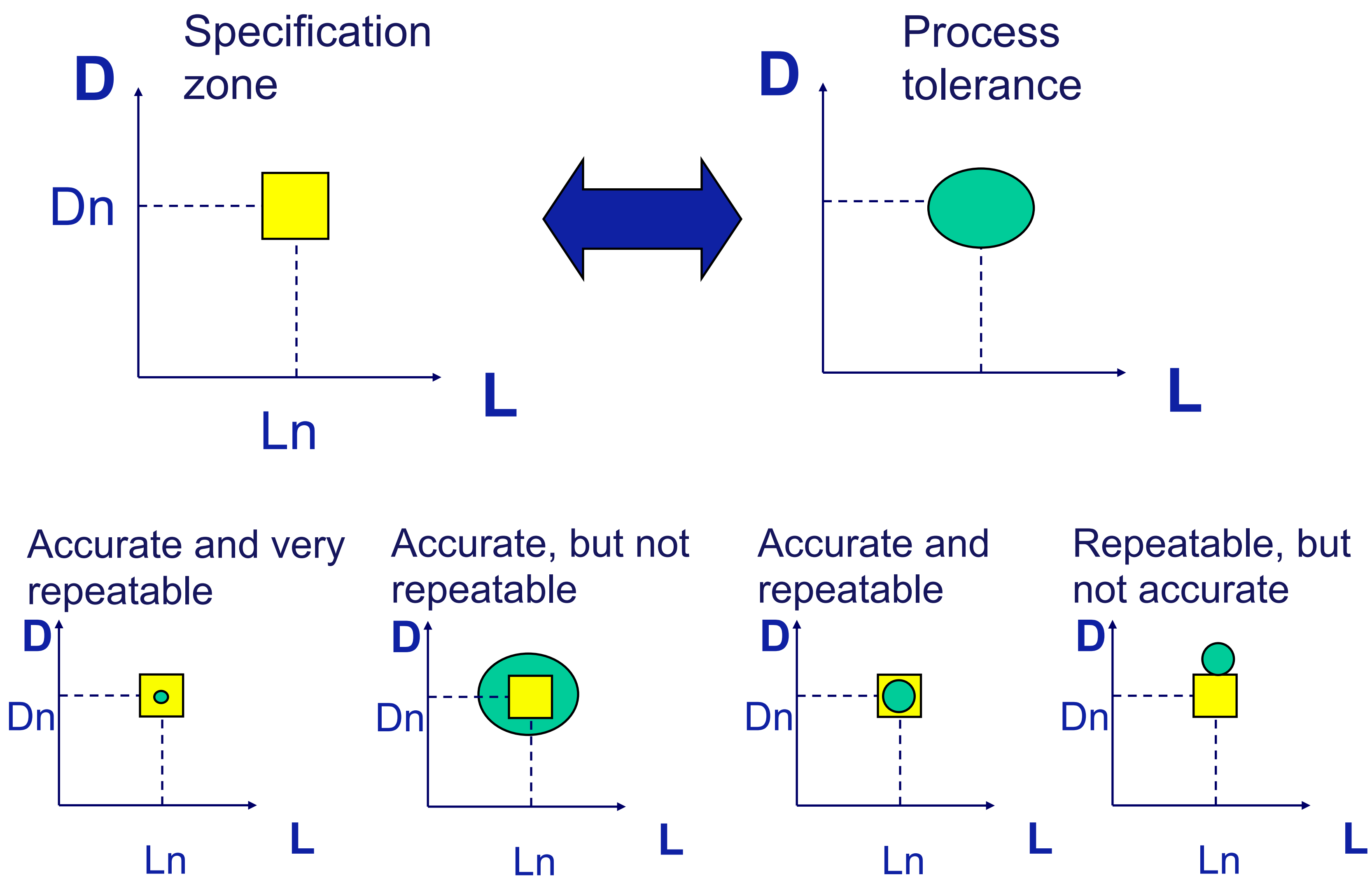
Data registration and presentation





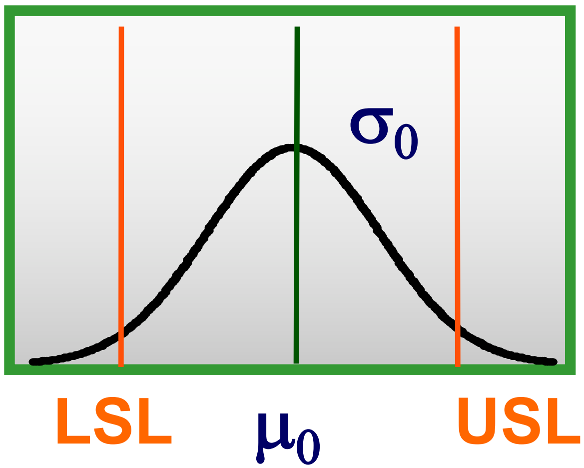
Systematic error vs Random error

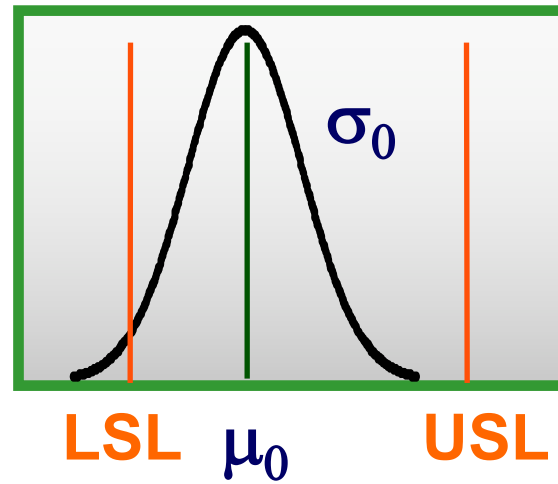
PROCESS OUTPUT VARIABILITY AND SPECIFICATIONS



PROCESS CAPABILITY

Two-sided specifications (Hp: Gaussian distribution of the results)



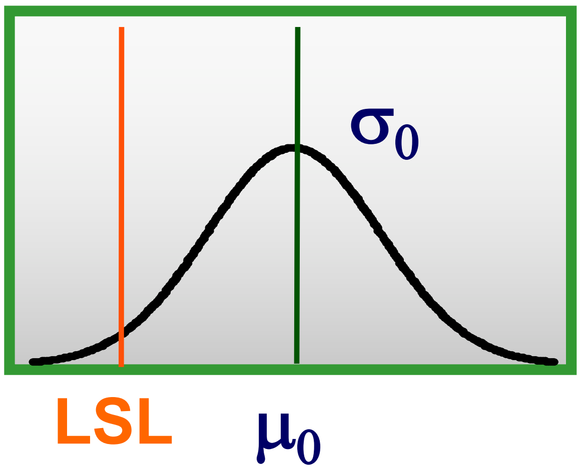


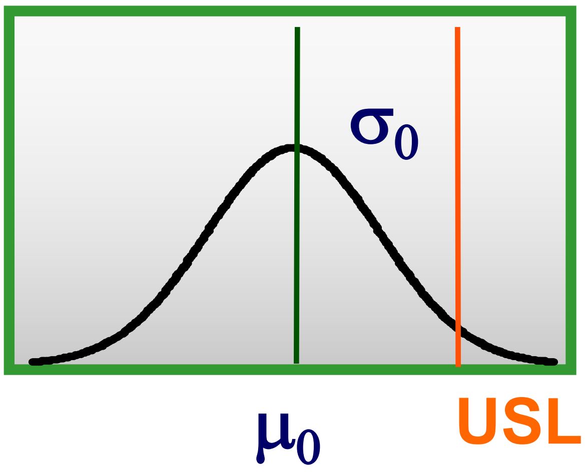
$$C_p = \frac{USL - LSL}{6\sigma_0} = PCR$$

PCR = Process Capability Ratio

PROCESS CAPABILITY

One-sided specifications (Hp: Gaussian distribution of the results)





$$C_L = \frac{\mu_0 - LSL}{3\sigma_0} = PCR_L$$

$$C_U = \frac{USL - \mu_0}{3\sigma_0} = PCR_U$$

Process Capability: PPM

PPM: parts-per-million

Number of defective parts per million of manufactured parts

Cp	One-sided specifications	Two-sided specifications
0,25	226.628	453.255
0,50	66.807	133.614
0,60	35.931	71.861
0,70	17.865	35.729
0,80	8.198	16.395
0,90	3.467	6.934
1,00	1.350	2.700
1,10	484	967
1,20	159	318
1,30	48	96
1,40	14	27
1,50	4	7
1,60	1	2
1,70	0,17	0,34
1,80	0,03	0,06
2,00	0,0009	0,0018

CMM: Probing Systems

- Contact probes
- Non-contact probes

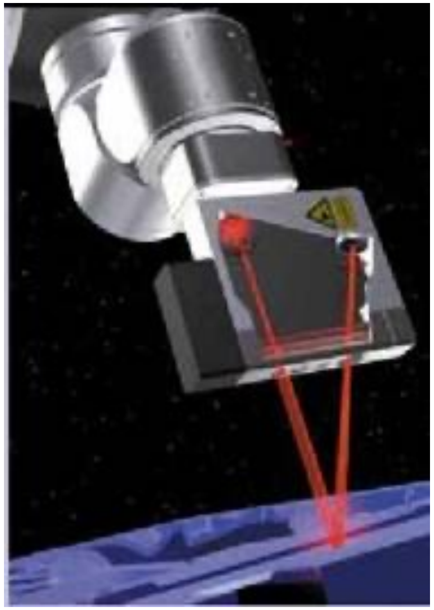
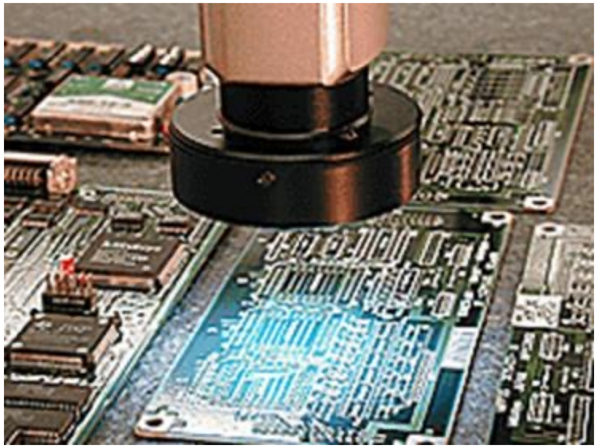
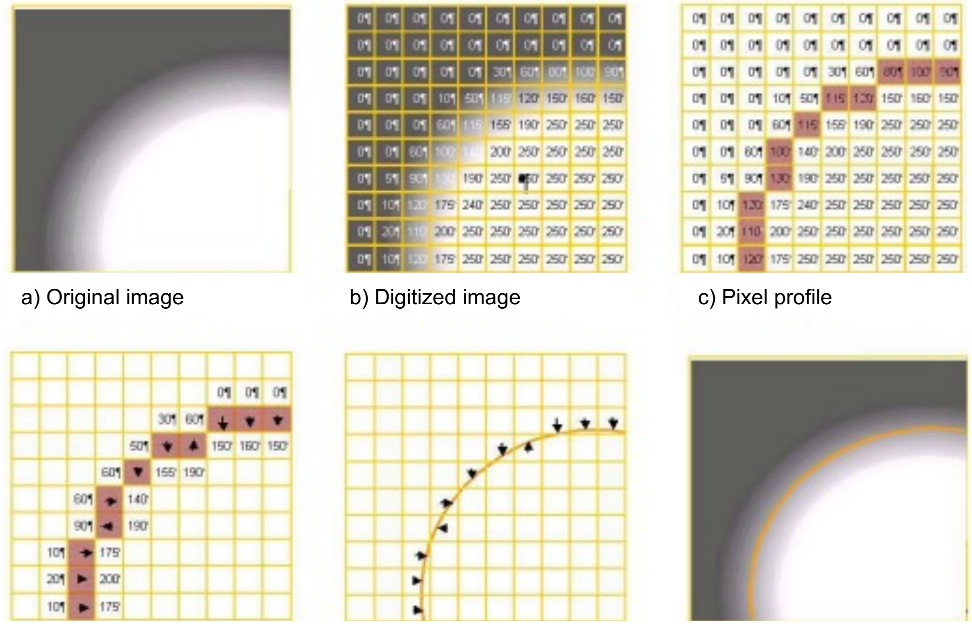


IMAGE PROBING

An image of the object to be measured is analyzed looking for profiles to be measured.

It can be combined with other techniques (often with auto-focus).

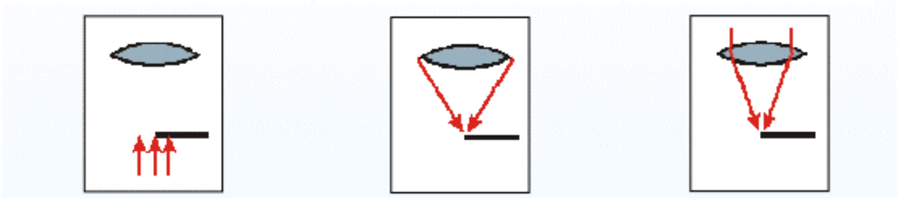


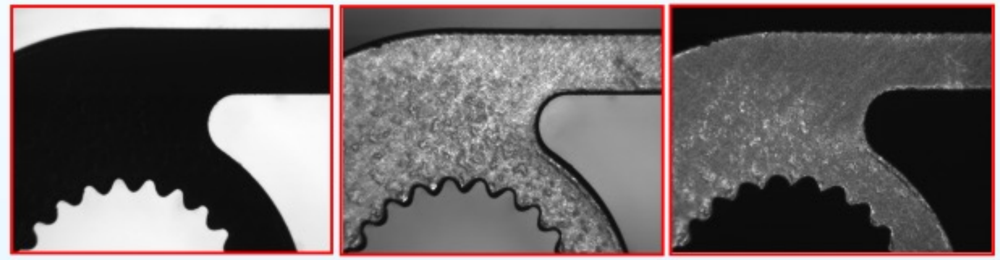


a) Original image b) Digitized image c) Pixel profile

d) Subpixeling e) Associated element f) Final result

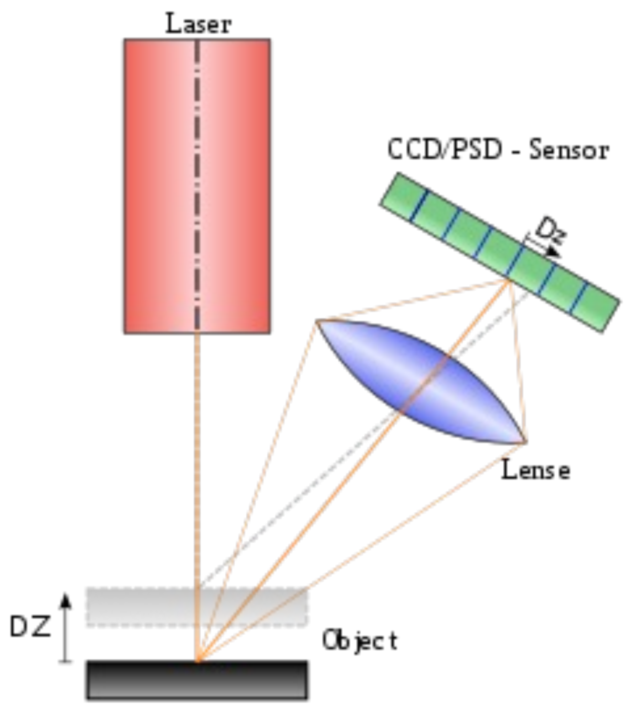
Diascopic illumination Episcopic ring light Episcopic coaxial light

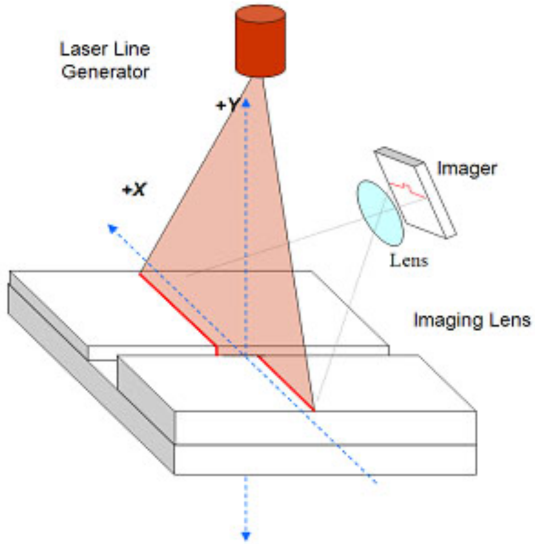


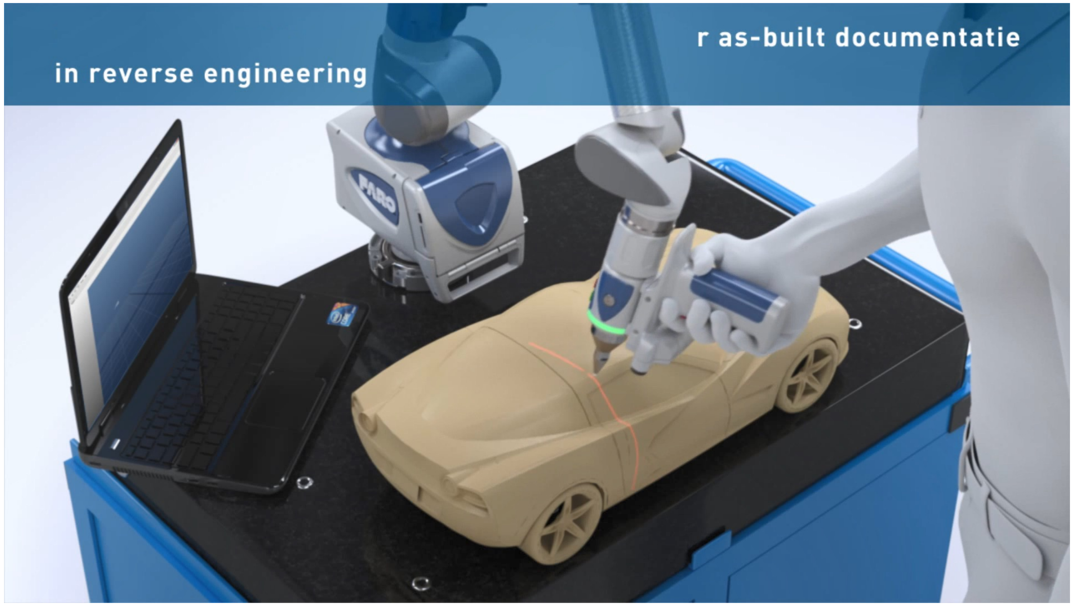


LASER TRIANGULATION

It projects a laser point (or line) on the surface to be measured, and it evaluates point distance by triangulation.

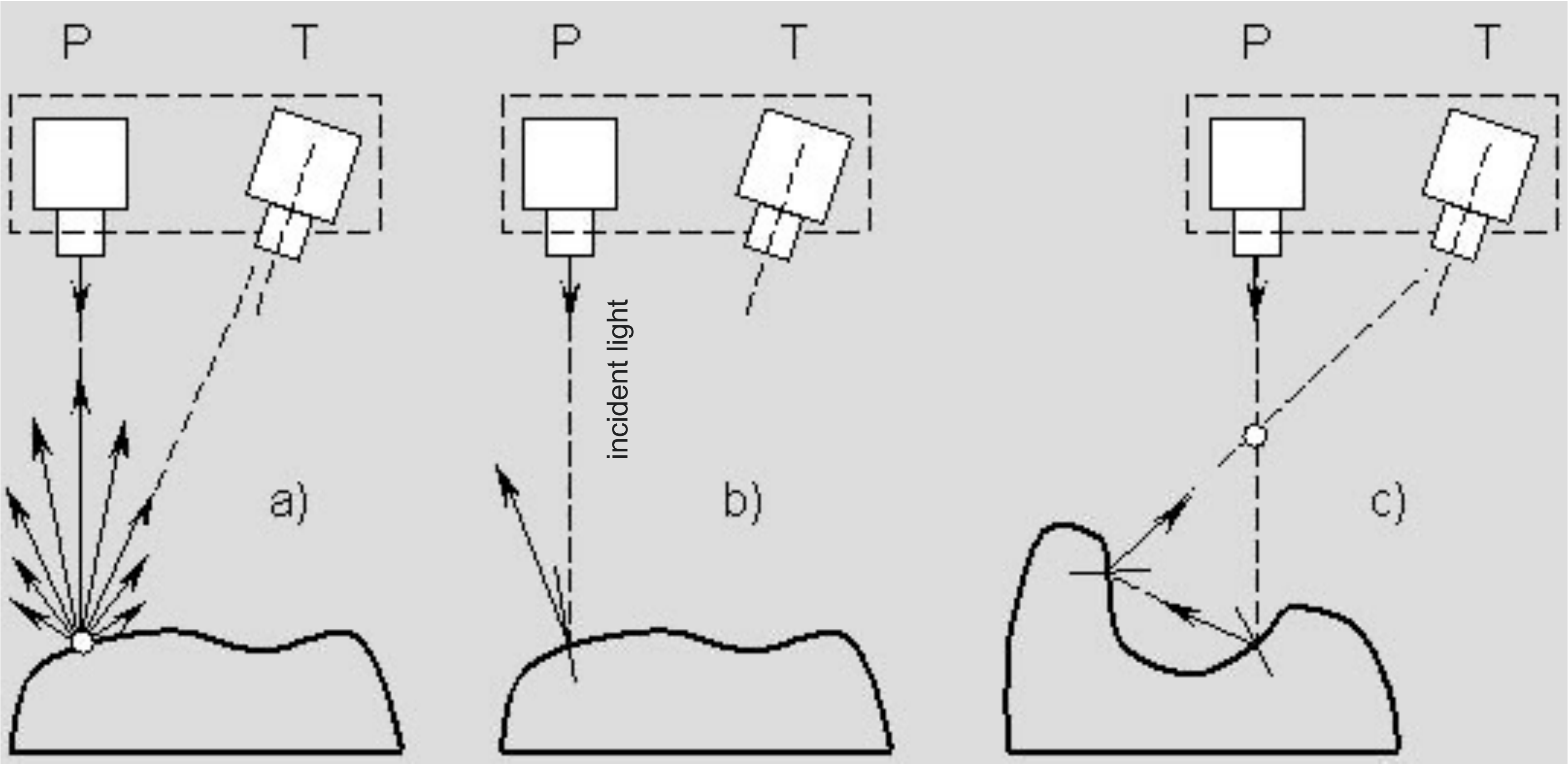






https://youtu.be/EyEKDAK1vjU

LASER TRIANGULATION

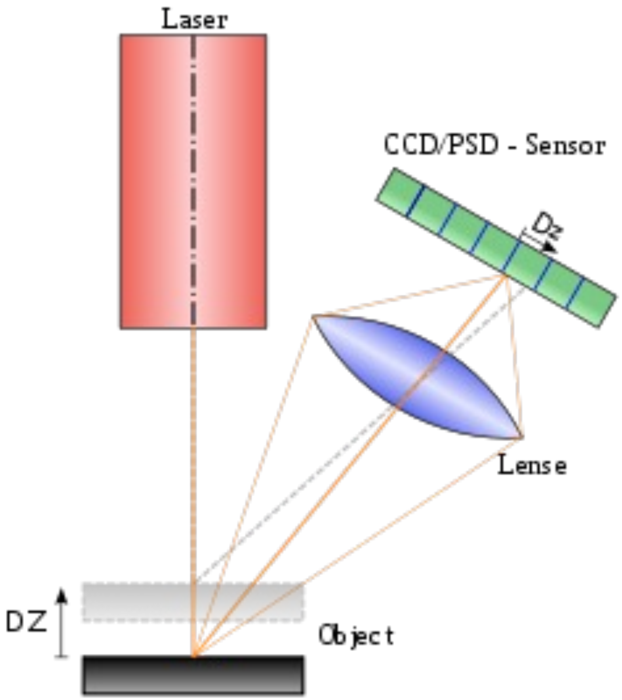


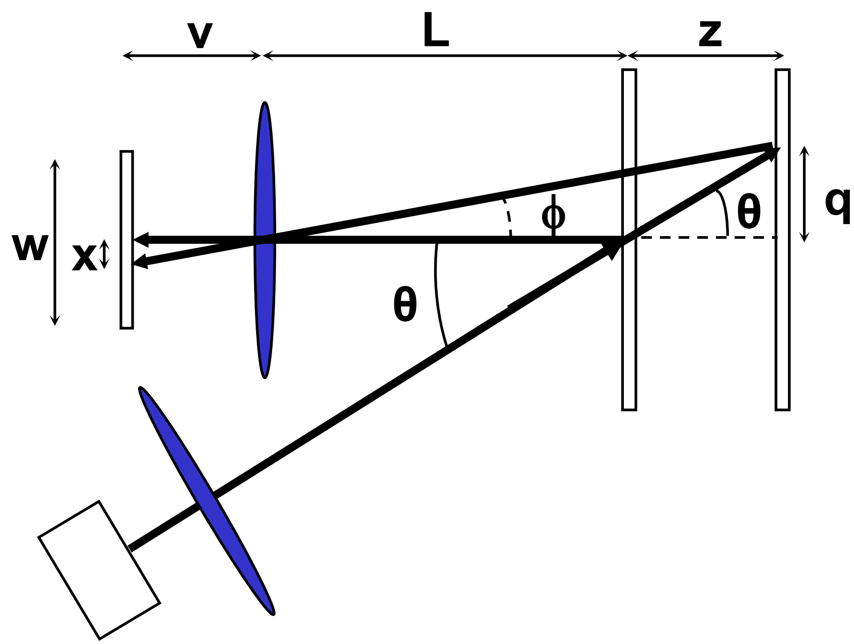
diffuse reflection

specular reflection

combined specular reflection

LASER TRIANGULATION





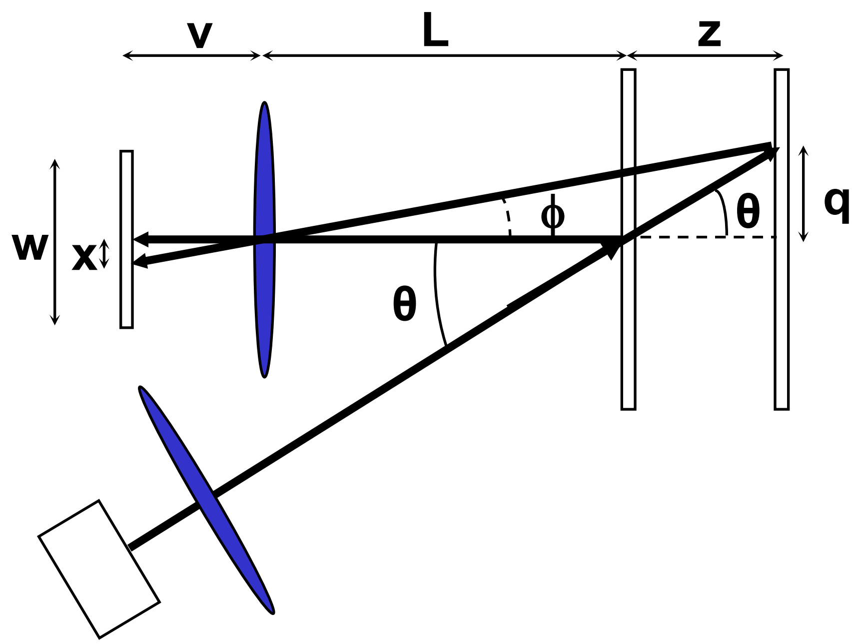
LASER TRIANGULATION

$$\tan\phi = \frac{q}{z+L}$$

$$\tan\theta = \frac{q}{z}$$

$$q = z\cdot\tan\theta = \left(z+L\right)\cdot\tan\phi$$

$$\Rightarrow\quad \tan\phi = z\cdot\frac{\tan\theta}{z+L}$$



If z << L : $\tan\phi \cong \dfrac{z}{L}\cdot\tan\theta$

$$x = v\cdot\tan\phi \cong \frac{v\cdot z}{L}\cdot\tan\theta \quad\Rightarrow\quad \boxed{z \cong \frac{x\cdot L}{v\cdot\tan\theta}}$$

LASER TRIANGULATION

Sensitivity *S* (output vs input) is equal to

$$S = \frac{x}{z} = \frac{v \cdot \tan\theta}{L} = m \cdot \tan\theta \qquad\qquad m = \frac{v}{L}$$

where *m* is the magnification, therefore $z = \dfrac{x}{m \cdot \tan\theta}$

Considering the width W of the CCD, we must have

$$x_{\min} \geq -W/2; \quad x_{\max} \leq W/2;$$

Therefore, the field of view (z_{max}-z_{min}) can be computed as

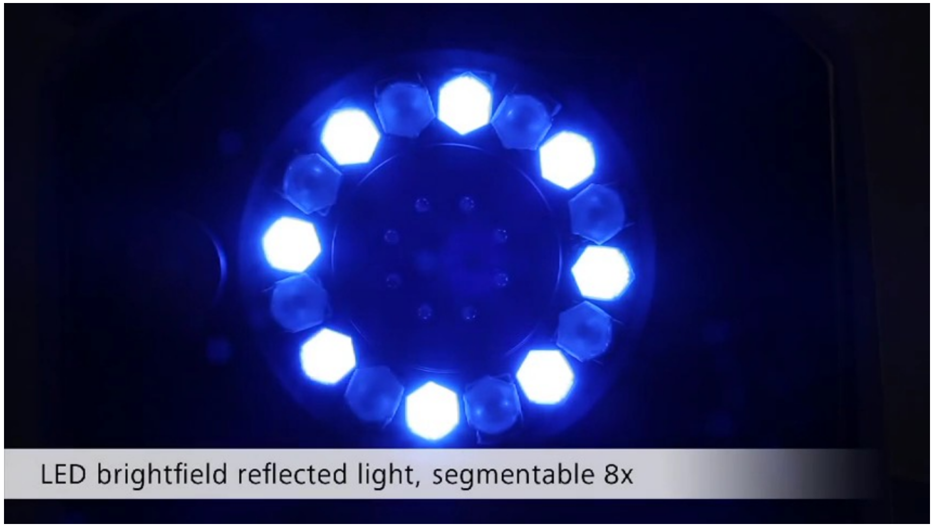
$$z_{\max} - z_{\min} = \frac{(x_{\max} - x_{\min})}{m \cdot \tan\theta} = \frac{W}{m \cdot \tan\theta}$$

Since the resolution is the smallest variation Δz that can be observed, considering a CCD capable of resolving Δx, we obtain:

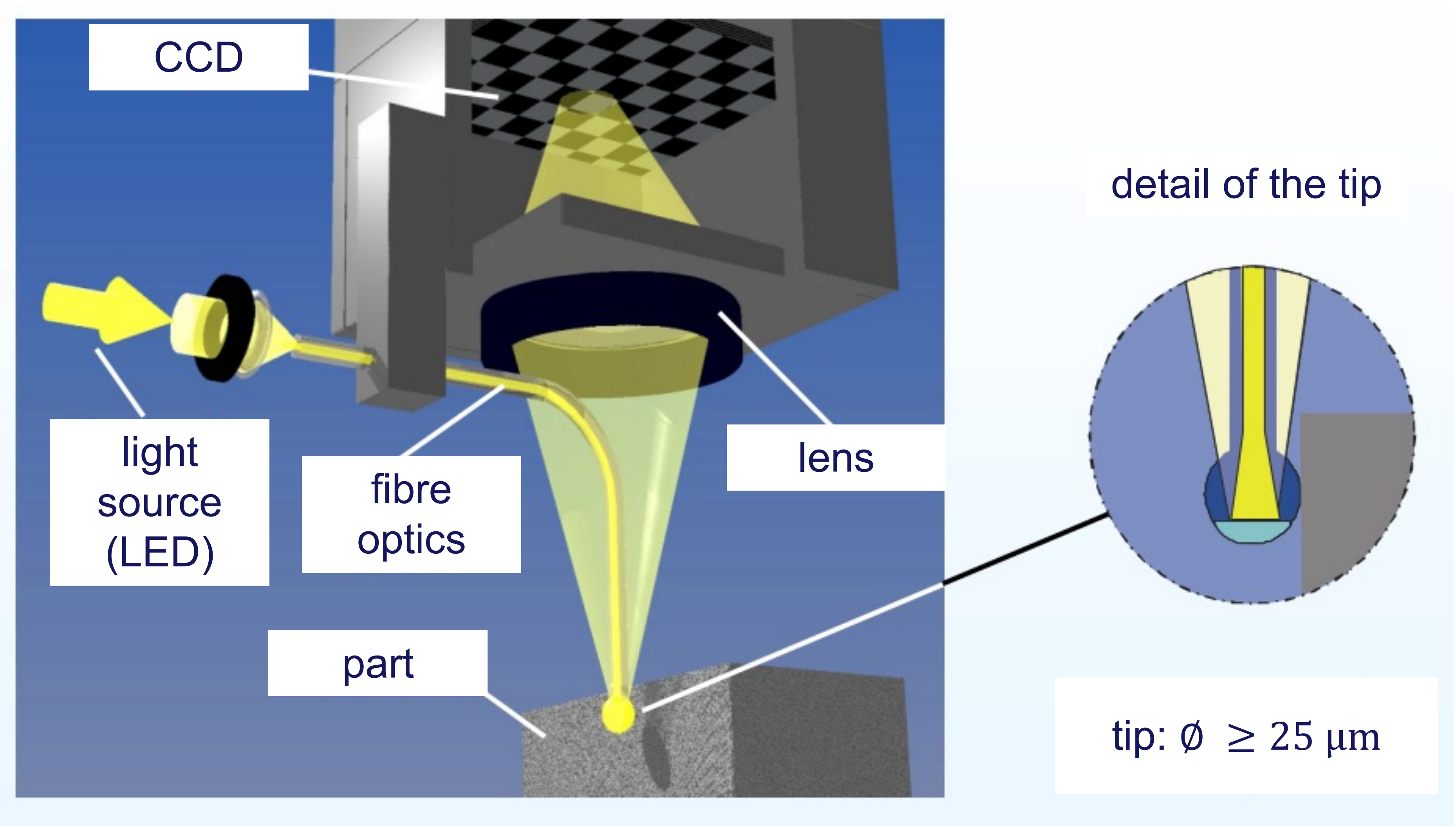
$$\Delta z = \frac{\Delta x}{m \cdot \tan\theta}$$

Multisensor Systems

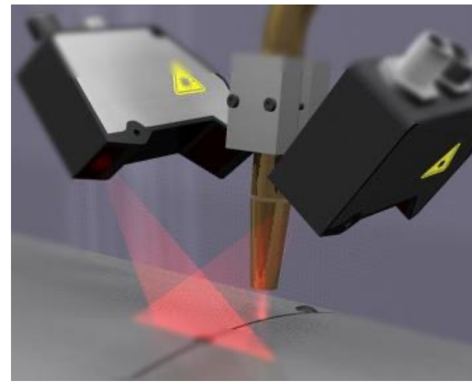
https://youtu.be/lOGBoB71oVg

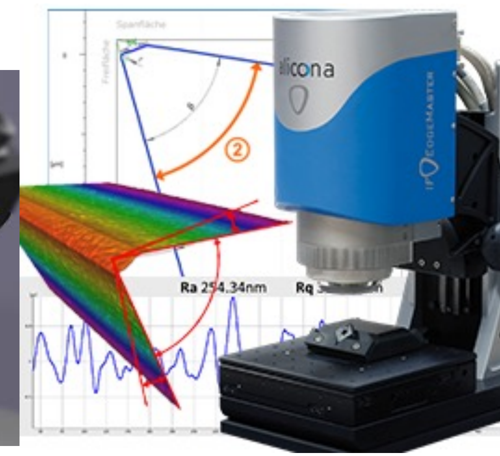


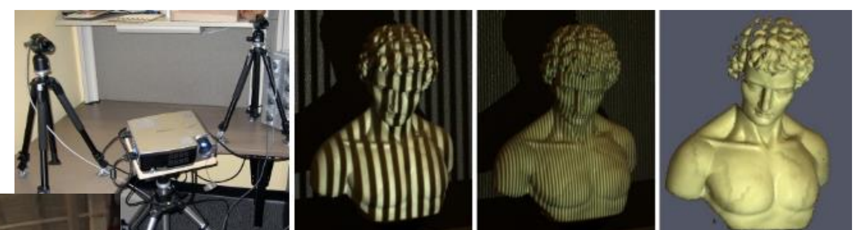
FIBER PROBE

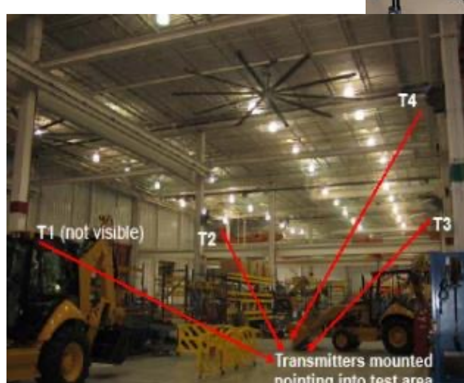

CCD
light source (LED)
fibre optics
lens
part
detail of the tip
tip: Ø ≥ 25 μm

Coordinate Measuring Systems









Coordinate Measuring Systems

Micro scale

- Micro CMM (Fiber Probe)
- Focus Variation Microscopy
- Confocal Microscopy
- Nano Computed Tomography
- Scanning Force Microscope (SFM)
- Atomic Force Microscope (AFM),
- Phase Shifting Interferometry
- Scanning white light interferometry
- Chromatic Length Aberration (CLA)
- …

Medium scale

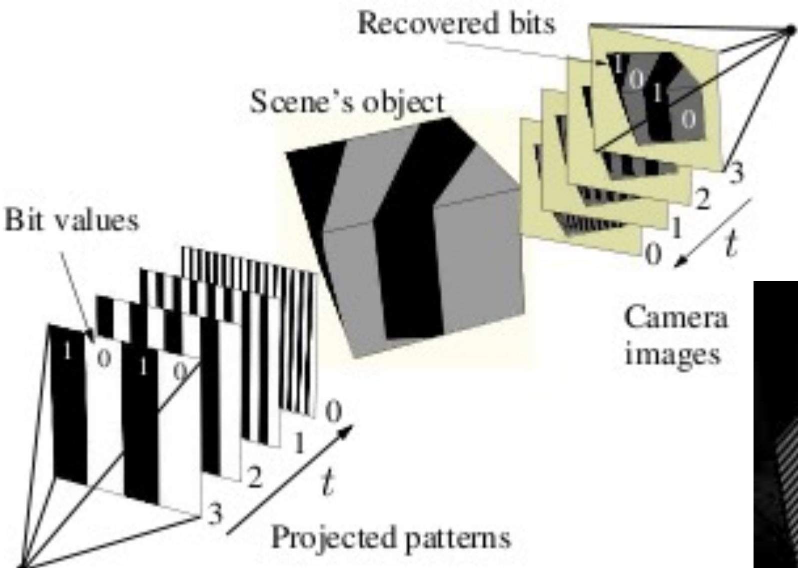
- CMM
- Image probing
- Laser Triangulation
- Micro Computed Tomography
- Focus Variation
- Photogrammetry
- Structured-light
- Conoscopic holography
- Ultrasound
- …

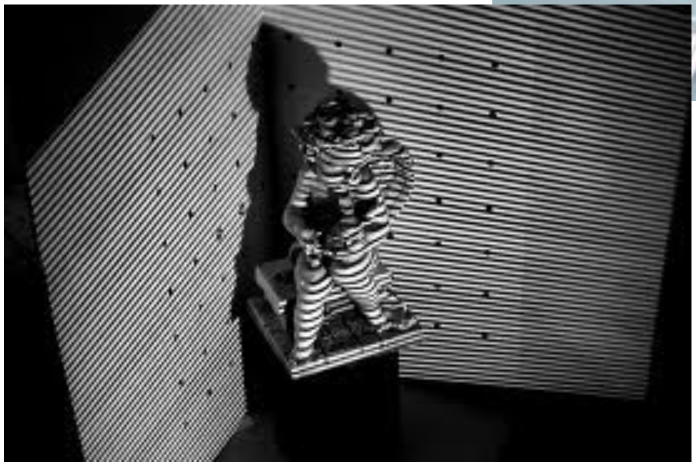
Large scale

- Large CMM
- Laser tracker
- iGPS
- …

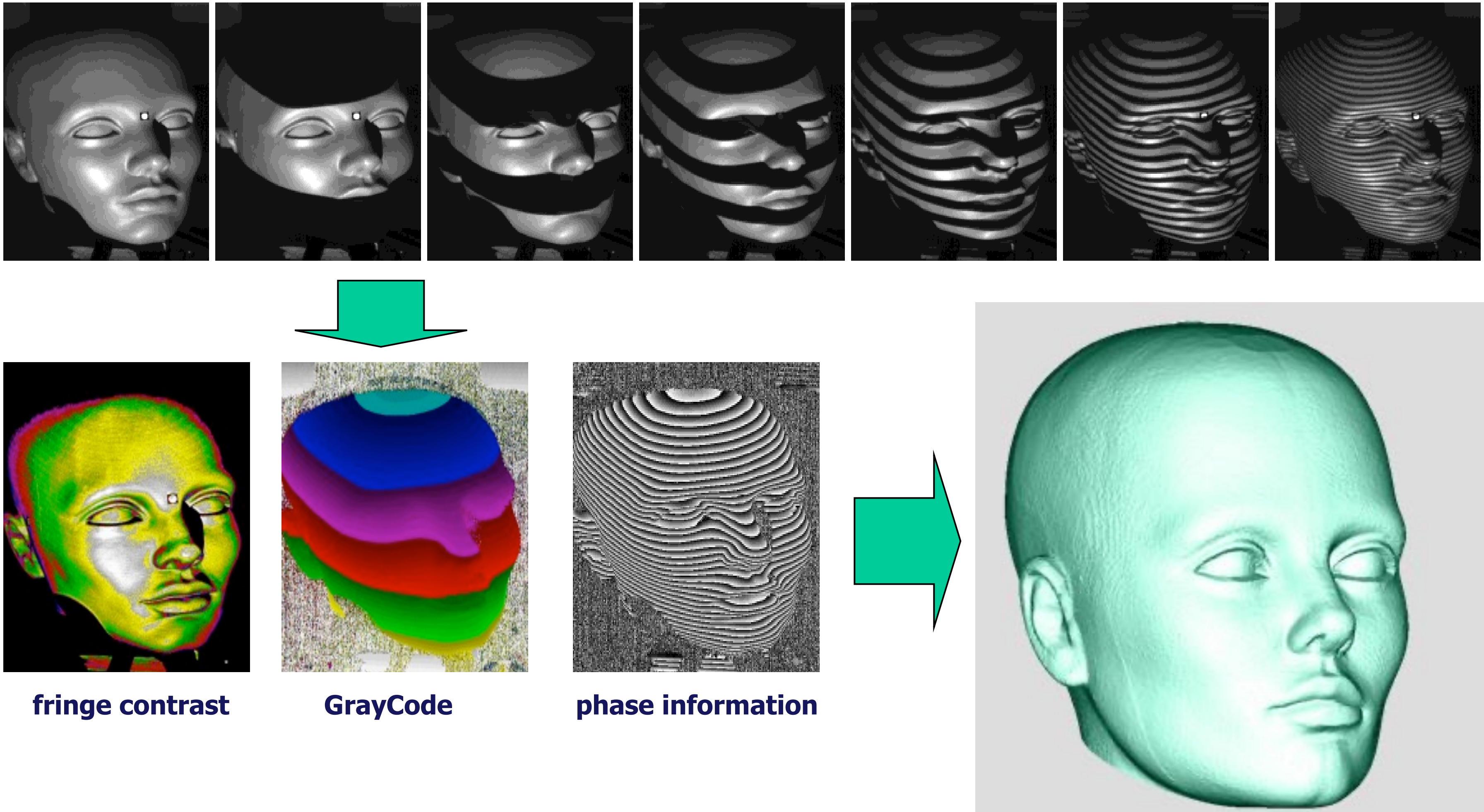
STRUCTURED LIGHT

It projects a series of light patterns on a surface, from the deformation of which it is possible to obtain a three-dimensional representation of the surface.





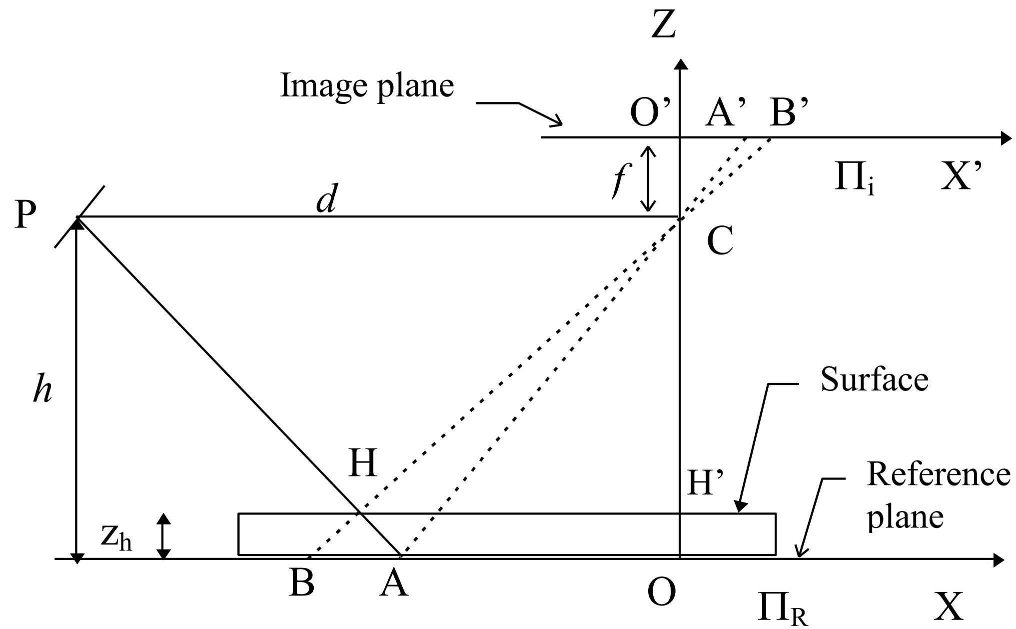
Structured Light



STRUCTURED LIGHT

Considering:

- H, a point on the surface;
- *P* and *C,* the focal points of, respectively, the projection system and the vision system;
- Π_R, the reference plane (XY plane);
- Π_i, the image plane, parallel to Π_R (hypothesis);
- O and O’, the origins of, respectively, the reference plane and the image plane;
- *f* , the known focal distance;
- *h,* the known distance of C and P from Π_R;
- *d,* the known distance between P and C.

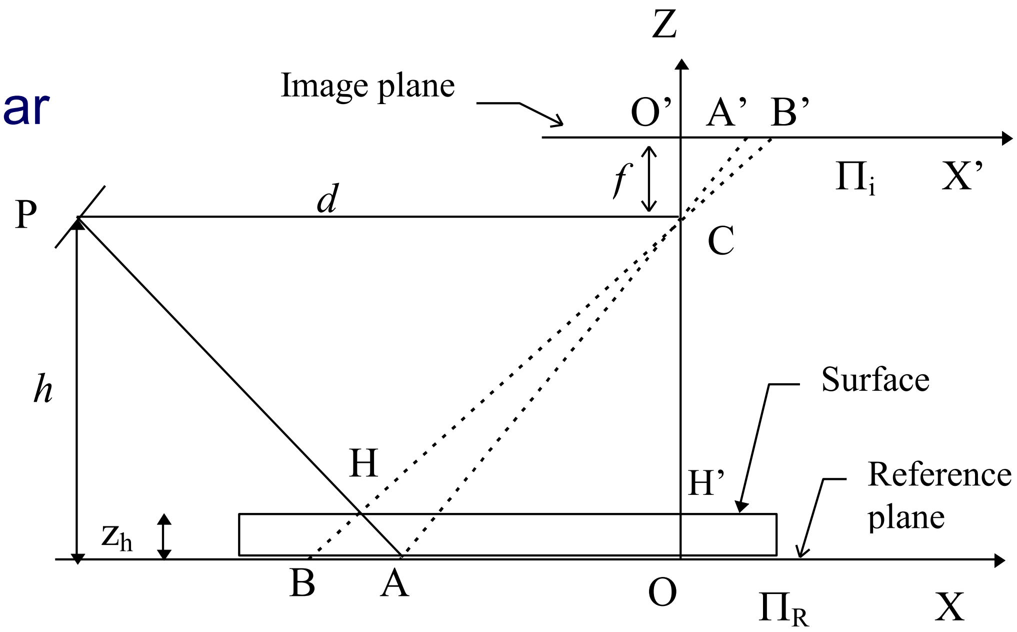


Structured Light

Triangles PHC and AHB are similar

$$\frac{h - z_h}{d} = \frac{z_h}{x_a - x_b}$$

$$z_h = \frac{h \cdot (x_a - x_b)}{x_a - x_b + d} \cong \frac{h}{d} \cdot (x_a - x_b)$$



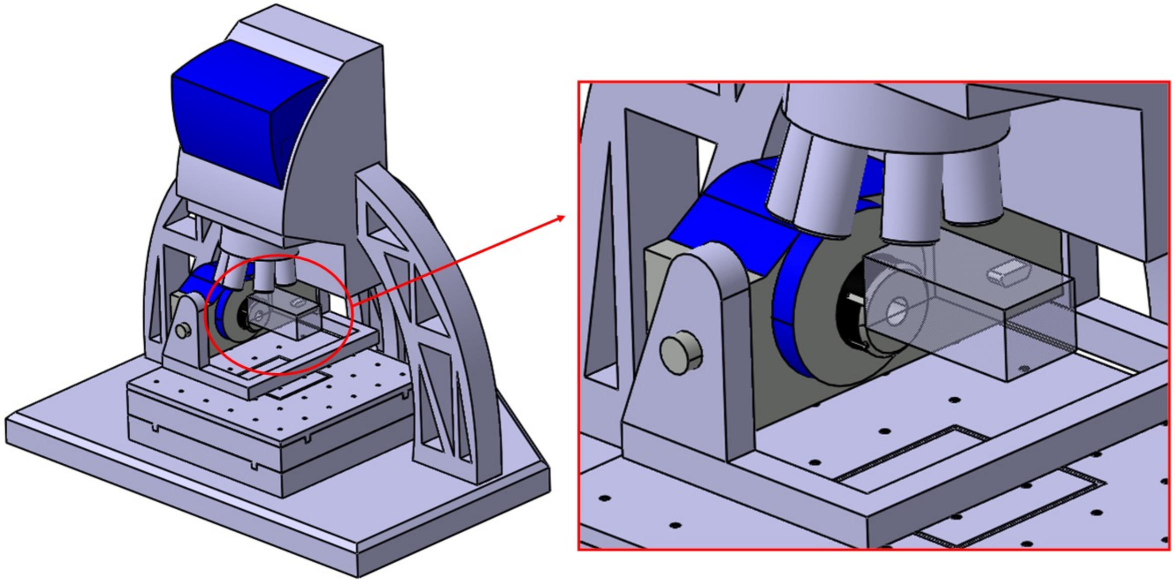
HCH' and B'CO' are similar

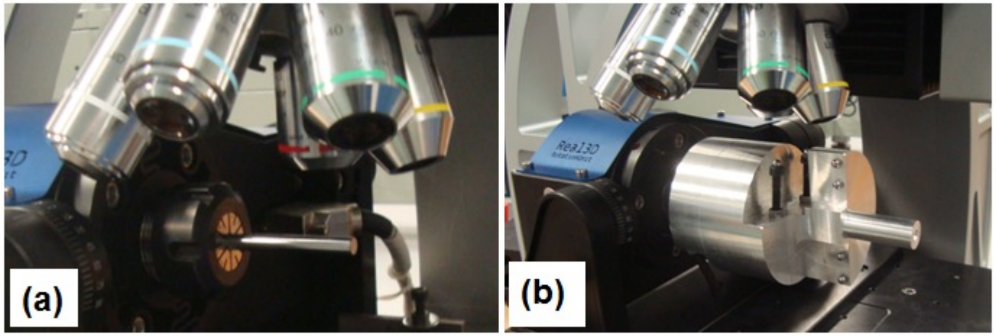
$$x_h = \frac{x'_b}{f} \cdot (h - z_h) = \frac{x'_b}{f} \cdot h \cdot (1 - \frac{z_h}{h}) \cong \frac{x'_b}{f} \cdot h$$

$$y_h = \frac{y'_b}{f} \cdot (h - z_h) \cong \frac{y'_b}{f} \cdot h$$

Focus Variation Principle

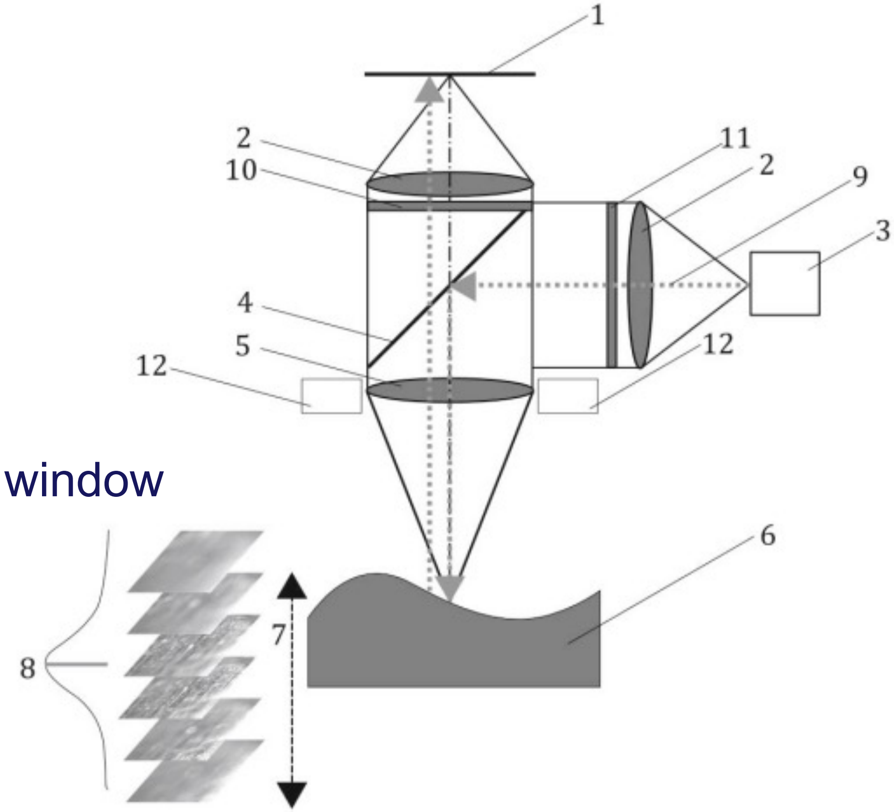
A microscope takes a series of images of a sample at different heights and evaluates the 3D topography from them.





Focus Variation Principle

1. CCD sensor
2. lenses
3. white light source
4. semi-transparent mirror
5. objective lens with limited depth of field
6. sample
7. vertical movement with driving unit
8. contrast curve calculated from the local window
9. light rays from the white light source,
10. optional analyzer
11. optional polarizer
12. optional ring light



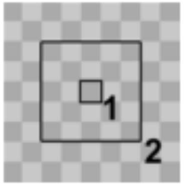
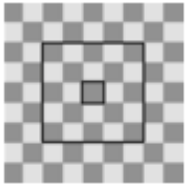
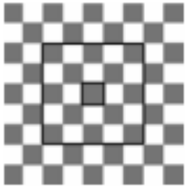
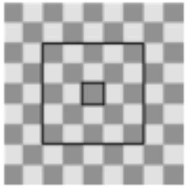
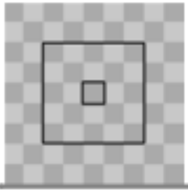
Focus Variation Principle

For each image I_z, for each point (x,y) in the image, a focus measurement is calculated:

$$F_z(x,y) = FM\left(reg_w\left(I_z, x, y\right)\right)$$

Note that this function works over a neighborhood of the considered point. The most common focus measurement is:

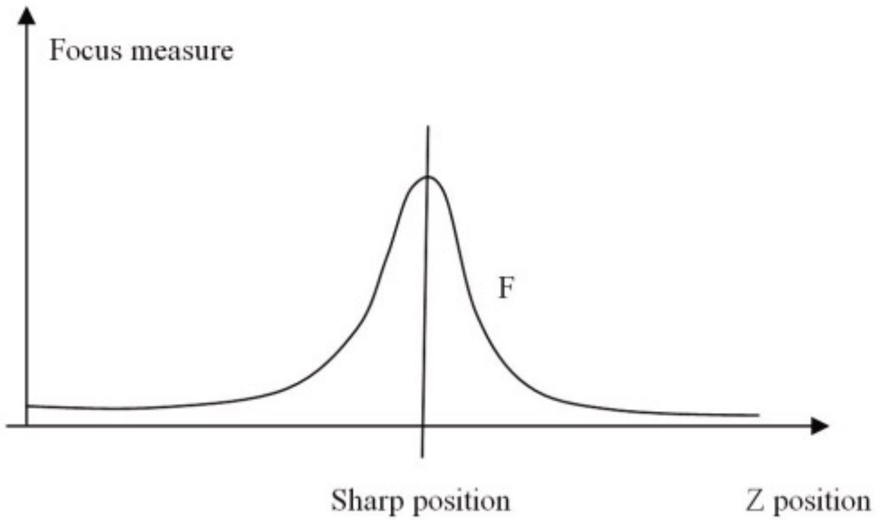
$$F_z(x,y) = \frac{1}{n^2} \sum_{i \in reg_w(I_z, x, y)} \left(GV_i - \overline{GV}\right)^2$$

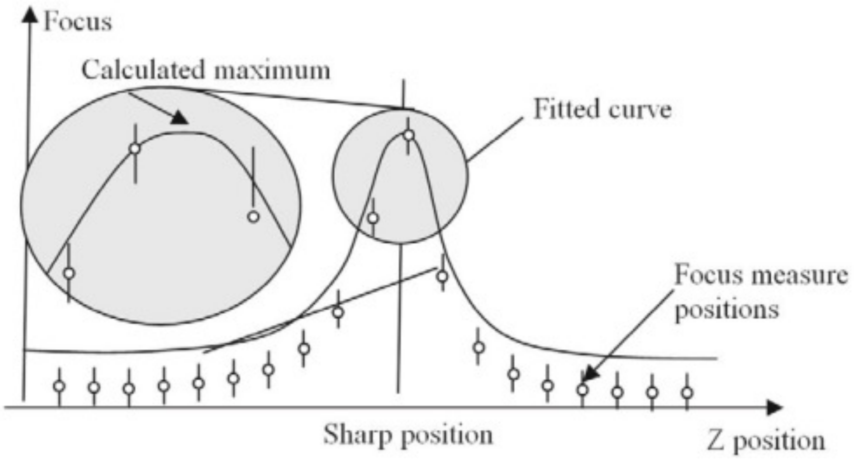
Scan position	Surface image	Standard deviation
Out of focus		10
Almost in focus		20
In focus		50
Almost in focus		20
Out of focus		10

Focus Variation Principle

Considering the different heights, a curve is obtained, whose maximum must be identified. To obtain the maximum, it is possible to consider:

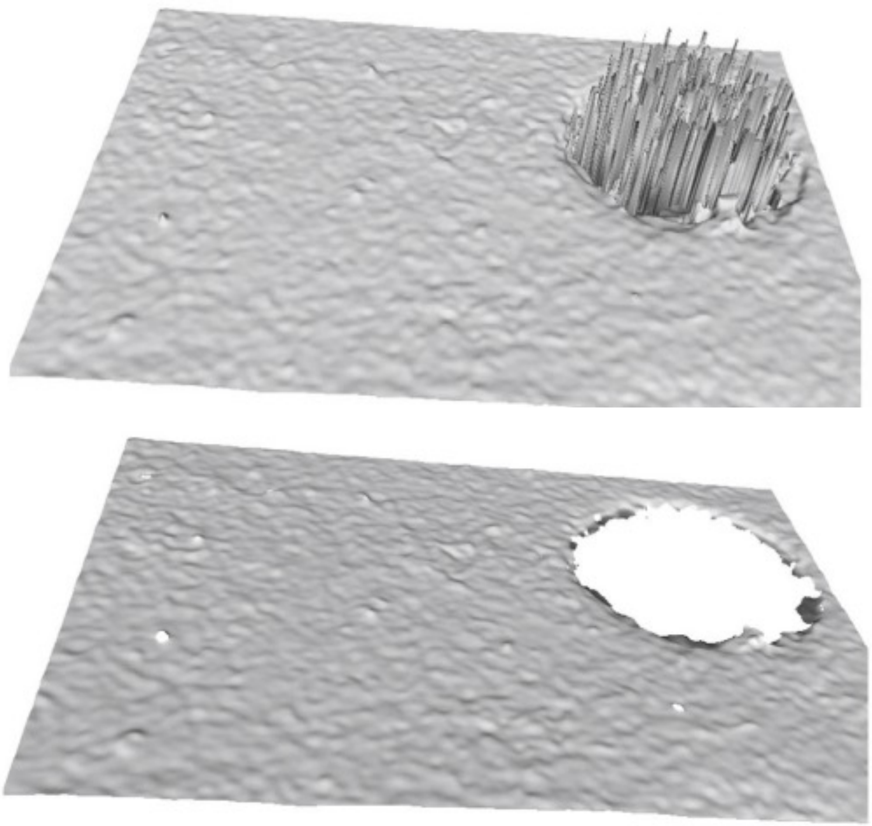
- Maximum value (fastest, least accurate)
- Polynomial fitting
- Point spread function (PSF) fitting (slowest, most accurate)

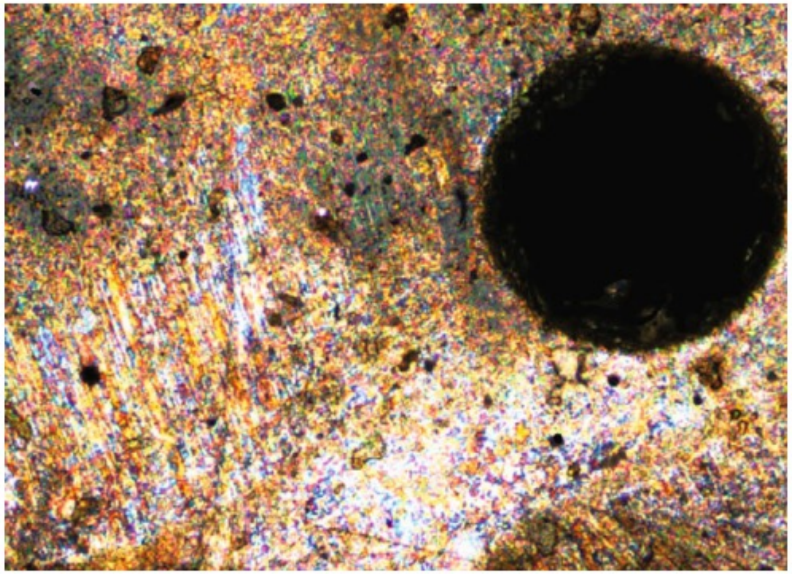




Focus Variation

The fitting quality of the curve allows the identification of badly scanned areas of the surface.

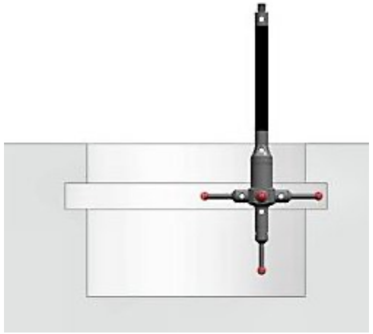




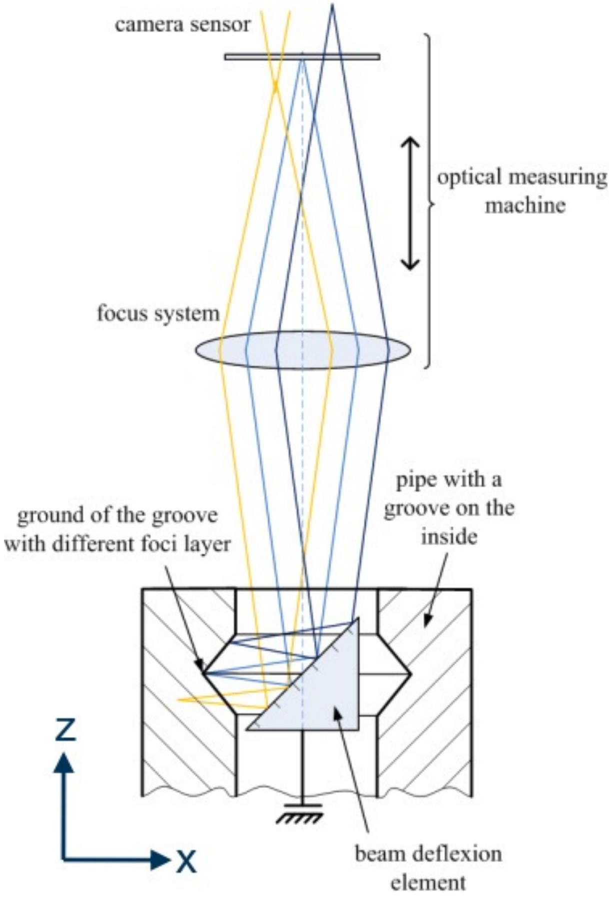
(a)

Contact and Non-contact Probes Limit

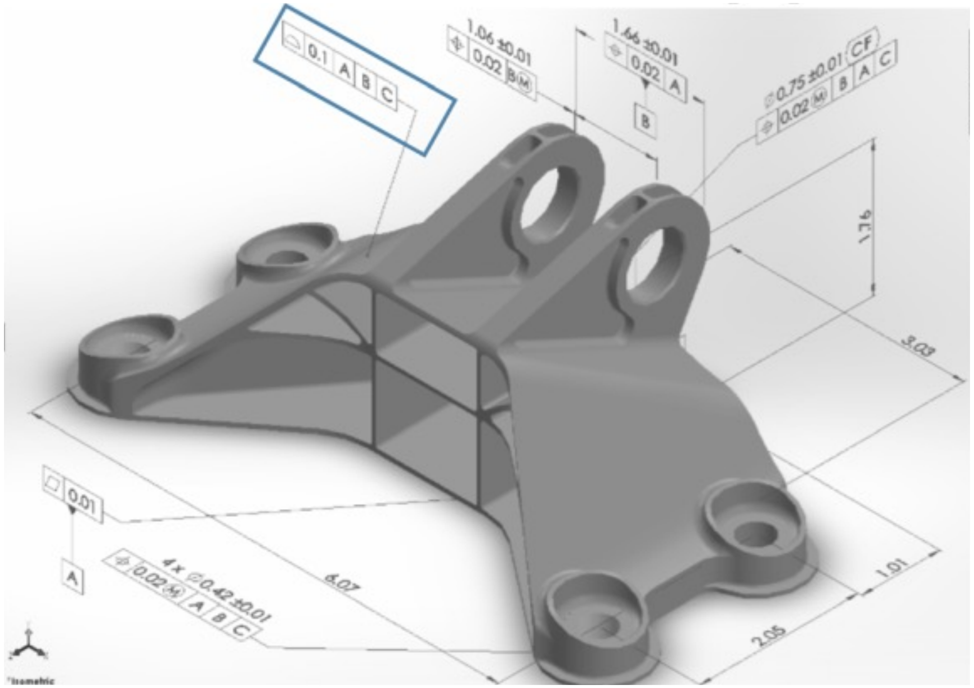
Contact and non-contact probes can measure only the surfaces they can touch or see

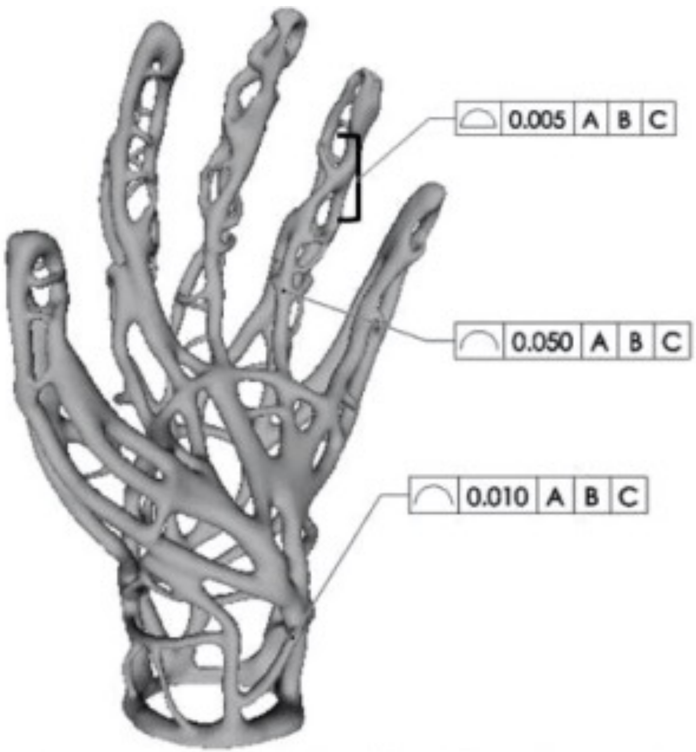


Accessibility issue !

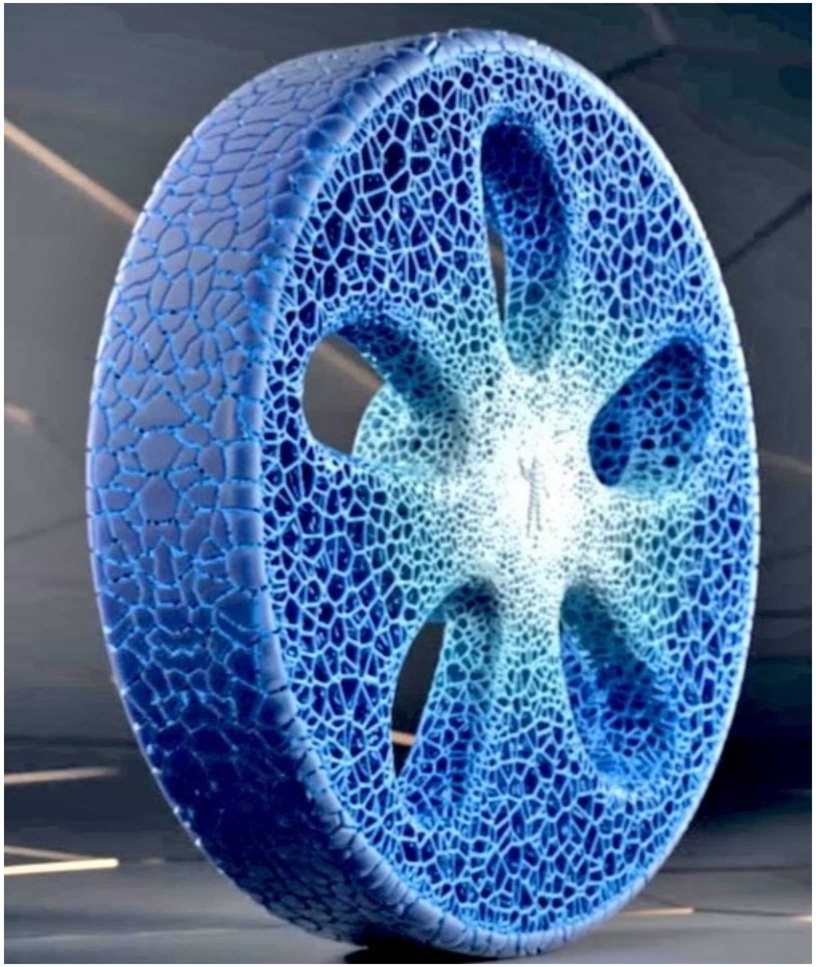


HOW TO MEASURE A COMPLEX PART?

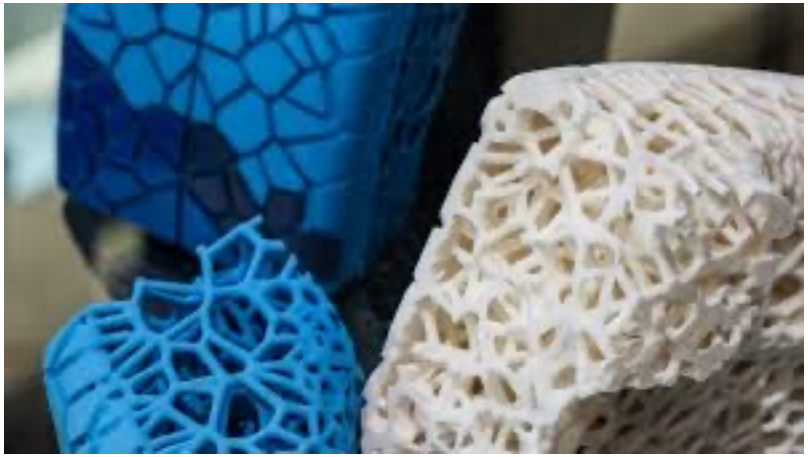




How to measure a complex part?



[Source: Michelin Vision]



Coordinate Measuring Systems

Micro Scale

- Micro CMM
- **Micro X-Ray CT**
- Focus Variation Microscopy
- Phase Shifting Interferometry
- Scanning white light interferometry
- Confocal Microscopy
- Atomic Force Microscope (AFM)
- Scanning Force Microscope (SFM)
- …

Meso Scale

- CMM
- Image probing
- Photogrammetry
- Structured Light
- Laser stripe
- **X-Ray CT**
- Ultrasound
- …

Large Scale

- Large CMM
- Laser tracker/tracer
- Indoor GPS
- (Laser stripe)
- (Structured Light)
- (Photogrammetry)

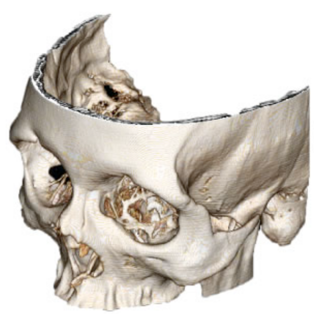
X-Ray Computed Tomography

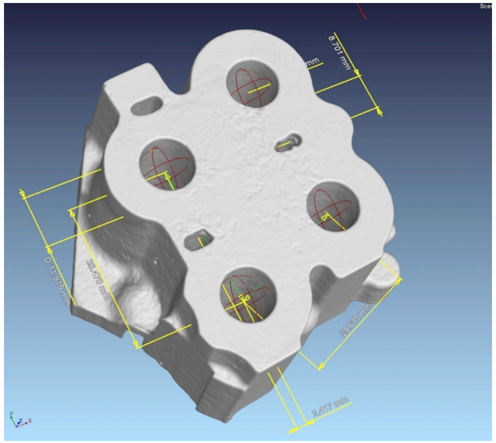
1970 1980 1990 2000

01/10/1971 First application in medicine.

CT application in non destructive testing.

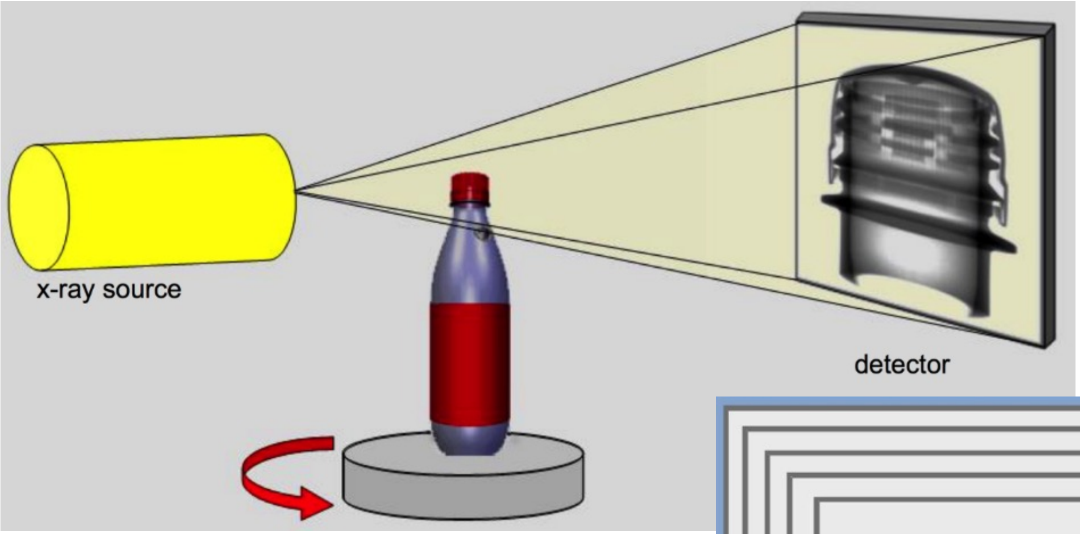
2006 CT as metrological device

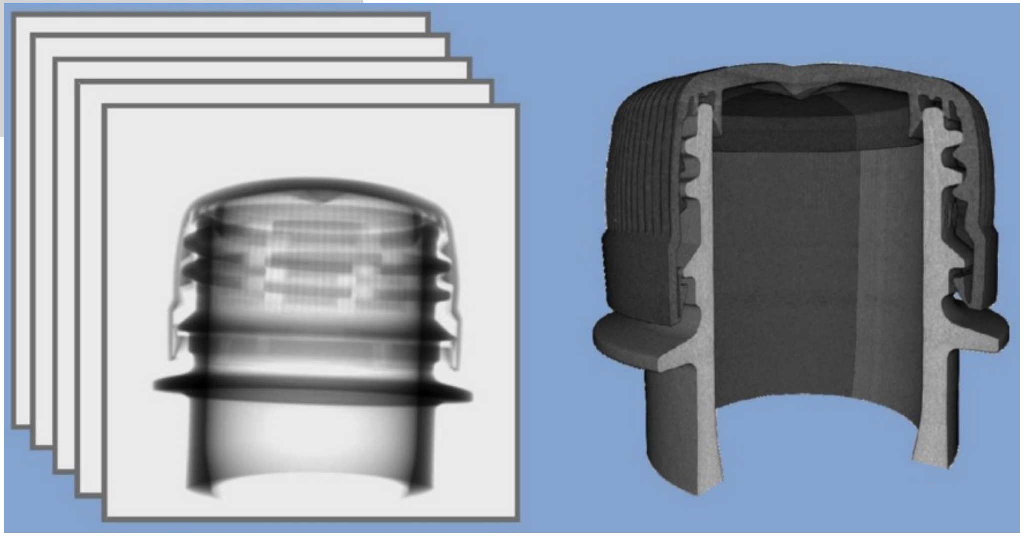




3D X-Ray Computed Tomography

By taking a series of X-Ray images of the object to be measured, according to different rotations, it builds a volumetric image of the same object.

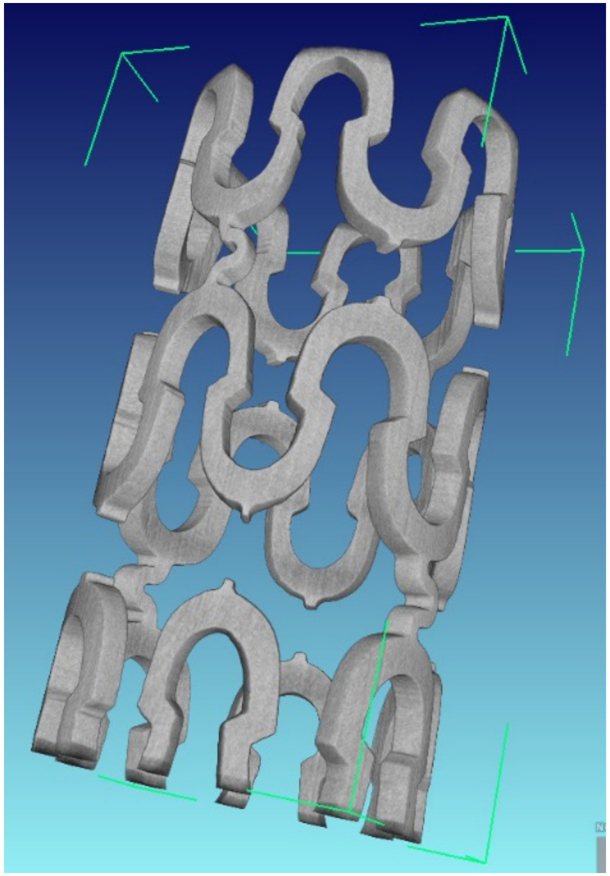


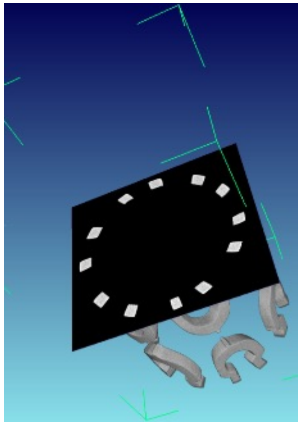


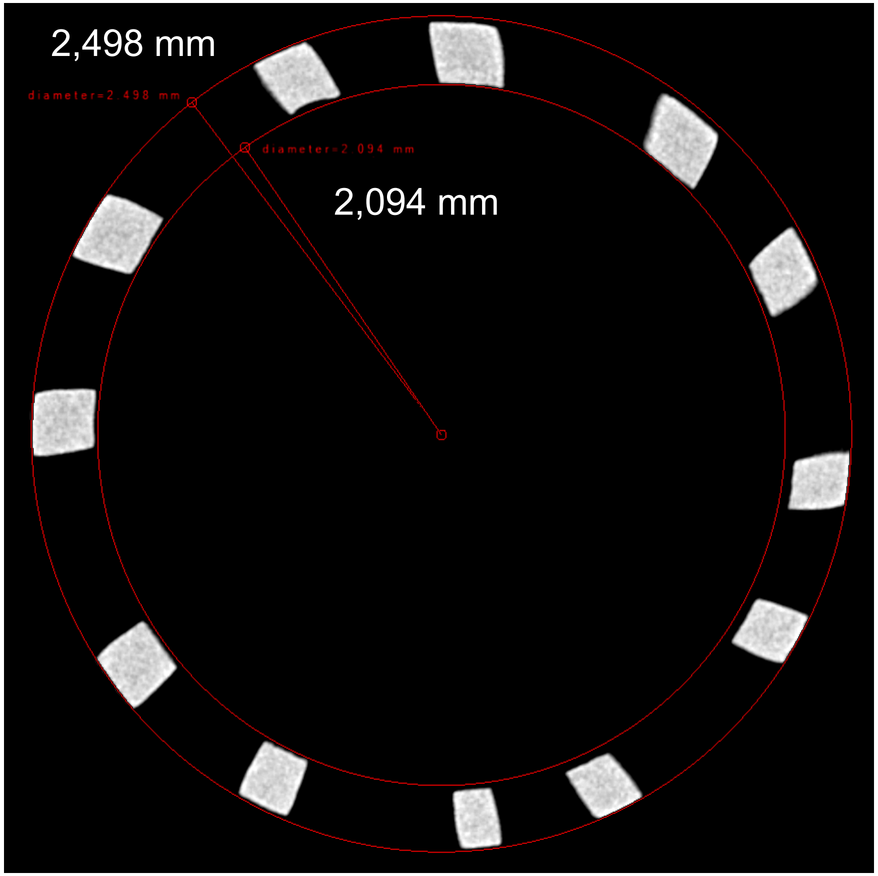
Magnesium Alloy Stent

Material: AZ31 (3%Al, 1% Zn)
Process: Microlaser cutting

$D_{internal}$: 2,1 mm
$D_{external}$: 2,5 mm

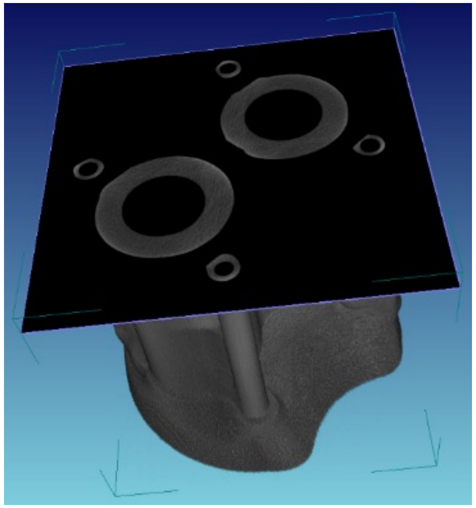


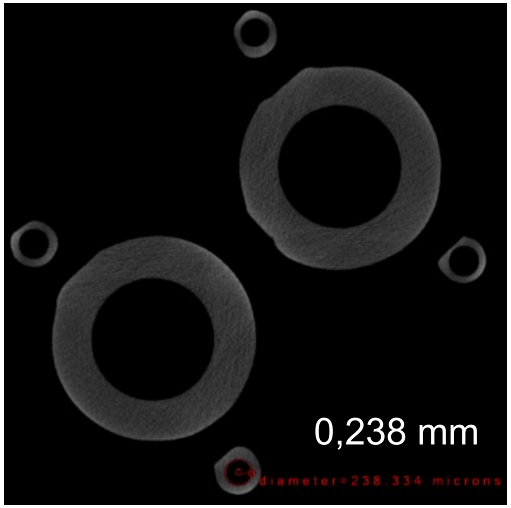


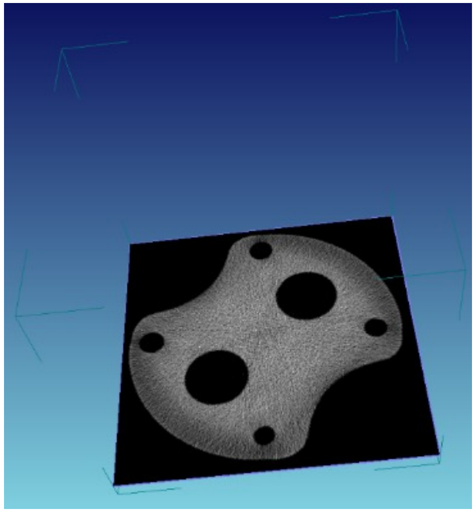


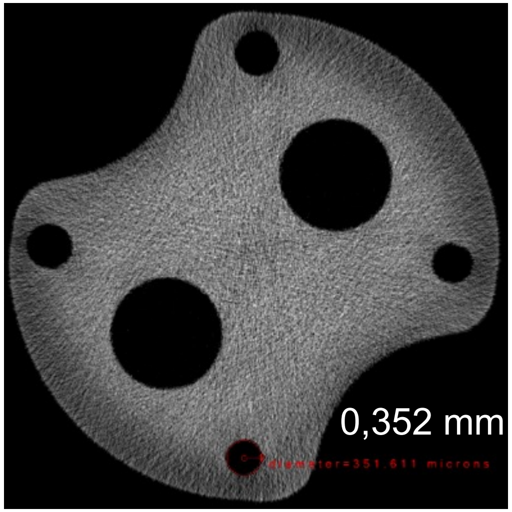
DIE COMPONENT TO EXTRUDE PLASTIC CATHETER

Material: STAVAX
Process: Micro-Milling/EDM

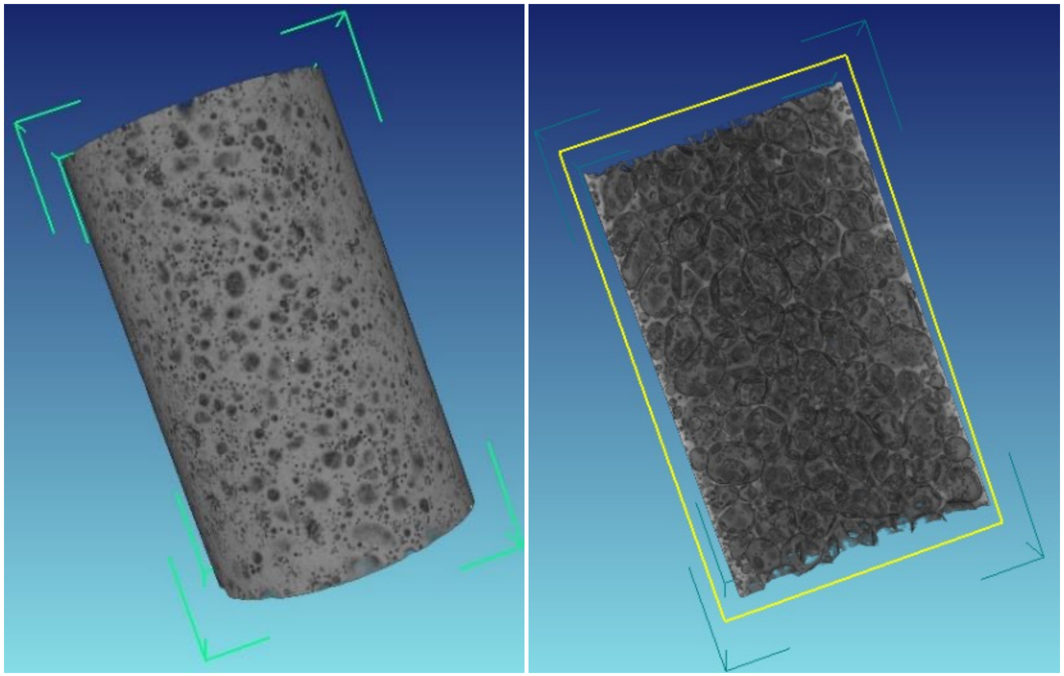


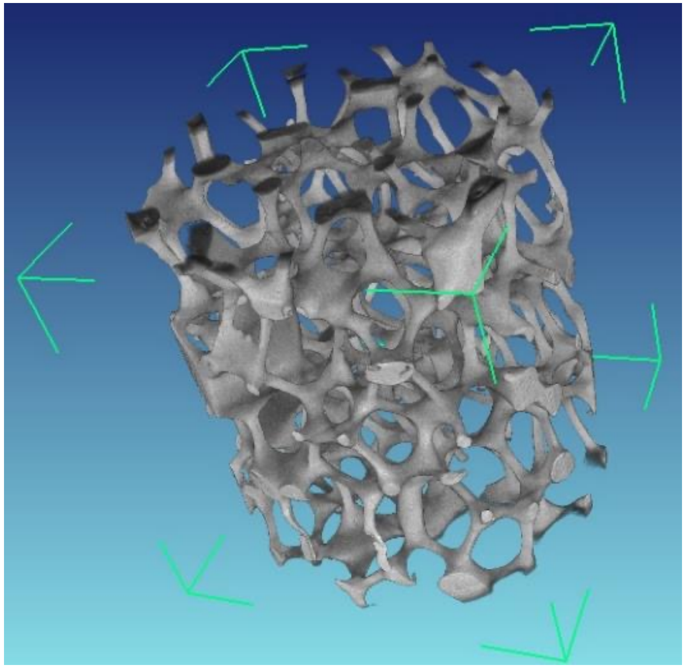






Aluminium Foam and Sponge

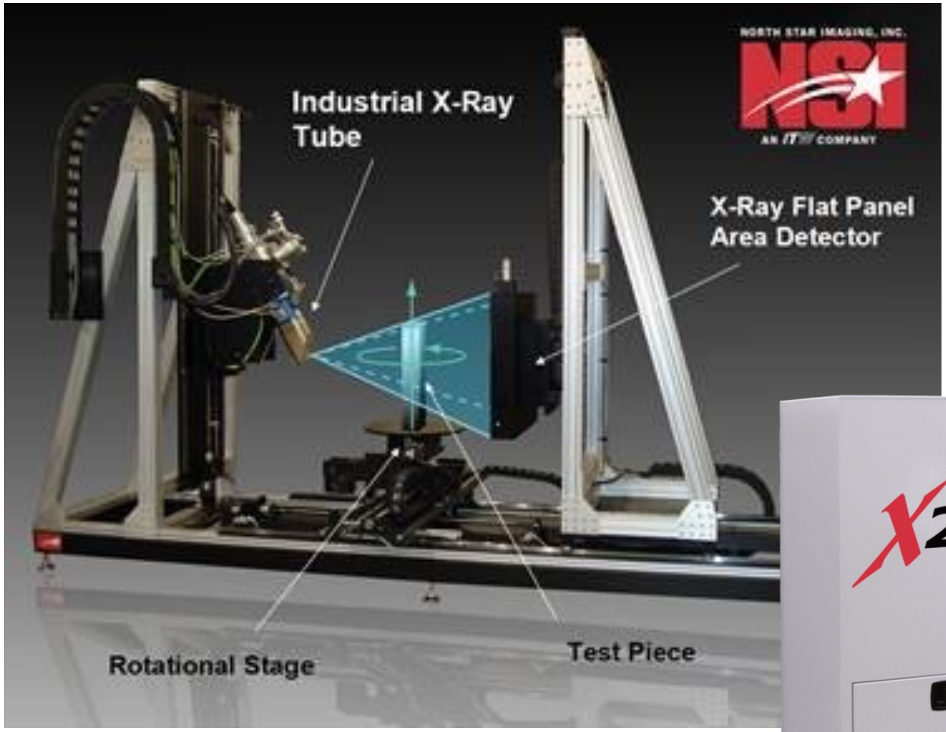




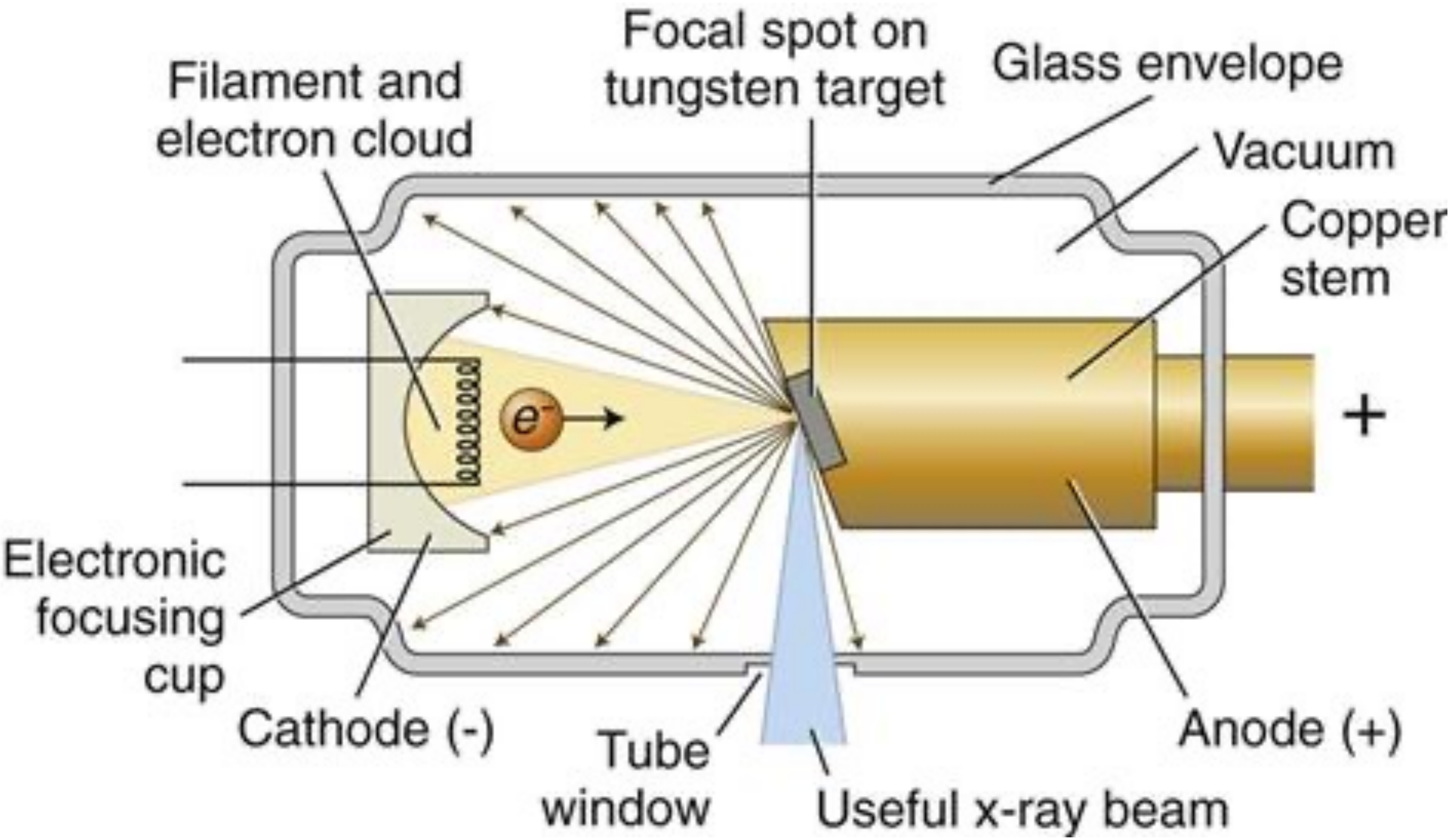
3D X-Ray Computed Tomography

https://youtu.be/ovDZtOeed14

X-RAY CT SCANNER STRUCTURE

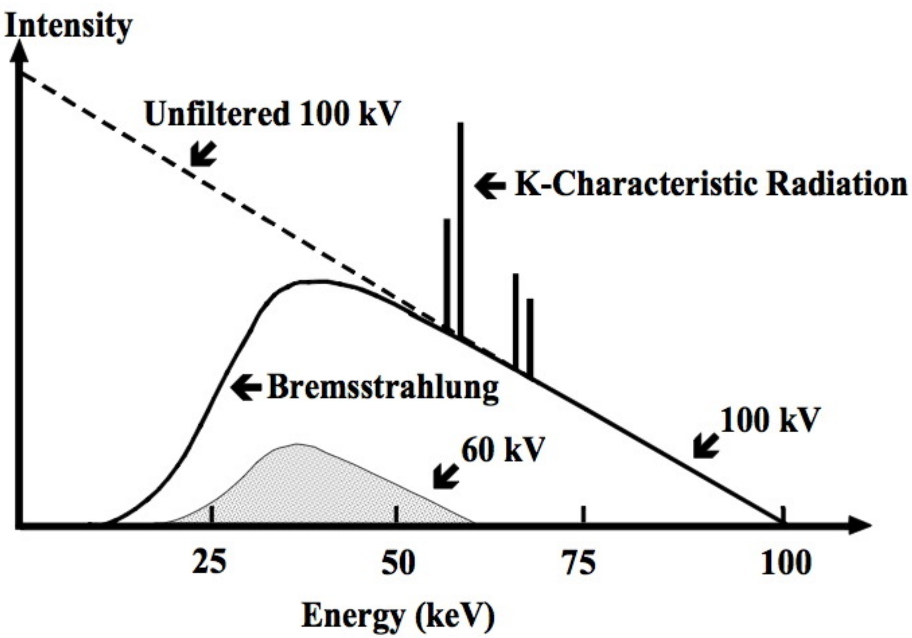


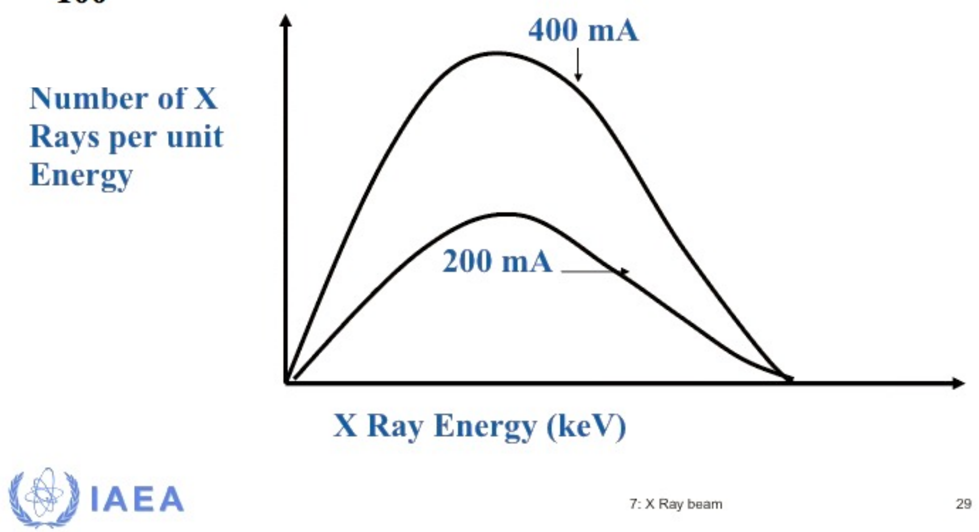
X-RAY SOURCE – REFLECTION TUBE



Pocketdentistry.com

X-RAY SOURCE – SPECTRUM



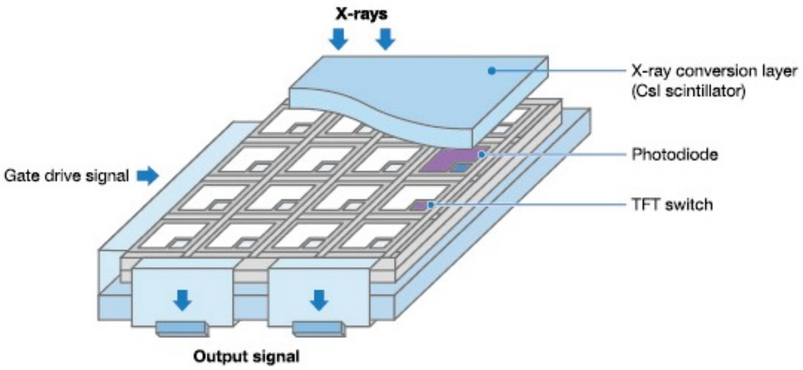


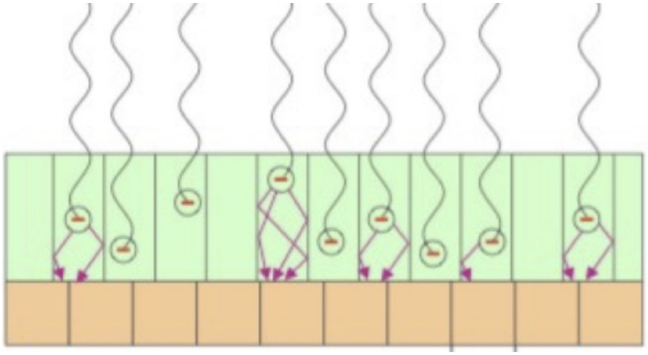
X-Ray Detector

Scintillator + Photodiode

Scintillator:

- Amorphous: larger pixel size, “infinite” life.
- Crystalline: smaller pixel size, 3-5 years life.

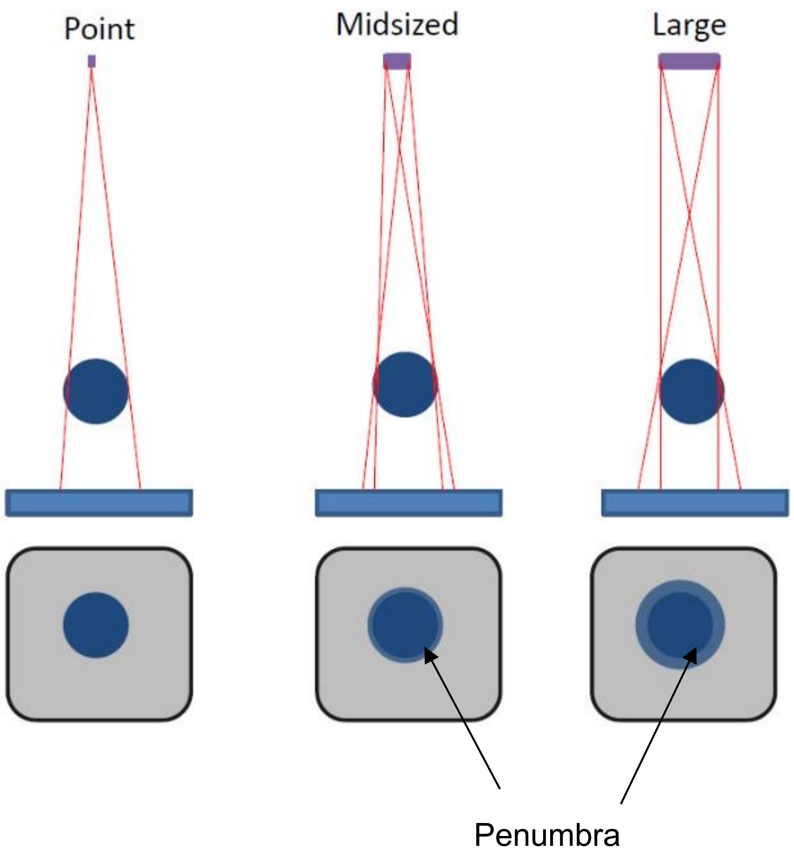


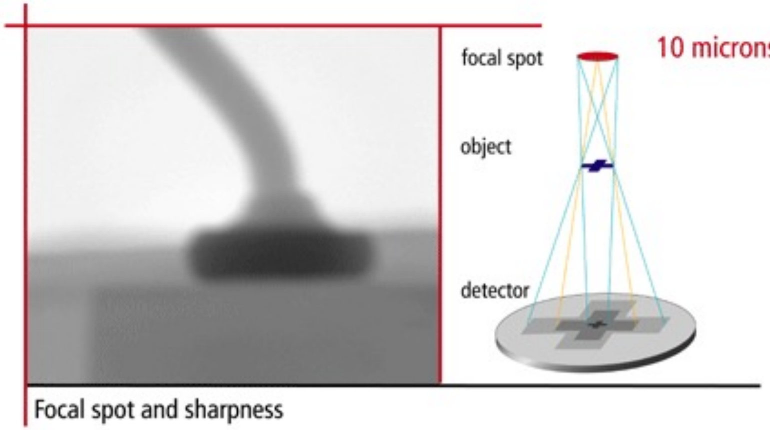


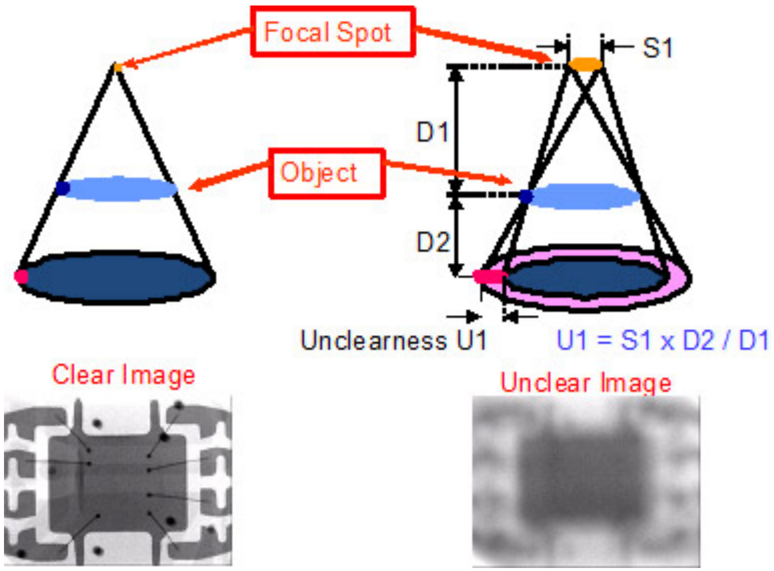
X-Ray Source and Material Thickness

	130 KV	150 KV	190 KV	225 KV
Steel/Ceramic	5 mm	8 mm	25 mm	40 mm
Aluminium	30 mm	50 mm	90 mm	150 mm
Plastics	90 mm	130 mm	200 mm	250 mm

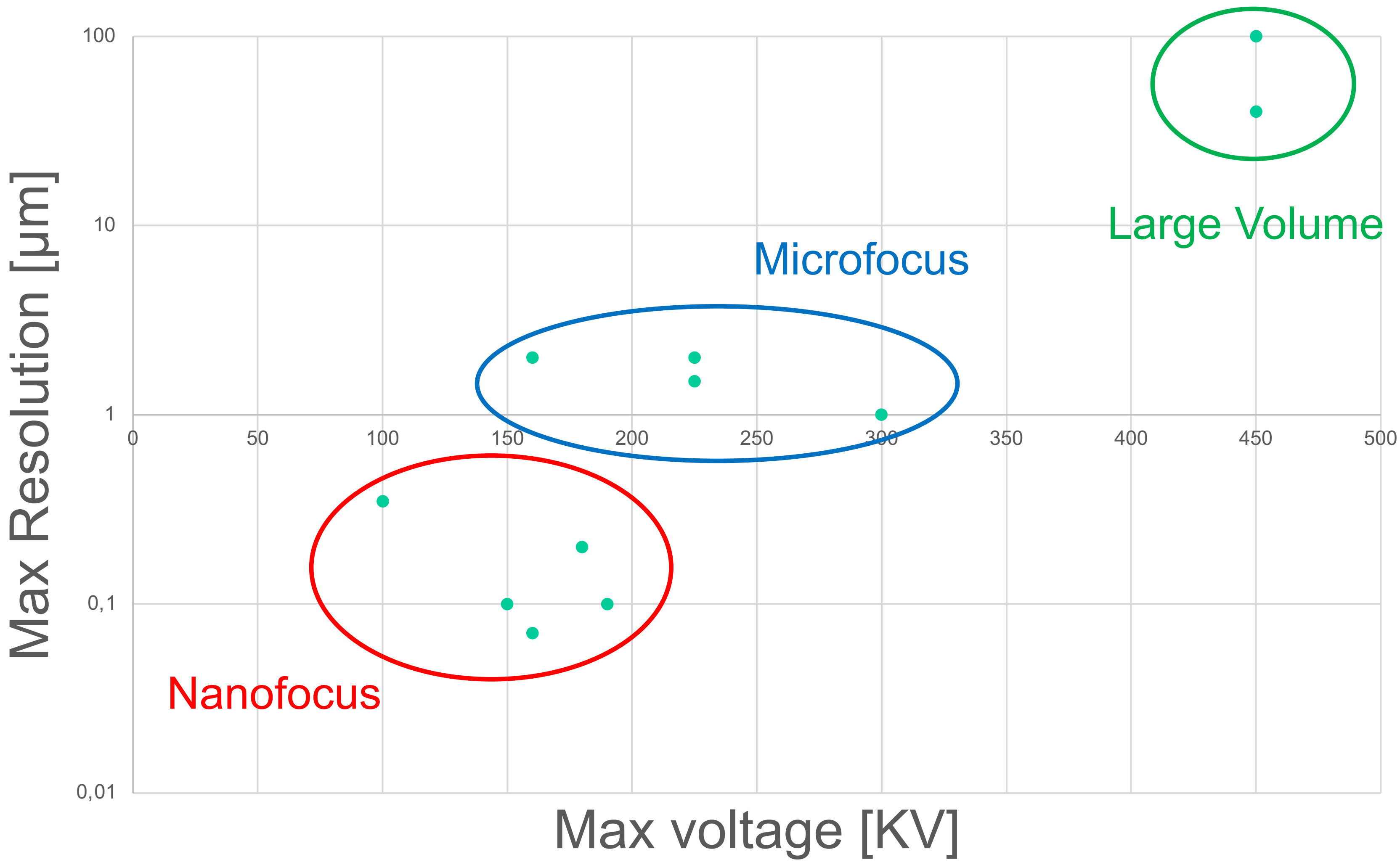
X-RAY SOURCE AND FOCAL SPOT



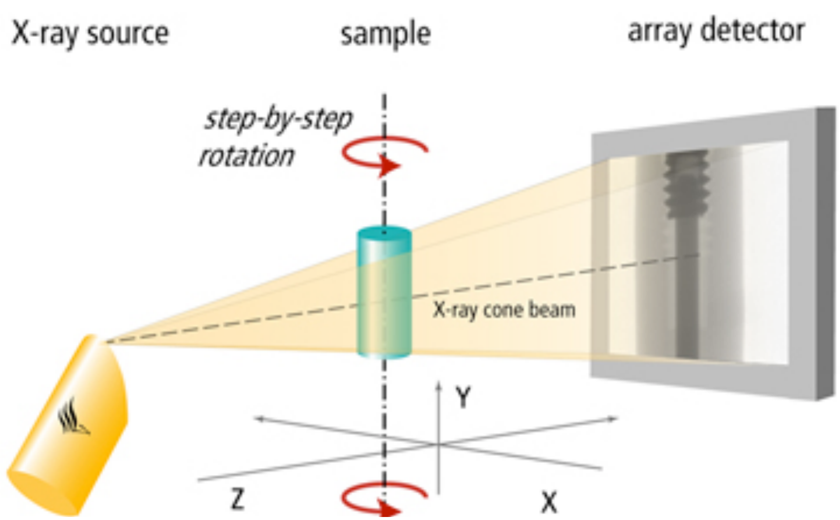




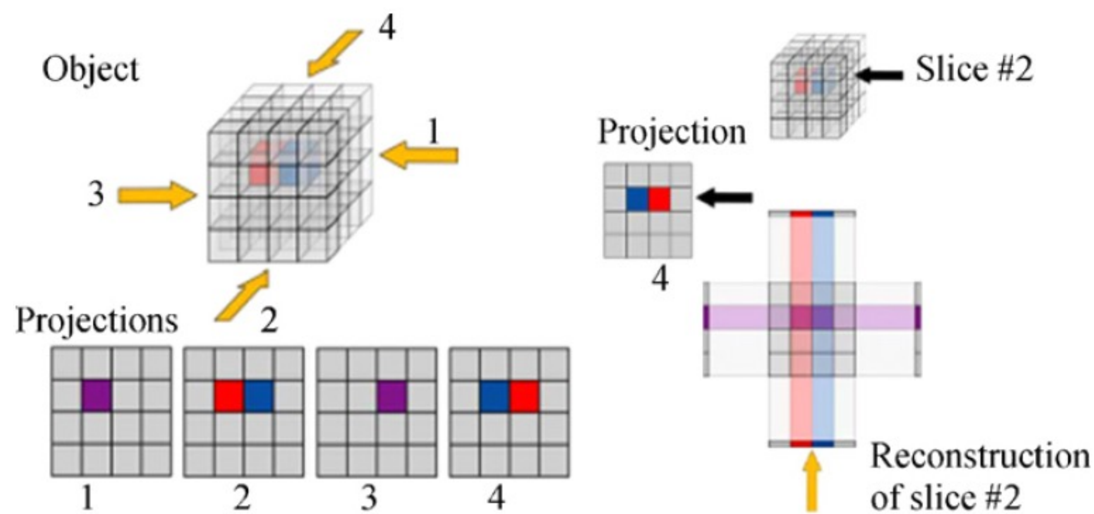
PENETRATION VS RESOLUTION



3D X-Ray Computed Tomography



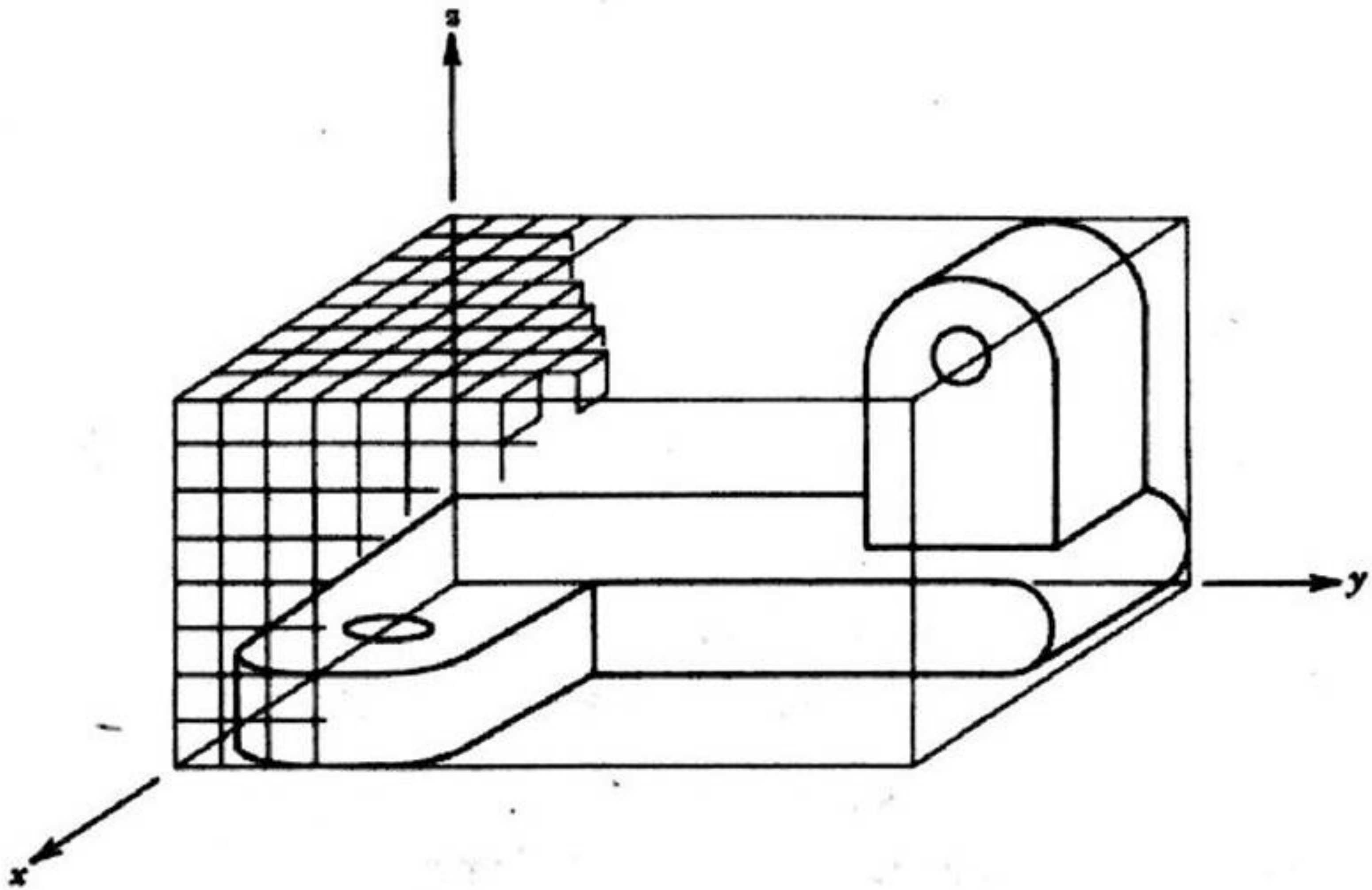
Calculating volume data by back projection of filtered radiographic images



CIRP Annals - Manufacturing Technology 60 (2011) 821–842

VOXEL-BASED REPRESENTATION

Spatial occupancy enumeration



VOXEL-BASED REPRESENTATION

Spatial occupancy enumeration

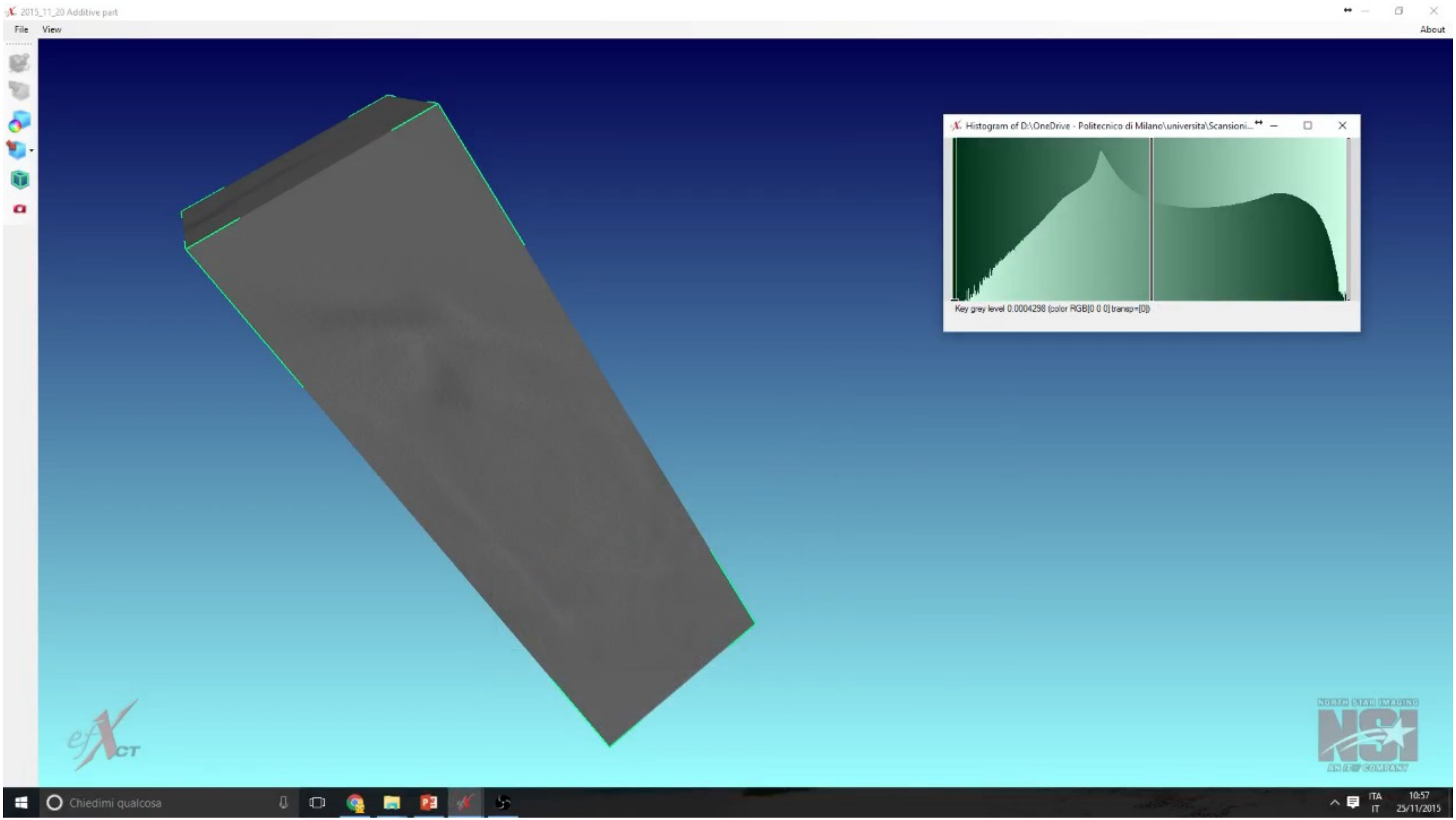
This scheme is essentially a list of spatial cells occupied by the solid.

The cells, also called ***voxels***, are cubes of a fixed size and are arranged in a fixed spatial grid (other polyhedral arrangements are also possible but cubes are the simplest).

Each cell may be represented by the coordinates of a single point, such as the cell's centroid.

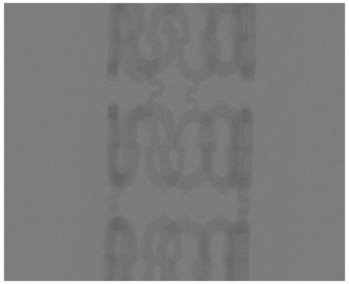
Spatial occupancy enumeration are unambiguous, but not unique solid representations and too verbose.

3D X-Ray Computed Tomography



https://youtu.be/SI6I4Ukd8uo

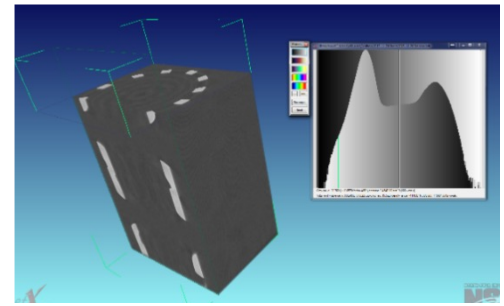
Voxel Representation and Geometric measurements

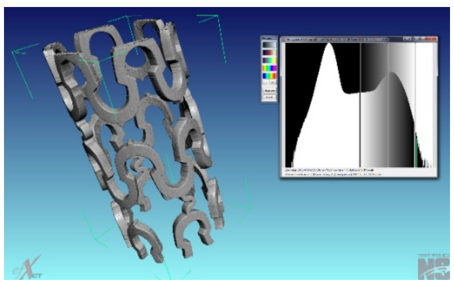


X-Ray Image Acquisition

↓

Volume Reconstruction





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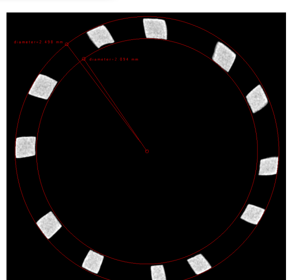
Boundary Points Determination

↓

CAD Model Comparison | Geometrical Verification | Reverse Modeling



3D X-Ray Computed Tomography: Uncertainty